

Recruiting Better Teachers? Evidence from a Higher Education Reform in Chile

Adriano De Falco
Lund University

Benjamin Hattemer
University of Helsinki

Sofía Sierra Vásquez
University of St. Gallen

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Abstract

This paper analyzes the impact of a recruitment policy in Chile designed to improve the quality of new teachers by incentivizing high-achieving and restricting low-achieving high school graduates from entering the teaching profession. We document that the reform effectively improved the average test scores of new teachers. Using a teacher value-added (TVA) model, we find that the reform increased TVA for mathematics but not for Spanish teachers. Finally, we find that higher test scores account for only a small part of the effect for mathematics, which instead appears to operate through selection on unobserved teacher characteristics.

Keywords: *educational policy, scholarship, teacher recruitment, teacher value-added*

JEL: I21, I22, J24, H52

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1 Introduction

Teacher quality plays a critical role in improving student learning and long-term outcomes (Rockoff, 2004; Chetty et al., 2014b; Bau and Das, 2020). Thus, how to design policies to improve teacher quality is a long-standing question. Among possible interventions, recruitment policies designed to attract more talented and motivated individuals into the teaching profession have become increasingly relevant. This is especially true in low- and middle-income countries, where teacher shortages and negative selection into the profession make recruitment a particularly promising lever (World Bank, 2017).

As a policy tool, recruitment policies offer a more sustainable path than strategies that rely on dismissing low-performing teachers once they are already in the classroom (Rothstein, 2015; Staiger and Rockoff, 2010). Yet evidence on their effectiveness remains limited. The core challenge lies in having reliable measures to *ex-ante* screen candidates on their potential to become effective teachers, and designing incentives strong enough to shift occupational choices. Still, even if both conditions are met, recruitment policies may generate unintended selection on unobservable qualities. Crucially, the direction of this selection is theoretically ambiguous, meaning it could either hinder or reinforce the effectiveness of the policy. Understanding whether recruitment policies can raise teacher quality, and through what channels, is therefore an open and consequential empirical question.

In this paper, we use rich administrative data from Chile that allows us to track individuals from their time as high school students all the way through higher education and into the labor market. Additionally, for those who choose to become teachers, we are able to observe not only their academic ability as students but also their performance as teachers. This provides unique information not only on who selects into teaching, but also on their effectiveness once in the classroom, making it an ideal setting to study recruitment into the teaching profession. We use these data to evaluate whether a reform targeting prospective college students succeeded in improving the average quality of newly trained teachers.

The reform was designed to reshape the academic ability composition of high-school graduates enrolling in undergraduate teacher training programs. To screen candidates, it relied on the national standardized university entrance exam, known as *Prueba de Selección Universitaria* (PSU). The reform incentivized the enrollment of high-achieving students by creating a scholarship for those enrolling in teacher training programs. The scholarship imposed requirements on both college students and higher education institutions: on the one hand, individuals who received the scholarship were required to work as teachers in

publicly funded schools for three of the seven years following college graduation; and on the other hand, universities were required to limit the number of low-achieving college students admitted to their programs so that high-achieving ones would be eligible for the scholarship.¹

The implicit assumption behind this reform was that high-achieving students become more effective teachers than low-achieving students. Evidence linking teachers' cognitive ability to student learning gains lends some credibility to this assumption, particularly in mathematics (Metzler and Woessmann, 2012; Bold et al., 2017; Bietenbeck et al., 2018; Jacob et al., 2018; Hanushek et al., 2019; Bau and Das, 2020). However, the assumption does not transfer easily across contexts or subjects: the relationship is mixed across countries, tends to be weaker or absent outside mathematics, and has never been established using pre-college test scores and teacher value-added measures directly (Araujo et al., 2016; Bardach and Klassen, 2020). More importantly, the reform's two-sided design makes selection on unobservables a first-order concern. The scholarship attracts individuals who would counterfactually have chosen not to enter the profession, raising the question of what unobserved characteristics kept them out. Similarly, the restrictions deter some low-achieving students from enrolling, raising the symmetric question of what distinguishes those who are deterred. In both cases, the reform directly shapes who enters teaching along dimensions that are unobserved, making the net effect on teacher quality theoretically ambiguous. Our empirical assessment of this policy answers two main questions: was this recruitment policy effective in increasing average teaching quality? And if so, what mechanisms can explain the estimated effects?

We begin by documenting the effectiveness of the reform in permanently shifting the academic ability distribution of newly trained teachers. As intended by the reform, we find that the scholarship encouraged high-achieving college students to enroll in teacher training programs, while the restrictions discouraged low-achieving ones. Tracking individuals through higher education and into the labor market, we show that these effects persisted over time, resulting in a shift in the academic ability distribution of active teachers. These results corroborate the recruitment effects of the reform previously documented by Gallegos et al. (2022). We extend their findings by showing that high-achieving teachers are more likely to work in publicly funded schools, and that these effects persist up to five years after graduation, well beyond the scholarship requirements.

Next, we turn to the central question of whether the reform improved teacher quality. We rely on a Teacher Value-Added (TVA) model using the standardized test scores of

¹Note that this was not a complete ban on low-achieving college students, as they could still enroll in these restricted slots at scholarship-eligible universities, or enroll at non-scholarship-eligible institutions.

approximately 430,000 primary school students taught by nearly 9,500 teachers. Our TVA specification controls for students' prior achievement and a rich set of student, classroom, and school characteristics to address any non-random sorting of teachers. To estimate the effect of the reform, we compare the TVA of teachers across PSU cohorts. This cross-cohort analysis leverages the fact that teachers from different PSU cohorts are observed teaching in the same year and grade. This allows us to net out any contemporaneous changes to the general education system. We additionally control non-parametrically for experience, accounting for the mechanical difference in seniority across cohorts.

Our main results show that the reform was successful in increasing the average TVA for mathematics teachers by 38% of a standard deviation. For Spanish teachers, we find a precise null effect. To our knowledge, this is the first paper to provide direct evidence that a recruitment policy targeting entry into teacher training programs can raise average teacher value-added. We complement these findings with a cost-benefit analysis. Even under conservative assumptions about how long the value-added gain persists and how long teachers remain in the profession, the benefits exceed the costs, often several times over. Because the scholarship is a one-time cost incurred at entry, rather than a payment that recurs every year a teacher stays, as with performance pay or higher salaries, these returns are achieved at comparatively low fiscal cost.

We then discuss the mechanisms behind the documented increase in teacher quality. We first examine the validity of PSU scores as a predictor of pre-reform teacher quality. PSU scores are positively correlated with TVA for mathematics teachers but not for Spanish teachers, which helps explain the heterogeneous effects across subjects. Strikingly, even for mathematics teachers, the observed increase in academic ability accounts for only around 10% of the overall effect on TVA. The reform therefore improved teacher quality largely through channels other than the one it was designed to operate on.

To rationalize the drivers behind the bulk of the effect, we develop a simple occupational choice framework à la Roy (1951). The model formalizes two competing forces: intrinsic motivation for the profession and the attractiveness of outside career options. It predicts that scholarships may raise teacher quality by attracting individuals whose skills are valued in outside occupations but who would also be effective teachers. Restrictions raise quality by screening out individuals at the margin who are also the least motivated candidates, provided that motivation is positively correlated with teaching effectiveness. We test these predictions by repeating our cross-cohort analysis of TVA, conditional on PSU, separately for the groups the reform targeted. Consistent with the model, the improvement in mathematics TVA appears only among the high-achieving teachers targeted by the scholarship and the low-achieving teachers targeted by the entry restrictions; those

in the middle of the PSU distribution, whom the reform did not directly affect, show no change. We complement these findings using teacher survey data and find suggestive evidence of the motivation channel: those facing restrictions exhibit higher motivation, while scholarship recipients show no evidence of lower motivation. Overall, the reform improved teacher quality primarily through selection on unobservable dimensions, with academic ability, as measured by PSU scores, playing only a minor role.

Related literature. Our paper contributes to two different strands of the literature.

First, we contribute to a growing literature on teacher recruitment policies. This literature has mostly studied the short-term effects of two types of interventions: entry barriers and financial incentives for prospective teachers. Entry barriers, often in the form of tests that screen candidates on content knowledge and attitudes, have been shown to improve teaching quality.² Nonetheless, they can have unintended negative effects on student learning if not properly designed (Goldhaber, 2007; Angrist and Guryan, 2008; Busso et al., 2024). Financial incentives take several forms, and their effect on who enters teaching hinges on their design. General salary increases for teachers have only modest effects on drawing higher-ability entrants into teaching (Ballou and Podgursky, 1995; Ortega, 2010). Pay schemes that reward teacher performance, whether at schools' discretion or through explicit performance contracts, attract higher-quality teachers, though the evidence to date speaks only to teachers moving across districts or schools rather than to new entrants choosing teaching over another career (Cabrera and Webbink, 2020; Biasi, 2021; Leaver et al., 2021).

We contribute to this literature by evaluating a recruitment policy with two distinctive features. First, it combines entry barriers and financial incentives, targeting opposite ends of the ability distribution: a scholarship for high-achieving students, and enrollment restrictions for low-achieving ones. This allows us to study how both channels operate within a single, internally consistent setting, rather than relying on evidence assembled across studies that differ in context, population, and design. Second, instead of targeting the hiring of already trained teachers, the studied reform targeted the earliest possible stage of recruitment, namely enrollment in teacher training programs. Existing studies have focused on the recruitment of trained teachers, i.e., individuals who have already acquired a certain level of human capital, which pre-conditions occupational sorting (Behrman et al., 2016). By studying effects at the enrollment stage, we characterize the individuals the policy induces into or out of the profession, that is, its full extensive-margin response. Recruitment studies of already-trained teachers cannot speak to this margin, and ignoring

²See Estrada (2019) for Mexico and Araujo Piedra et al. (2025) for Ecuador.

it can lead to a significant underestimation of the long-term benefits of such interventions (De Ree et al., 2018). Moreover, this literature typically evaluates recruitment policies by whether they shift the observable composition of entrants, test scores, credentials, or experience, rather than by their effect on teaching quality itself. We instead estimate the reform's effect directly on teacher value-added, a measure of teaching effectiveness based on student outcomes rather than on teachers' own test scores, and one that nets out student sorting, allowing us to test whether a shift in observable academic ability actually translates into better teaching.

Second, we contribute to the literature on the determinants of teacher quality. There is broad consensus that teacher quality is hard to predict *ex ante*, which makes finding reliable screening devices a central and unresolved challenge. Characteristics observable at the time of hire, including credentials, degrees, and personality traits, are weak predictors of teacher effectiveness (Staiger and Rockoff, 2010; Chingos and Peterson, 2011; Rockoff et al., 2011; Bau and Das, 2020). Standardized exam scores, such as college entry and certification exams, are a partial exception and have been shown to predict effectiveness in some settings, though this evidence does not directly relate pre-college test scores to teacher value-added (Clotfelter et al., 2006; Goldhaber et al., 2017; Jacob et al., 2018; Bardach and Klassen, 2020). We contribute to this literature by directly relating a pre-college standardized test to teacher value-added: we show that college entry exams are at best a modest predictor of value-added for mathematics teachers, and uninformative for Spanish teachers.

Our paper is most closely related to Gallegos et al. (2022), who study the recruitment effects of the same reform and document a robust positive pre-reform correlation between pre-college academic achievement and teacher productivity, as measured by graduation, labor market outcomes, exit exam scores, and on-the-job evaluations. The authors then use this correlation to project the reform's likely effects on teaching quality. We extend this analysis in three important ways. First, we directly estimate the reform's effect on teacher value-added using large-scale administrative test-score records for the students taught by pre- and post-reform cohorts. Rather than extrapolating from a pre-reform correlation, we allow for this relationship to vary as a result of the incentives and restrictions imposed by the reform. Second, we use TVA as our outcome, a measure which is based on student performance rather than teachers' own test scores, and which nets out the effect of sorting of students to teachers and of teachers to schools.³ This lets us capture actual differences in teaching effectiveness and document heterogeneous effects across subjects. Third, we go

³See Cohen and Goldhaber (2016); Steinberg and Garrett (2016); Bacher-Hicks et al. (2019); Bacher-Hicks (2022).

beyond the cognitive ability channel: while Gallegos et al. (2022) interpret the correlation between entrance scores and productivity as evidence that screening on ability alone can improve teacher quality, we show that ability explains only a small share of the reform’s effect, with the bulk operating through selection on unobservables that a pre-reform correlation cannot capture. Our results are in line with the literature showing a weak correlation between observable characteristics and value added, and challenge the idea that college entry exams may serve alone as a tool to increase teaching quality. Indeed, we attribute most of the effect of the reform to unintended selection effects.⁴

Road map. The rest of the paper proceeds as follows. Section 2 presents the institutional setting and the reform we focus on. Section 3 presents the data and results on enrollment, graduation, and employment at schools. In Section 4, we estimate teacher value-added and quantify the change in teacher quality due to the reform. Section 5 discusses the results, presenting a theoretical framework to interpret our findings and examining the mechanisms behind the effects on teacher quality. Finally, in Section 6 we conclude.

2 Background

In this section, we present some relevant features of the Chilean institutional setting in the years surrounding the reform and discuss some key aspects of it.

2.1 Chilean institutional setting

The higher education system. In Chile, there are three types of higher education institutions (HEIs): vocational institutions, technical institutes, and universities. Moreover, universities can be categorized based on whether they are part of a network called CRUCH (Consejo de Rectores de Universidades Chilenas). They are generally considered of higher prestige and are able to attract better students on average. To access most universities, students have to take a nationwide standardized test which, between 2004 and 2020, was called *Prueba de Selección Universitaria (PSU)*.

⁴A recent working paper by Pal (2025) also studies the same reform, using it to estimate a structural model of higher-education admissions in which compositional changes in entrance scores are the only channel through which the reform affects teacher quality. Pal (2025) also reports reduced-form evidence on the reform’s effect on teacher quality, based solely on changes in the composition of entrance scores across programs that did and did not admit BVP enrollees. This approach also assumes that changes in the PSU composition are the only relevant ones. Our paper predates Pal (2025) and was first made publicly available in 2024 (<https://dx.doi.org/10.2139/ssrn.4874361>).

The PSU was offered once a year in December. It consisted of four components: mathematics and Spanish, which were compulsory, and science and/or history. For each component, the raw scores were standardized at the national level to produce a normal distribution of scores ranging from 150 to 850, with a mean of 500 and a standard deviation of 110. Admission to each program was determined by the number of places available and the number of applicants. The score used to determine admission was calculated as a weighted average of high school GPA and all sub-components of the PSU test, with program-specific weights. Programs would rank students based on the program-specific score, and fill the available spots based on their ranking.⁵

In 2011, annual tuition fees posted by HEIs were high relative to other OECD countries. For this reason, Chilean higher-education students relied heavily on financial aid. The Chilean government provided access to loans and scholarships, covering up to 90% of the tuition fees. Therefore scholarships were a useful policy tool to attract individuals into specific university programs.⁶ Access to most financial aid was determined by the weighted average of the PSU test scores obtained in mathematics and Spanish.

The teaching profession. In order to work in the school system, teachers need to have a higher education degree in teaching.⁷ These degrees are level-specific and can only be granted by some institutions: for preschool and elementary teachers, this can be a university or a vocational institution, while the degree to become a high-school teacher can only be granted by universities. At the time of the reform, 86% of higher-education students enrolled in a teacher training program were enrolled in a university program and 14% were enrolled in vocational institutions.

While at the time of the reform there was no shortage of teachers in Chile (OECD, 2010), individuals who pursued teacher training were historically negatively selected in terms of their PSU score (Gallegos et al., 2022). Figure 1 plots the distribution of PSU scores for students enrolling in universities in 2010 by field of study. The figure shows that students enrolling in teacher training programs performed significantly worse than students enrolling in other fields, scoring on average almost one standard deviation below the mean of all university enrollees taking the PSU.

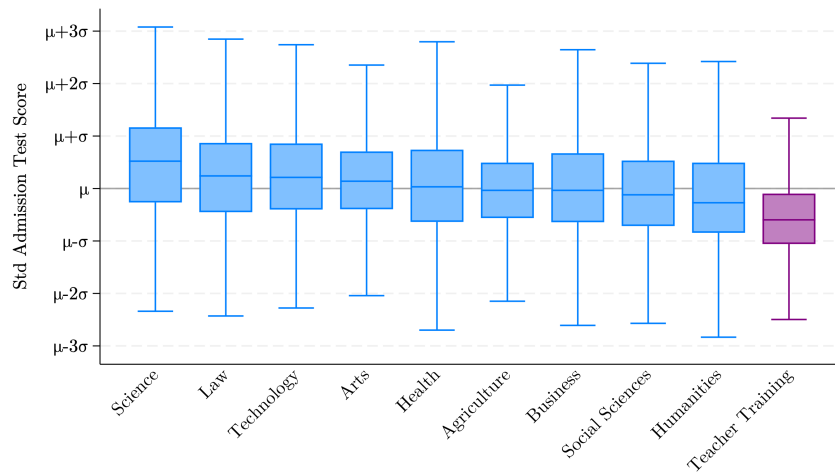
After graduation, new teachers can decide to apply for a teaching job. In Chile, there

⁵See Kapor et al. (2024) for a detailed description of the algorithm matching students to programs.

⁶Recent papers highlight how, in our setting, students are credit-constrained (Solis, 2017) and price-sensitive to tuition changes (Bucarey et al., 2020; De Falco and Reichlin, 2025).

⁷There are a few rare exceptions, such as teachers recruited through Enseña Chile, a program modeled similarly to Teach For America, which places university graduates in underserved schools. However, they represent only a negligible share of the teaching workforce (De Gregorio and Neilson, 2021).

Figure 1: PSU scores of higher-education students enrolled in HE in 2010 by field of study



Note: The figure plots the distribution of PSU scores for high school graduates enrolling in different university programs in 2010. The gray horizontal line represents the mean test score of all university enrollees who took the PSU.

are three types of schools determined by their funding and administration authority: *public* schools which are publicly funded and managed, representing 46.2% of the schools in 2010; *voucher* or *subsidized* schools which are publicly funded, but privately managed, representing 47.7% of the schools; and fully *private* schools, representing 5.4% of the schools. The recruitment of teachers varies by type of school. For public schools, municipal authorities organize the hiring of teachers. Vacancies are publicly posted, and a commission reviews applications by rating applicants based on professional performance, seniority, and other formal training. Wages are set nationally and are low compared to other occupations.⁸ In the privately managed schools, schools have discretion in deciding how and whom to hire. There is, therefore, little scope for public schools to try to hire higher-quality teachers. In privately managed schools, however, wages can be negotiated and are generally more competitive than in public ones.

2.2 The 2011 reform

This negative selection into teacher training programs motivated the Ministry of Education to implement a reform in 2011 aimed at changing the distribution of PSU takers pursuing a teaching career. More precisely, the reform introduced a scholarship, called the *Beca*

⁸Tincani (2021) shows that non-teaching salaries in Chile are roughly 60% higher, conditional on the level of education.

Vocación de Profesor (BVP) for high-achieving college students who enrolled in an *eligible* teacher training program.

The BVP offered several benefits, most notably a full tuition waiver, which was awarded solely on the basis of the average PSU score in mathematics and Spanish. Thus, all individuals who scored one standard deviation above the mean (i.e. above 600 points) and who enrolled in an eligible program, were offered the opportunity to study for free.⁹ In addition, individuals who scored above 700 points could receive a monthly stipend of 80,000 Chilean pesos (almost 45% of the 2011 national minimum wage), and those who scored above 720 points were given the opportunity to study one semester abroad. Given the standardized distribution of PSU scores, around 20% of all PSU takers were potentially eligible for the tuition waiver and about 5% for the additional benefits. Our discussion of the impact of the reform will focus on the tuition waiver, mainly because the effects of the additional benefits are hard to detect due to the small number of eligible students. Individuals scoring just below the 600 threshold were eligible for different types of financial aid depending on their family income, choice of institution, and cohort of study. Appendix Table B1 summarizes financial aid eligibility by these three dimensions.

The BVP required its beneficiaries to work in publicly funded schools for at least three years in the seven years following college graduation. This requirement was reduced to two years if the school was located in a rural area. Individuals who did not meet this requirement had to repay the scholarship.

Finally, the introduction of the BVP imposed some requirements on universities. To be considered *eligible*, programs had to meet two conditions. First, the program had to be *accredited*¹⁰, and second, they could only admit up to 15% of their applicants with a PSU score below the standardized mean (i.e., below 500 points). In practice, this meant that low-scoring PSU takers had fewer places available to enroll in teacher training programs, as they could either try to enroll in the few slots offered by eligible programs or enroll in non-eligible programs. Overall, in our study period around 80% of the programs of CRUCH universities and 15% of the programs of non-CRUCH universities accepted BVP recipients, effectively limiting the access to their programs to applicants scoring

⁹Note that while the standard deviation of each individual component is 110, the standard deviation of the combined mathematics and Spanish scores is around 100. This cutoff was set to be slightly lower (580) for students who were eligible for an additional scholarship, called *Beca Excelencia Académica* (BEA), which was awarded to students coming from public and subsidized schools, who were in the top 10% of students in their respective high schools.

¹⁰Since 2006, Chile has had an accreditation system for higher education institutions, designed to regularly evaluate and guarantee the quality of the programs offered by these higher education institutions. The accreditation process is carried out by an agency of the Ministry of Education. While accreditation is generally voluntary, this is not the case for medicine and teacher training programs, which since 2009 must go through the process.

below the mean. Additionally, almost 70% of the eligible programs did not accept any applicant scoring below 500. The other 30% of eligible programs accepted on average 7% of candidates scoring below 500.

3 Recruitment

In this section, we document the effectiveness of the reform in terms of incentivizing (or discouraging) prospective college students from the top (bottom) of the PSU distribution to enroll in a teacher training program and subsequently pursue careers in publicly funded schools. To quantify the impact of the reform, we exploit the local variation of its two components in a Regression Discontinuity (RD) framework and provide causal estimates of their local impact.

3.1 Data

We use rich administrative data on the universe of PSU test takers and HEI enrollees between 2009 and 2014, provided by the Chilean Ministry of Education (MINEDUC) and the PSU administrative body (DEMRE). In addition to PSU scores, DEMRE provided information on the sociodemographic characteristics of test takers, including gender, high school GPA, and parental education. We merge this information with MINEDUC data on enrollment, graduation, and a rich set of program characteristics, including institution type, field of study, and data on application and take-up of financial aid. Finally, we combine this with administrative records on the universe of teachers working in primary and secondary education, also provided by MINEDUC. This results in a unique panel dataset that tracks PSU takers through their university years and into their teaching careers up to 2024, allowing us to identify key aspects of their higher-education outcomes and subsequent career trajectories as teachers.

3.2 Empirical strategy

We carry out a regression discontinuity analysis on first-time PSU takers between 2011 and 2014. This restriction avoids including individuals who may retake the test in order to improve their scores, either to become eligible for the BVP or to expand their choice set of institutions. We estimate the following model

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 D_i \times PSU_i + \beta_3 PSU_i + v_i \quad (1)$$

where Y_i is our outcome of interest, D_i is either an indicator of BVP eligibility or of being below the restriction cutoff for eligible institutions. We allow Y_i to depend linearly on the PSU score (PSU_i), with different slopes above and below the relevant threshold. Our parameter of interest, β_1 , captures the difference in outcomes for individuals just affected by the scholarship eligibility or the restrictions, relative to those marginally unaffected. For both cutoffs, we apply a triangular kernel-weighting around the threshold. For the BVP eligibility cutoff, bandwidths are optimally chosen following Calonico et al. (2019). For the 500-point cutoff, bandwidths are set to 24 in order to avoid capturing the effects of an important policy for prospective college students in that period, namely the student loan program, which was available for those scoring above 475 points.¹¹ Summary statistics of the analyzed sample are presented in Appendix Table B2.

Identification. For the sharp RD estimate β_1 to identify the causal effect of the scholarship and the restrictions, standard identification assumptions must hold. Most importantly, potential outcomes must be continuous around the cutoff. This assumption could be violated if PSU takers were able to manipulate their scores. Results from a McCrary test (McCrary, 2008), presented in Appendix Figure A1, provide no evidence of manipulation. We complement this with a balance test, where we estimate model (1) for some sociodemographic characteristics. Results in Appendix Table B3 show that being marginally eligible is not significantly correlated with observables.

As mentioned above, individuals who are marginally below the cutoff may retake the test in subsequent years to improve their scores. Appendix Figure A2 plots the probability of becoming eligible in the second, third, and fourth years after high school graduation as a function of the first-time PSU score. Only about 10% of retakers manage to improve enough to change their eligibility status. For this reason, our results would remain virtually unchanged under a fuzzy RD design.

3.3 Results

Table 1 reports the RD parameters at the 600 cutoff (Panel A) and the 500 cutoff (Panel B). For enrollment in teacher training programs, individuals marginally eligible for the scholarship are 27% more likely to enroll (1.9 p.p.), while those below the 500 cutoff are 36% less likely (2.9 p.p.).¹² These effects are large relative to the baseline but modest in absolute

¹¹See Solis (2017) for a discussion of the loans programs.

¹²In Appendix Figures A3 and A4 we show heterogeneous results by type of institution. Results are mostly driven by the most prestigious universities (CRUCH), which completely prevent students scoring below the 500 cutoff from enrolling and are more attractive to BVP-eligible test takers.

terms, as teacher training enrollees are a small share of PSU takers. Column (2) shows results for graduation, defined as obtaining a degree from a teacher training program within eight years of the first PSU attempt: the effect is a 13% increase at the 600 cutoff and a 17% reduction below 500.¹³

Table 1: Regression Discontinuity results

	(1) Enrolled	(2) Graduated	(3) Employed	(4) Public	(5) Voucher	(6) Private	(7) Rural
<i>Panel A. 600 cutoff</i>							
RD Estimate	0.019*** (0.003)	0.008** (0.003)	0.009*** (0.003)	0.004*** (0.001)	0.006*** (0.002)	0.000 (0.001)	0.001 (0.001)
Bandwidth	43	47	53	62	56	56	62
Observations	125,458	135,437	152,497	178,390	162,619	162,619	178,390
Baseline Mean	0.068	0.061	0.047	0.014	0.030	0.008	0.004
Percentage Change	27.4	13.1	19.3	29.0	20.4	5.7	23.3
<i>Panel B. 500 cutoff</i>							
RD Estimate	-0.029*** (0.005)	-0.013*** (0.004)	-0.012*** (0.003)	-0.005*** (0.002)	-0.009*** (0.002)	0.000 (0.001)	-0.002* (0.001)
Bandwidth	24	24	24	24	24	24	24
Observations	113,468	113,468	113,468	113,468	113,468	113,468	113,468
Baseline Mean	0.082	0.075	0.054	0.021	0.036	0.003	0.008
Percentage Change	-35.7	-16.7	-21.4	-22.6	-24.6	13.1	-22.6

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table shows Regression Discontinuity parameters where all specifications are estimated using weighted local linear regressions and consider high school graduates taking the PSU between 2011 and 2014. For the 600 cutoff, the RD Estimate corresponds to the effect of being marginally *above* 600 or marginally eligible for the scholarship; bandwidths are chosen optimally according to Calonico et al. (2019). For the 500 cutoff, the RD Estimate corresponds to the effect of being marginally *below* 500 or marginally affected by the restriction; bandwidths are set to 24 to avoid capturing the effect of the loan eligibility cutoff at 475 points. Standard errors are clustered at the PSU test score level and reported in parentheses. *Observations* correspond to the number of observations with non-zero weight given the chosen bandwidth. The *Baseline Mean* refers to the outcome mean for marginally unaffected students. The *Percentage Change* is computed as the coefficient over the Baseline Mean (times 100). *Enrolled* is an indicator for individuals enrolling in a teacher training program in the year after high school graduation. *Graduated* is an indicator for individuals graduating from a teacher training program within eight years of high school graduation. *Employed* is an indicator for individuals observed working in any school within 2 years of their graduation from a teacher training program. *Public*, *Voucher*, *Private*, and *Rural* are indicators for individuals observed working in each particular type of school within 2 years of their graduation from a teacher training program. Results are conditional on taking the college admissions test.

Our estimates for enrollment and graduation align with those in Gallegos et al. (2022), who also document effects on employment shortly after graduation, albeit only for the

¹³Although we pool all cohorts, Appendix Figure A5 documents some heterogeneity across cohorts. For the BVP treatment, effects are particularly pronounced in 2011, most likely due to the rollout of more generous financial aid from 2012 onward (De Falco and Reichlin, 2025)

2011 cohort. We go further in two directions that are central to interpreting the effect of the policy. First, we track where the new teachers are employed, distinguishing public, voucher, and private schools. Even among publicly funded schools, public and voucher schools differ markedly in the socioeconomic composition of the students they serve, and hence in their attractiveness to teachers; how the induced teachers sort across them is therefore informative in itself. Second, we follow employment up to five years after graduation, beyond the three-year period the scholarship mandates, to see whether the induced sorting persists once the requirement ends.

Columns (3) to (7) of Table 1 report the effect on the probability of being employed in a school within two years of graduation, overall and by school type. Scholarship-eligible takers are about 19% more likely to be employed relative to the baseline, an effect concentrated in public and voucher schools, that is, the publicly funded sector the reform targets. Consistent with the differences in attractiveness noted above, public schools have a lower baseline (1.4% vs 3%); yet the effect on public schools is larger in relative terms (29% vs 20%), even if smaller in absolute size. Symmetrically, those constrained by the restriction are about 21% less likely to be employed, again in public and voucher schools. Importantly, this sorting is not a short-lived response to the scholarship's requirements: Appendix Figure A6 shows that the employment effects persist up to five years after graduation, beyond the three-year mandate, for both public and voucher schools.

Robustness and sensitivity. We perform a series of robustness checks to test the validity of our results. Supporting the identification of the RD parameters, Appendix Table B4 and Appendix Figure A7 show that the RD estimates are robust to different local polynomials and bandwidth choices.¹⁴ Additionally, Appendix Figure A8 shows that the documented discontinuities were not present in the years prior to the reform. We test formally whether these discontinuities existed in Appendix Table B5, where we estimate model (1) for cohorts before the reform.

Effects away from the cutoff and change in PSU distribution. The RD estimates are local to the cutoffs and do not speak to the change in the PSU distribution of the whole population of enrollees. We therefore complement them with an across-cohort comparison of the 2009–2010 and 2011–2012 cohorts, replicating Table 1 and Figure A6 in Appendix Table B6 and Appendix Figure A9. The estimates are broadly consistent with the RD

¹⁴Not surprisingly, given that observables are balanced at the cutoff, the results remain virtually unchanged when adjusting for observable characteristics (such as cohort fixed effects or student covariates) and are robust to alternative choices of standard errors. We remain conservative and report clustered standard errors.

results. This before-after design rests on a stronger assumption, the absence of underlying trends in the outcomes. Appendix Figure A10 shows that pre-trends are limited, especially for the scholarship-eligible. Still, this design is less well identified than the RD, so we treat these results as suggestive.¹⁵ After 2011, teacher training enrollees had higher PSU scores on average: the mean rose by 10 points, about 15% of its overall variation, with similar increases at the 10th and 90th percentiles (Appendix Figure A12). Notably, alongside the shift in composition toward higher PSU scores, the reform was associated with an overall reduction in the number of individuals entering and completing teacher training (Appendix Figure A13).

Taken together, we argue the reform increased the academic ability of new teachers, especially in publicly funded schools. Whether this translated into an increase in their average performance as teachers is the question we seek to answer in the remainder of this paper.

4 Teacher quality

In this section, we quantify the change in teacher quality induced by the reform, as measured by the value added by teachers to their students' test scores. The analysis focuses on elementary mathematics and Spanish teachers. We begin by describing the data and sample, followed by the empirical model and the procedure for estimating the standard deviation of teacher value added, which serves as a benchmark for interpreting the policy effects. We then present the main results on the change in teaching quality induced by the reform.

4.1 Data and population of interest

We complement the data described in Section 3 with administrative data provided by the Ministry of Education (MINEDUC) covering the universe of Chilean students enrolled in primary education. The data include information on students' average GPA, gender, and date of birth, as well as the school, grade, and classroom in which they are enrolled each calendar year. In addition, we use individual-level results of a standardized test called SIMCE, provided by the *Agencia de Calidad de la Educación*, for the years 2012 to 2018. This test is conducted annually for all Chilean students in a specific grade. Appendix

¹⁵See Appendix Figure A11 for a non-parametric representation of the probability of working in different types of school, as a function of PSU. The probabilities are computed conditional on graduating from a teacher training program, to show how the graduates are distributed in different school types, and how this changed after the reform.

Table B7 summarizes the grades tested each year. Information on parental education and family income quintiles is collected by the *Agencia de Calidad de la Educación* through a survey. Crucially, we can merge these data with information on each student's teachers, including their years of experience, PSU test scores, and higher education enrollment and graduation.

We restrict our sample of interest to primary school teachers of mathematics and Spanish. We apply two further sample restrictions closely related to our identification strategy. First, we limit the sample to teachers for whom PSU test score data are available. Information on PSU takers is available starting in 2004. Thus, we focus on seven teacher cohorts who took the PSU before the reform - i.e. whose decisions were not affected by it - and four cohorts who took it afterward. Second, we focus on teachers whose students have complete information on all variables included in the value-added model, which we describe in the next section. This restriction implies that our analysis focuses only on 6th- and 8th-grade teachers, since we lack data on both mathematics and Spanish lagged test scores for students in other grades (see Appendix Table B7). For these students, lagged test scores are measured two years before the current test. Taken together, this means that our population of interest is the universe of mathematics and Spanish teachers trained between 2004 and 2014 who taught in 6th and 8th grade in a year when SIMCE was administered.

Appendix Tables B8 and B9 provide some summary statistics of our sample of teachers. Consistent with the findings discussed in the previous section, teachers who pursued their training after the reform and obtained a PSU score above 600 are more likely to work in public and subsidized schools. Unsurprisingly, teachers who enrolled after the reform have, on average, less than one year of teaching experience, compared to roughly 2.5 years for those who enrolled before. Finally, Appendix Figure A14 shows the distribution of PSU scores before and after the reform for teachers in our sample. On average, standardized PSU scores increased by 0.31 for math teachers and by 0.40 for Spanish teachers after the reform.

4.2 Estimation of teacher quality

We begin by estimating the standard deviation of the teacher effects. This provides an insight into the heterogeneity among teachers in terms of their impact on student learning. More precisely, it gives a measure of teacher effectiveness in terms of the return to test score from being taught by a teacher whose impact is one standard deviation higher. Moreover, it serves as a useful benchmark against which to contrast our results on the change in TVA estimates induced by the reform.

4.2.1 The variance of teacher effects

Consider the following model

$$Y_{isjgtc} = \lambda Y_{i,t-1} + \gamma \mathbf{X}_{i,t} + \theta_{cjst} + \theta_s + \theta_{js} + \alpha_t + \mu_g + v_{isjgtc} \quad (2)$$

where Y_{isjgt} is the (mathematics or Spanish) test score of student i , in school s , taught by teacher j , in grade g , at time t , and in classroom c . The variable $Y_{i,t-1}$ corresponds to the lagged value of student i 's test score, and $\mathbf{X}_{i,t}$ is a vector of individual characteristics measured at time t . They include indicators for being female, belonging to a specific family income quintile, and parental education. Parameters θ_{cjst} , θ_s and θ_{js} represent respectively classrooms, schools and teachers effects. Finally, α_t and μ_g are year and grade fixed effects, and v_{isjgtc} are idiosyncratic random shocks. Estimating the variance of θ_{js} , defined as σ_{js}^2 , comes with some complications. In our setting, many parameters from model (2) are not identifiable. For example, as long as teachers teach in one school and one class, it is not possible to separate the teacher effect from the school or classroom effects. The empirical variance of $\hat{\sigma}_{js}^2$ is therefore an upward biased estimator for the true variance σ_{js}^2 . In order to produce an unbiased estimator for σ_{js}^2 we start by estimating the following value-added model

$$Y_{isjgtc} = \lambda Y_{i,t-1} + \gamma \mathbf{X}_{i,t} + \delta_{cjst} + \alpha_t + \mu_g + v_{isjgtc} \quad (3)$$

where δ_{cjst} combines classroom, teacher, and school effects. It can be shown using model (2) that the variance of the school effects can be expressed as $\sigma_s^2 = Cov(\delta_{cjst}, \delta_{cj'st})$. That is, the covariance of classroom effects between classes taught by different teachers, during the same year. Using a similar argument, the covariance of classroom effects taught by the same teachers identifies the sum of the school and teacher variances, that is $Cov(\delta_{cjst}, \delta_{c'jst}) = \sigma_s^2 + \sigma_{js}^2$ (Kane and Staiger, 2008).¹⁶

Appendix Figure A15 shows the frequency with which we observe different teachers in the same school, teachers in more than one classroom and those observed in more than one year. Note that within the same year we do not often observe different teachers in the same school. However, our data have the advantage that across years, in 40% of the cases we observe different teachers working in the same school, and 55% of the teachers are observed in more than one classroom. This implies that we can estimate σ_s^2 as the covariance of $\hat{\delta}$ for two different teachers, in the same school, regardless of the year ($Cov(\hat{\delta}_{cjst}, \hat{\delta}_{cj'st'})$). Similarly, we estimate $\sigma_s^2 + \sigma_{js}^2$ as the covariance of $\hat{\delta}$ for different

¹⁶The assumption behind this result is the independence of the error term v_{isjgt} and $v_{isjgt'}$. This would be arguably violated if teachers teach the same students across years. In our sample, this happens in less than 1% of the cases.

classrooms taught by the same teacher, in the same school, regardless of the year.¹⁷ We estimate σ_{js}^2 and σ_s^2 separately for mathematics and Spanish.

Table 2: Effect of 1 SD increase in school and teacher effects

	With weights		Without weights	
	Math	Spanish	Math	Spanish
	(1)	(2)	(3)	(4)
School	0.22	0.12	0.22	0.13
Teacher	0.10	0.11	0.09	0.11

Note: The table reports the effect of receiving a one standard deviation better school and teacher on students' subject-level test scores. School variance σ_s^2 is computed as $Cov(\hat{\delta}_{cjs}, \hat{\delta}_{c'js})$, and the teacher variance σ_{js}^2 is computed as the difference between $Cov(\hat{\delta}_{cjs}, \hat{\delta}_{c'js})$ and $Cov(\hat{\delta}_{cjs}, \hat{\delta}_{c'js})$, where $\hat{\delta}_{cjs}$ is the OLS estimate of δ_{cjs} in model (3). In columns (1) and (2), we use the sum of the sizes of both classrooms involved as analytical weights when computing the covariance. In columns (3) and (4), we exclude the weights such that each classroom pair has the same weight when computing the covariance.

Table 2 summarizes the results.¹⁸ The standard deviation of teacher effects is 0.10 in math and 0.11 in Spanish. In student test score terms, moving a student from a teacher at the 5th percentile of the value-added distribution to one at the 95th percentile would raise the student's test scores by 0.33 SD in math and 0.36 SD in Spanish. These estimates are similar to those found in the United States (Bacher-Hicks, 2022; Chetty et al., 2014a). However, they appear somewhat smaller in magnitude when compared to research conducted in developing countries, such as the studies by Bau and Das (2020) and Buhl-Wiggers et al. (2022). Closer to our setting, Urrea (2018) and Barrios-Fernández and Riudavets-Barcons (2024) use Chilean SIMCE data and report similar estimates for the standard deviation of Spanish teacher effects, and larger ones for math. The higher estimates for math may reflect that, by design, our sample is restricted to younger teachers.

¹⁷Similar procedures have been implemented in the literature. See Araujo et al. (2016); Bau and Das (2020).

¹⁸Appendix Table B10 shows that our estimates remain almost unchanged when the covariance of $\hat{\delta}$ is computed using different classes for the same teacher, in the same school, in the same year, and when we estimate $\hat{\delta}$ in model (3) without including $X_{i,t}$.

4.2.2 Teacher value-added and the 2011 reform

To estimate the impact of the reform on teaching quality, we estimate a value-added model that compares teachers trained across different cohorts. Consider the following simple value-added model:

$$Y_{isjgtc} = \lambda_0 + \lambda_1 Y_{i,t-2} + \gamma_1 \mathbf{X}_{it} + \gamma_2 \mathbf{C}_{ct} + \gamma_3 \mathbf{S}_{st} + \tau_j + \alpha_t + \mu_g + \epsilon_{isjgtc}, \quad (4)$$

where Y_{isjgt} is the (mathematics or Spanish) test score of student i , in school s , classroom c , taught by teacher j , in grade g in year t . The term $Y_{i,t-2}$ refers to the student's lagged test scores in both subjects, measured two years earlier. $\mathbf{X}_{it}$ is a vector of student-level covariates (female, family income quintile, and parental education). $\mathbf{C}_{ct}$ captures classroom characteristics (class size, share of female students, average lagged test scores for both subjects, the distribution of students by family income quintile, and the distribution of parental education). $\mathbf{S}_{st}$ is a vector of school-level covariates (indicators for public, subsidized, technical, and rural schools). α_t and μ_g denote year and grade fixed effects, respectively. The coefficient τ_j recovers the teacher value-added (TVA) of each individual teacher, conditional on the included covariates.

We now turn to how individual teacher quality might have been affected by the reform. Our parameter of interest is the difference in τ_j across teacher cohorts. We model this as

$$\tau_j = \Phi_0 + \sum_{\substack{y=2004 \\ y \neq 2010}} \Phi_y \mathbb{1}\{PSUcohort_j = y\} + e_j, \quad (5)$$

where $\mathbb{1}\{PSUcohort_j = y\}$ are indicators for the PSU cohort of teacher j . Plugging this expression into equation (4) yields our final estimating equation:

$$Y_{isjgtc} = \lambda_0 + \lambda_1 Y_{i,t-2} + \gamma_1 \mathbf{X}_{it} + \gamma_2 \mathbf{C}_{ct} + \gamma_3 \mathbf{S}_{st} + \alpha_t + \mu_g + \sum_{\substack{y=2004 \\ y \neq 2010}} \Phi_y \mathbb{1}\{PSUcohort_j = y\} + \eta_{jt} + \epsilon_{isjgtc}, \quad (6)$$

which combines the standard value-added framework with our cohort comparison. Here, Φ_y captures the difference in teacher quality for each cohort relative to the 2010 baseline cohort. Alternatively, we estimate model

$$Y_{isjgtc} = \lambda_0 + \lambda_1 Y_{i,t-2} + \gamma_1 \mathbf{X}_{it} + \gamma_2 \mathbf{C}_{ct} + \gamma_3 \mathbf{S}_{st} + \alpha_t + \mu_g + \phi_1 \mathbb{1}\{AfterReform_j\} + \eta_{jt} + \epsilon_{isjgtc}, \quad (7)$$

where we include a single indicator for teachers having enrolled in a teacher training program in 2011 or later.

In models (6) and (7), η_{jt} additionally controls non-parametrically for teachers' years of experience. This is crucial, since comparing teachers trained before and after the reform, even after residualizing by student, classroom, and school characteristics, would otherwise omit one of the most important determinants of teacher quality: teaching experience. Appendix Figure A16 illustrates this point by showing the non-parametric relationship between TVA estimates and years of experience, which follows a clear upward trend and peaks at around five years. Mechanically, teachers who enrolled in teacher training programs before 2011 had, on average, more years of teaching experience at their disposal. However, Appendix Figure A18, which plots the distribution of years of experience by study cohort, shows that there is considerable variation in teaching experience within each cohort.

We estimate models (6) and (7) by weighting observations by the inverse of the number of students taught by each teacher, and cluster the standard errors at the teacher-by-year level. Reweighting ensures that each teacher contributes equally to the estimation, avoiding giving more weight to those with more students.

Identification. Identification of the parameters of interest relies on a conditional independence assumption. In other words, once we adjust for all the variables in the value-added model and years of teaching experience, any remaining bias in the comparison of teachers trained across PSU cohort would be negligible. It is important to stress that this comparison is carried out controlling for students' baseline test scores and a rich set of student, classroom and school characteristics. Crucially, this is a within-year and within-grade comparison. Any confounders that vary over time, such as the introduction of new policies, are absorbed by year fixed effects, as long as they do not differentially affect teachers belonging to different PSU cohorts.¹⁹ As highlighted in Appendix Figure A17, there is substantial variation within the year of entering the labor force across PSU cohorts, providing variation to identify our parameters of interest.

4.3 Results

Table 3 presents our main findings. Column (1) shows that, after accounting for teaching experience and a rich set of student, classroom and school covariates, mathematics teach-

¹⁹Among the reforms worth mentioning, there are the *Sistema de Desarrollo Docente* (Law 20.903) and the *Ley de Inclusión Escolar* (Law N 20.845). These reforms introduced changes in the salary structure for public school teachers and in the allocation of students to schools, respectively.

ers who enrolled in college after the reform are, on average, more effective in improving students' test scores. The size of the effect is about 38% of the standard deviation of math teachers' value added. Since the value-added estimates are expressed in standardized student test score units, this corresponds to an increase of approximately 0.04 SD in students' math test scores. By contrast, column (2) shows that the estimated effect of the reform for Spanish teachers is an order of magnitude smaller and statistically indistinguishable from zero. The two coefficients are statistically different. In fact, testing the null of equal post-reform effects in a fully interacted version of equation (7) yields a p-value of 0.051.

Table 3: Effect of the reform on TVA

	(1) Math teachers	(2) Spanish teachers
Post-reform cohort	0.038** (0.016)	-0.004 (0.014)
# Students	206,094	225,584
# Teachers	4,386	4,966
Δ Std PSU	0.320	0.387

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table reports OLS estimates for the coefficients of interest in model (7). Δ Std PSU corresponds to the change in the average standardized teacher test scores. Standard errors, clustered at the teacher-by-year level, are shown in parentheses.

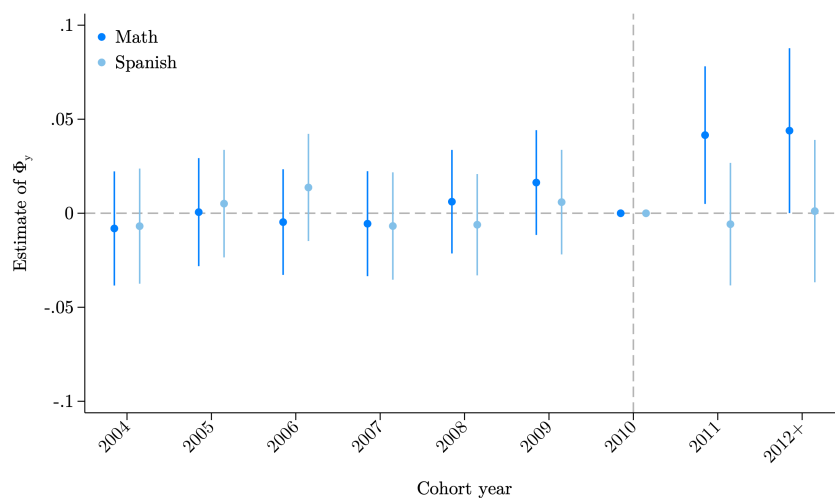
It is interesting to note that this heterogeneity in effects across subjects cannot be explained by differences in the change in the composition of teachers. As shown in Table 3, both groups of teachers experienced an increase in PSU scores of approximately one-third of a standard deviation, and this increase was even larger for Spanish teachers.

Trends in teaching quality and other policies. A potential threat to the identification strategy is the presence of trends in teaching quality across cohorts of university enrollment. If, for example, teacher training programs are becoming more effective over time, our estimates would be biased. Figure 2 plots the estimates of Φ_y from model (5). To increase precision, we pool cohorts 2012, 2013 and 2014 due to the small number of teachers trained after 2012 in our sample. As the figure shows, there were no significant differences in teaching quality across cohorts before 2010. For math teachers, the positive effects appear only after 2011, the year of the reform's implementation.

Beyond cohort trends, a natural concern is that other changes in the Chilean higher education system after 2011 may differentially affect the cohorts we study. Two events

are worth mentioning. First, the expansion of the centralized application system in 2012 (Kapor et al., 2024). The fact that the change in TVA is already visible in 2011 and remains similar in 2012 suggests this is unlikely to drive our results. The second event is the introduction of free tuition in 2015. However, this falls outside our sample period: we restrict the sample to 2014 precisely to avoid confounding from the free tuition reform, which effectively eliminated the BVP's effect on enrollment in teacher training programs (Castro-Zarzur et al., 2022). Finally, potential contemporaneous events in 2011, for example the student protests documented in González and Prem (2024), may also raise concerns. We address these in Appendix F.2 with additional robustness checks that suggest the effect we observe is not driven by other contemporaneous shocks.

Figure 2: Average change in TVA by cohort



The figure reports OLS estimates of Φ_y from model (6). Confidence intervals at the 10% significance level are shown, with standard errors clustered at the teacher-by-year level.

Sensitivity to TVA estimates and sample selection. We further test the robustness of our results in several ways. In Appendix Table B11, we test the sensitivity of our estimates to different specifications. We estimate model (7) controlling for more restrictive sets of covariates (including GPA, share of days at school) and less restrictive (including only school or class characteristics) sets of covariates. Results remain fairly stable. Despite controlling non-parametrically for years of experience in the specification, there may be concerns about the model being estimated using teachers who studied before the reform with more years of experience than those in the treated cohorts, namely, those who studied after. The relevant comparison, in fact, is at low levels of experience, since

teachers in the treatment groups have, on average, less than one year of experience. As shown in Appendix Table B11 and Appendix Figure A19, the results remain essentially unchanged when restricting the sample to teachers with fewer than six, four, or three years of experience.²⁰ Finally, newer cohorts have had less time to appear in the dataset, and if higher-value-added teachers are more likely to complete their degrees on time, this could generate cohort differences in who is observed in the data. To assess whether this type of sample selection is driving our results, we re-estimate the model, restricting the sample to individuals who finish their studies on time and whom we first observe six or seven years after taking the PSU. As reported in Appendix Table B12, math teachers from post-reform cohorts exhibit higher TVA, while the estimates for Spanish remain close to zero, even in this subsample of teachers.

Heterogeneity The BVP aimed not only to attract potentially better teachers into the profession, but also to match them with more vulnerable schools. The small proportion of rural and private schools does not allow us to test for heterogeneity of effects using these natural sample divisions. Therefore, we test whether the reform was more effective in the targeted schools by analyzing the heterogeneity of effects by school vulnerability. We define vulnerable schools using the IVE-SINAE index. This measure is calculated annually by the Ministry of Education for all publicly funded schools in order to identify children who, without appropriate support, are at high risk of dropping out of the school system. The IVE-SINAE corresponds to the percentage of students at risk in a given school (Cornejo, 2005).²¹ We define high-vulnerability schools as those with an IVE-SINAE above the median (78% of vulnerable students).

Results in Appendix Table B13 show that the average change in math TVA is similar for both types of school (columns (1) and (2)). Similarly to our main results by subject, the heterogeneity does not reflect differential changes in the average PSU of the relevant group of teachers.

²⁰The reform may also have affected the returns to experience. We cannot directly test this hypothesis because very few teachers trained after the reform have more than one year of experience. However, this limitation is not critical for our analysis, since we non-parametrically compare teachers before and after the reform.

²¹The IVE-SINAE identifies three priority groups: 1) children of medium priority, i.e. children living below the poverty line; 2) children of high priority, i.e. children living below the poverty line with additional risk factors for dropping out of school; and 3) children of very high priority, i.e. children at risk of extreme poverty. The IVE-SINAE index of the school is calculated as the sum of these three priority groups.

4.4 Cost-benefit analysis

A natural question is whether the benefits of the reform justify its costs. We focus on the monetary costs, which are those of the scholarship. In Appendix D we present a back-of-the-envelope cost-benefit analysis for mathematics teachers, and Table 4 summarizes the resulting benefit-cost ratios. On the cost side, the scholarship covers tuition of approximately 9.4 million CLP per completed degree. Since it is paid to all enrollees, while only about 60% of them become employed teachers (and dropouts receive only partial tuition), the cost of bringing one teacher into the profession through the BVP is about 13.3 million CLP. Only 29% of post-reform math teachers received the BVP, however, so spread across all post-reform math teachers the effective cost per teacher is about 3.84 million CLP.

On the benefit side, we convert the value-added gain into the present value of students' lifetime earnings. Since no estimate of how teacher value-added maps into students' earnings is available for Chile, we use the one estimated for the United States by Chetty et al. (2014b), who find that a one standard deviation increase in teacher value-added raises the present value of lifetime earnings by 1.34%, and apply it to the lifetime earnings of Chilean students. We construct these from CASEN (*Encuesta de Caracterización Socioeconómica Nacional*), Chile's main household survey. We then make a conservative adjustment to this elasticity: we assume that the returns to cognitive skills in Chile are lower than in the United States, and scale the elasticity down using Hanushek et al. (2015), who estimate returns to numeracy of 18% per standard deviation at the OECD average against 28% in the United States. This gives an effective elasticity of $1.34\% \times 18/28 \approx 0.86\%$.

Across a range of assumptions on discounting, persistence of the value-added effect, and teacher retention, the benefits exceed the cost of the reform (Appendix D). The benefit-cost ratio is 15.3 when the value-added effect is treated as permanent over a full career, and ranges from 2.7 to 10.4 for teachers who remain in the profession between 3 and 15 years (Table 4).

This analysis is conditional on mathematics teachers, the group for which we detect a value-added effect. Because the reform funds teachers across all subjects, a more conservative approach would also account for the cost of the scholarship paid to teachers whose subjects show no improvement. For example, including the cost for Spanish teachers roughly doubles the effective cost per teacher and halves the benefit-cost ratios. Even then, the ratio remains above one across the scenarios.

Table 4: Benefit-cost ratio of the reform

	Benefit-cost ratio
<i>Full 30-year career, value-added effect:</i>	
Constant (permanent)	15.3
Linear fade to zero over 30 years	9.2
Exponential fade	7.7
<i>Value-added effect held constant; teacher stays in the profession for</i>	
3 years	2.7
5 years	4.3
10 years	7.7
15 years	10.4

Note: Benefit-cost ratios for mathematics teachers at a 5% discount rate. The cost per teacher (3.84 million CLP, compounded to the teacher's entry date) accounts for the share of enrollees who become employed teachers and for the BVP take-up rate among post-reform math teachers.

5 Discussion and Mechanisms

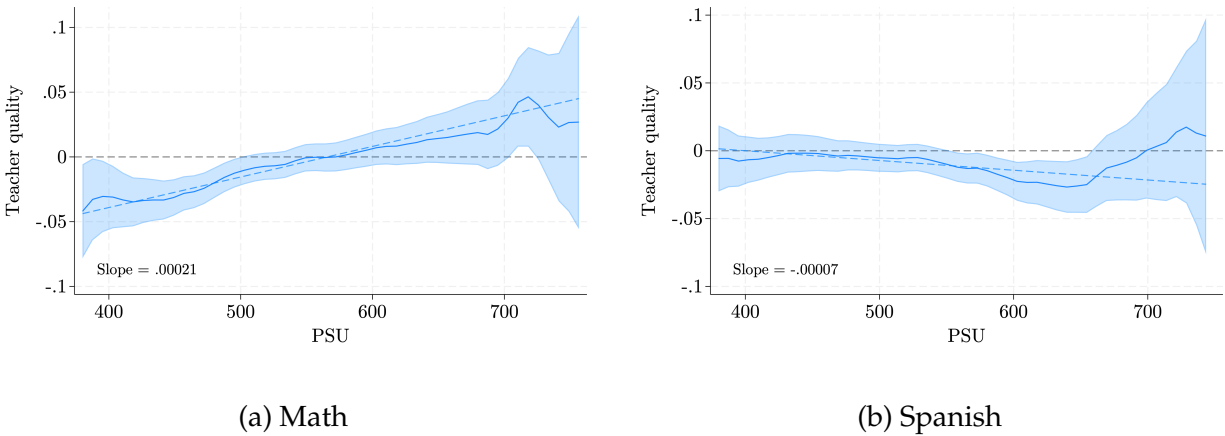
The reform increased the average TVA for mathematics teachers by 38% of a standard deviation, but had no effect for Spanish teachers. In this section, we investigate the mechanisms behind these results. We begin with two key observations: first, PSU scores are a good predictor of teacher quality for mathematics but not for Spanish; and second, for mathematics, the increase in PSU scores alone is far too small to explain the observed gains in TVA. We then discuss a theoretical framework, based on occupational sorting, to rationalize the findings and test some of its implications.

5.1 PSU as a screening device for teacher quality

We now test the reform's implicit assumption about the predictive power of PSU scores for an individual's teacher quality. In Figure 3, we plot the non-parametric fit alongside the linear fit for a teacher's value-added (estimates of τ_j from model (4)) as a function of their PSU score, for mathematics and Spanish teachers who began their studies before the 2011 reform. The figure shows that before the reform, PSU scores were predictive of teacher quality for mathematics teachers in primary school, but not for Spanish teachers. Specifically, a one standard deviation increase (100 points) in PSU score was associated with a 0.02 increase in TVA for mathematics teachers. For Spanish primary school teachers, however, this coefficient is smaller in magnitude and not statistically different from zero.

While these results are in line with the findings by Gallegos et al. (2022), they challenge

Figure 3: Pre-reform local polynomial fit of TVA on PSU



Note: The figures show the non-parametric relationship between TVA and PSU described by a local polynomial regression together with their linear fit, for teachers that started their studies before the 2011 reform. The shaded area represents the 90% confidence interval. On the left-hand side, the focus is on the mathematics teachers before the reform, while on the right-hand side, the focus is on Spanish teachers. The numbers in the left-bottom corner refer to the coefficient of the linear fit.

the idea that PSU scores can serve as an effective recruitment screening device to enhance students' learning outcomes across all subjects. Gallegos et al. (2022) use several measures of teacher quality, namely, graduation from college, college exit exams, government evaluation, employment, wages, and students' achievement gains. While the authors provide interesting insights on the correlation between PSU and these measures, our measure allows us to disentangle the role of sorting - between and within schools - in partially explaining this correlation. This is crucial when thinking of measuring teacher productivity as sorting might amplify the predictive power of pre-college achievement.²²

Several studies have investigated the role of teachers' cognitive ability in explaining their performance, using TVA as an outcome. Its predictive power differs depending on the country studied and how cognitive ability is measured. Most papers have analyzed the role of teachers' contemporaneous content knowledge, as measured by standardized tests. Focusing on Peru, Metzler and Woessmann (2012) find that a one standard deviation increase in teachers' test scores, increases their students' math test scores by about 0.06, but they find no correlation for Spanish teachers. For lower income countries, Bau and Das (2020) and Bietenbeck et al. (2018) find that a one standard deviation increase in test scores is associated with roughly a 0.25-0.3 increase in value-added, regardless of the subject. When measuring teachers' cognitive ability using proxies different than content

²²The authors also use measures of students' achievement gain. However, they are limited to a small number of teachers (around 100), which also limits the richness of potential controls in the estimation of the value-added model. Additionally, they only focus on mathematics.

knowledge, correlations are lower. For example, Araujo et al. (2016) find that one standard deviation increase in teachers' IQ is associated with a 0.04 higher TVA across subjects. Papers studying the role of pre-college test scores, such as the one we study, are rare and never related to value-added measures (Bardach and Klassen, 2020).

A candidate explanation for why PSU predicts teaching effectiveness in mathematics but not in Spanish is that the PSU places greater weight on mathematical proficiency, so that variation in PSU scores maps more directly onto differences in mathematical ability than in language ability. Appendix Figure A20 is consistent with this hypothesis: the relationship between PSU scores and the share of correct answers is steeper for mathematics. At a PSU of 500, test-takers answer roughly 20% of math questions correctly but about 40% of language questions; at a PSU of 600, the figures are approximately 45% and 60%, respectively. However, when we directly correlate teachers' share of correct answers in each PSU subject with their TVA (Appendix Figure A21), both math and Spanish correct answers are significant predictors of mathematics TVA with similar slopes, while neither predicts Spanish TVA. If the math-specific content of the PSU were the key mechanism, we would expect math correct answers to be a stronger predictor than language correct answers. The fact that they are not, points to general cognitive ability, rather than math-specific knowledge as the relevant dimension along which PSU scores predict teaching effectiveness in mathematics.

Even though PSU scores are predictive of mathematics TVA, the observed increase in academic ability can account for only a small share of the reform's overall effect on teacher quality. The average teacher's test scores increased by 0.32 of a standard deviation, and given our estimates of the correlation between PSU and TVA, we would predict an increase of around 0.01 in value-added. Therefore, the role of PSU scores appears marginal in explaining the overall effect of the reform. A natural question is where the remainder of the effect comes from. If the reform's incentives—scholarships and restrictions—changed who selects into teaching beyond their PSU score, the incentivized entrants and disincentivized stayers may differ systematically in unobserved traits that also affect teaching effectiveness. We formalize this idea in the next subsection.

5.2 Selection beyond academic ability: a theoretical framework

We develop a simple Roy-style occupational choice model, building on Roy (1951) and the “motivated agents” literature (Besley and Ghatak, 2005; Dal Bó et al., 2013), to characterize how the reform may have changed the composition of teachers along unobserved dimensions. Here we summarize the key elements; the full model is developed in Appendix E.

An individual's productivity in teaching and in the outside sector is $P_T = \kappa_T A + \epsilon_T$ and $P_O = \kappa_O A + \epsilon_O$, where A is academic ability (measured by PSU), ϵ_T and ϵ_O are residual ability in each sector, and κ_T, κ_O govern how strongly academic ability translates into productivity. The residual ϵ_T is the object of interest: it captures all dimensions of teaching quality beyond what PSU measures, so the reform can raise teacher productivity both directly, by increasing A , and indirectly, by changing the composition of ϵ_T among those who enter.

Individuals choose teaching when its utility exceeds that of the outside option. The utility of teaching increases with intrinsic motivation ϵ_M , while the outside option rewards productivity, with r_O measuring how much a unit of productivity is paid outside teaching. The policy shifts the net payoff from teaching, raising it for scholarship recipients (scores above 600) and lowering it for those facing the restriction (below 500). Comparing the two options, and netting out the terms that depend only on A , the decision to teach is governed by a selection index

$$Z = -r_O \epsilon_O + \epsilon_M. \quad (8)$$

The full specification of utilities and the policy term is given in Appendix E. The index reflects two competing forces: a *motivation* channel (ϵ_M), which pulls motivated individuals toward teaching, and an *outside pull* channel ($r_O \epsilon_O$), which draws productive individuals toward better-paid non-teaching careers. An individual enters teaching when Z is high enough, that is, when motivation outweighs the pull of outside options. The reform shifts this entry threshold: scholarships lower it for individuals scoring above 600, making teaching attractive to those who would not have entered otherwise; restrictions raise it for those scoring below 500, pushing out individuals at the margin. Under joint normality, the resulting change in average residual teaching quality can be written as

$$\Delta \mathbb{E}[\epsilon_T] = \frac{\text{Cov}(\epsilon_T, Z)}{\text{Var}(Z)} \Delta \mathbb{E}[Z]. \quad (9)$$

The sign of $\Delta \mathbb{E}[\epsilon_T]$ therefore depends on $\text{Cov}(\epsilon_T, Z)$, which can be decomposed as

$$\text{Cov}(\epsilon_T, Z) = \sigma_T (\rho_{T,M} \sigma_M - r_O \rho_{T,O} \sigma_O), \quad (10)$$

where $\rho_{T,M}$ and $\rho_{T,O}$ are the correlations of residual teaching ability with motivation and outside-sector ability, respectively. The first term captures a *motivation* channel: when motivation is positively correlated with teaching effectiveness ($\rho_{T,M} > 0$), it pushes the covariance upward. The second term captures an *outside pull* channel: when the traits that make someone a good teacher are also rewarded outside ($\rho_{T,O} > 0$), it pushes the

covariance downward. Whether the reform increases or decreases average teaching quality beyond the direct effect of PSU scores depends on which of these two forces dominates. For example, if the outside pull is strong among high-ability individuals, scholarships draw back into teaching people who would have been effective teachers but were lured by better-paid outside careers, raising average quality. Similarly, if motivation is the dominant force among low-ability individuals, restrictions screen out the least motivated candidates, who are also the least effective, again raising average quality. However, the opposite is also possible: scholarships could attract individuals with low intrinsic motivation and decrease quality (Ashraf et al., 2020; Deserranno, 2019). Which scenario prevails is ultimately an empirical question.

5.3 Selection beyond academic ability: results

Table 5: Before-After comparison beyond academic ability - teacher quality

	Math teachers		Spanish teachers	
	(1)	(2)	(3)	(4)
Post-reform cohort	0.034** (0.016)		-0.001 (0.015)	
Post-reform cohort $\times$ PSU < 500		0.065* (0.036)		0.004 (0.027)
Post-reform cohort $\times$ PSU $\in [500, 600)$		0.001 (0.021)		-0.014 (0.019)
Post-reform cohort $\times$ PSU ≥ 600		0.065** (0.026)		0.030 (0.027)
# Students	206,094	206,094	225,584	225,584
# Teachers	4,386	4,386	4,966	4,966

Note: * $p < 0.1$ ** $p < 0.05$, *** $p < 0.01$. The table shows OLS estimates for the coefficients of interest in model (7). We additionally control for a polynomial of order two of PSU score. Standard errors are clustered at the teacher-by-year level, and shown in parentheses.

In Section 5.1, we argued via a back-of-the-envelope calculation that the observed increase in PSU scores can account for only a small share of the reform's effect on TVA. We now confirm this formally by re-estimating model (7) controlling for PSU scores. Column (1) in Table 5 shows that, even after adjusting for academic ability, math teachers who studied after 2011 exhibit a higher TVA compared to those who studied before the reform.²³

²³A formal Gelbach (2016) decomposition of the post-reform coefficient with and without controlling for PSU yields an indirect (PSU-mediated) effect of 0.004 (s.e. 0.001, $p = 0.005$), or 9.1% of the total math effect,

In column (2), we interact the post-reform dummy with each subgroup of PSU scores based on the restrictions and incentives imposed by the reform. We observe that math teachers scoring below 500 and above 600 are associated with a 0.065 higher value-added for post-reform cohorts, while teachers scoring between 500 and 600, who were not directly targeted by either the scholarship or the restrictions, show no change in TVA. This is consistent with our framework, in which the reform improves teacher quality among the groups it targeted, plausibly through selection on unobservables. The change in the relation between teacher quality and PSU scores is also illustrated in Figure 4, where we plot the non-parametric fit of the TVA estimates τ_j and PSU scores (see Appendix Figure A22 for a non-parametric representation). For mathematics teachers, we observe an increase in the share of high-value-added teachers among those scoring below 500, no differences in teacher quality for those scoring between 500 and 600, and a steeper positive relationship for teachers scoring above 600. Consistent with our previous findings, the relationship between TVA and PSU is not altered for Spanish teachers.

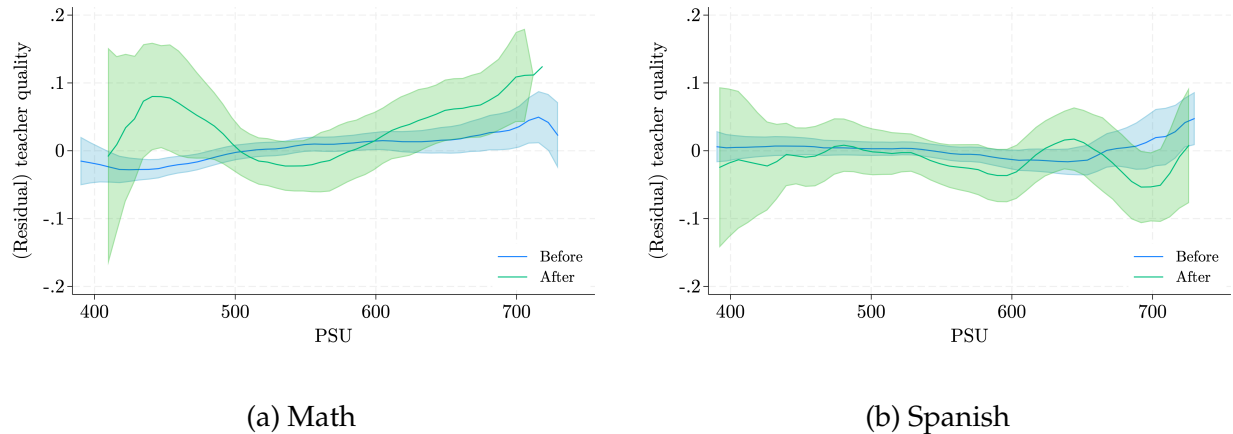
Why might different channels dominate at different points in the PSU distribution? A plausible explanation is that the returns to outside-sector productivity are heterogeneous across academic ability levels. Individuals with high PSU scores have access to a wider set of higher education programs (Appendix Figure A23) and more lucrative career options, making the outside pull relatively stronger. For low-PSU individuals, outside options are more limited, so intrinsic motivation becomes the dominant force in the decision to teach.

We test the robustness of our results in the same spirit as we proceed for the average effect. Namely, we test whether they are sensitive to different value-added specifications, or to restricting the years of experience of teachers. Results, shown in Appendix Table B14, remain stable across specifications. Moreover, Appendix Figure A24 shows the effect not being driven by any specific cohort before 2011.

It is important to stress that finding no differences for teachers not directly targeted by the reform (those scoring between 500 and 600) also has implications for identification, which relies on differences across cohorts. For math teachers, the absence of changes in teaching quality over the long pre-reform window is suggestive that the results are driven by the reform. Since the mechanisms we discuss apply only to teachers scoring below 500 or above 600, the fact that the TVA does not change for those scoring between 500 and 600 after the reform aligns with the idea that the results are not confounded by other contemporaneous events or reforms.

with an approximate 95% upper bound of 13.5%. The remaining 91% reflects changes in unobserved teacher quality.

Figure 4: Before and after local polynomial fit of TVA on PSU



Note: The figures show the non-parametric relationship between the residual TVA and PSU described by a local polynomial regression together with their linear fit, for teachers that started their studies before the reform (2004-2010) and after the reform (2011-2014). The shaded area represents the 90% confidence interval. The *(residual) teacher quality* measure is obtained as the residual from a non-parametric regression of TVA on years of experience. Panel (a) shows the relationship for mathematics teachers, while Panel (b) shows it for Spanish teachers.

Suggestive evidence on motivation. The theoretical framework also yields predictions about intrinsic motivation. A natural concern is that scholarships might attract less motivated individuals, generating a trade-off between academic ability and intrinsic motivation. The model shows that this trade-off arises only when the motivation channel dominates at high ability levels; if instead the outside pull dominates, incentivized teachers need not be less motivated. More specifically, the framework predicts that individuals facing restrictions should exhibit higher levels of motivation if they are of high value-added, while for scholarship recipients it yields no clear prediction.²⁴

To provide suggestive evidence on this prediction, we use on-the-job survey data on a sub-sample of primary school math and Spanish teachers. These data are regularly collected by the Chilean Ministry of Education during the days in which the standardized test for their students takes place. While the teacher survey takes place every year, the specific questions asked change regularly. In 2018 and 2019, teachers were asked some questions about their satisfaction, dedication and motivation towards their role as teachers. Our main variable of interest is an indicator variable for teachers strongly agreeing to the statement “*If I could decide, I would choose this job again*”. Appendix Table B15 shows that our measure of motivation is positively correlated with teacher quality: strongly agreeing with the statement is associated with a 0.033 higher TVA.

Appendix Table B16 presents the average change in motivation after the reform, con-

²⁴See Appendix E for a simple proof.

trolling for a second-order polynomial of PSU, as well as years of experience, year, gender, and additional school covariates. School covariates adjust for the possible effect of sorting happening after graduation. Column (1) shows that math teachers enrolling before and after the reform are not differentially likely to strongly agree with the statement. However, column (2) reveals some heterogeneity across PSU scores, as it presents evidence of positive selection based on motivation for teachers who scored below 500 points: those enrolling after the reform are almost 21 p.p (36%) more likely to strongly agree with the statement. For teachers scoring above 600, we do not find evidence of negative selection on motivation induced by the reform. While motivation is measured ex-post and adding school controls may not fully address sorting, the pattern is consistent with the TVA results, and the effect on motivation appears precisely among teachers scoring below 500, the subgroup for which the framework predicts positive selection on intrinsic motivation.

5.4 Competing explanations

We discuss potential confounders to our interpretation. We examine two channels of teacher sorting that may account for the observed effects even after adjusting for PSU: differential exposure to peer composition and other characteristics of the program in higher education training programs, and differential exposure to colleagues and other characteristics of the schools where teachers work.

Peer effects in college and training programs. As discussed in Section 3, the reform altered the sorting of high school graduates into teaching training programs. Teachers studying with the BVP were more likely to pursue more prestigious higher education programs, whereas those scoring below 500 faced restrictions to access them. If teachers are trained more effectively in these programs, we might expect this to explain part of the estimated effect. In Appendix Table B17, we re-estimate model (4) adding indicators for whether the institution is a public or private CRUCH university, the average PSU score of the same-cohort enrollees in the program, the proportion of enrollees who took the PSU, and the program size. Including these controls leaves the post-reform coefficient substantially unchanged.²⁵ We complement this with a direct test of peer effects in training in Appendix F, where we exploit within-program across-cohort variation in peer composition while conditioning on the teacher’s own PSU and the other controls of the

²⁵Consistent with this evidence, Gallegos et al. (2022) estimate the value-added of Chilean teacher colleges using an RD design on the centralized admissions system and find no evidence that any teacher college differentially contributes to teacher effectiveness.

main TVA specification, and find non-detectable and close to zero effects of peer PSU and program-cohort size on teacher value-added.

Colleagues and other school characteristics. The reform also changed the types of schools that some teachers sort into, as a consequence of its incentives to work in public and rural schools. This may influence TVA in two ways. First, improving the test scores of students from different backgrounds may be more or less difficult: in Appendix Figure A25 we show that TVA is negatively correlated with the share of vulnerable students taught. Second, newly hired teachers may be exposed to different colleagues, who may contribute to their TVA (Jackson and Bruegmann, 2009). In columns (5) to (7) of Appendix Table B17, we adjust for the vulnerability index of the school where the teacher works, the number of colleagues, their average PSU scores, the percentage of teachers without PSU scores, and their average years of experience.

Including these factors marginally changes the post-reform coefficient. The effect remains largely unexplained after accounting for all of them.

6 Conclusion

This paper examines the impact of a Chilean educational reform that sought to improve the quality of teachers by changing the composition of students who enrolled in higher education teacher training programs. The reform used the national standardized university entrance exam (PSU) to set entry standards for teacher education programs. It limited the admission of students who scored below the mean while offering scholarships to incentivize the enrollment of those who scored more than one standard deviation above the mean.

Our analysis shows that this reform was successful in attracting new teachers from the top of the PSU score distribution and reduced the entry of lower-scoring individuals into the teaching profession. To test how this translated into teaching quality, we focus on mathematics and Spanish primary school teachers and estimate the effect based on a Teacher Value-Added (TVA) framework using students' test scores. We show that the reform increased the average teacher quality for mathematics teachers, but not for Spanish teachers. Moreover, we estimate that the change in academic ability explains only about 10% of the positive effect of the reform on TVA.

We furthermore show that after the reform, both graduates who were offered the scholarship and those who experienced admissions restrictions are associated with higher TVA relative to similar teachers from earlier cohorts. We rationalize these findings through

the lens of an occupational sorting model à la Roy (1951), that predicts that scholarships can draw in high-ability individuals who would otherwise sort into outside options, while restrictions can screen out candidates with low motivation.

The improvement in teacher quality also appears to justify its cost. Across a range of assumptions on discounting, persistence of the value-added effect, and teacher retention, the implied gains for the students taught by more effective teachers exceed the cost of the scholarship.

What makes these results notable is that the reform succeeded through a mechanism different from the one that motivated it. The policy relied on test scores on the premise that academic ability predicts teaching quality, yet that channel accounts for only a small part of the gain. Most of it comes from how the scholarship and the admission restrictions changed who chose to enter or remain in teaching, selecting them on unobserved traits.

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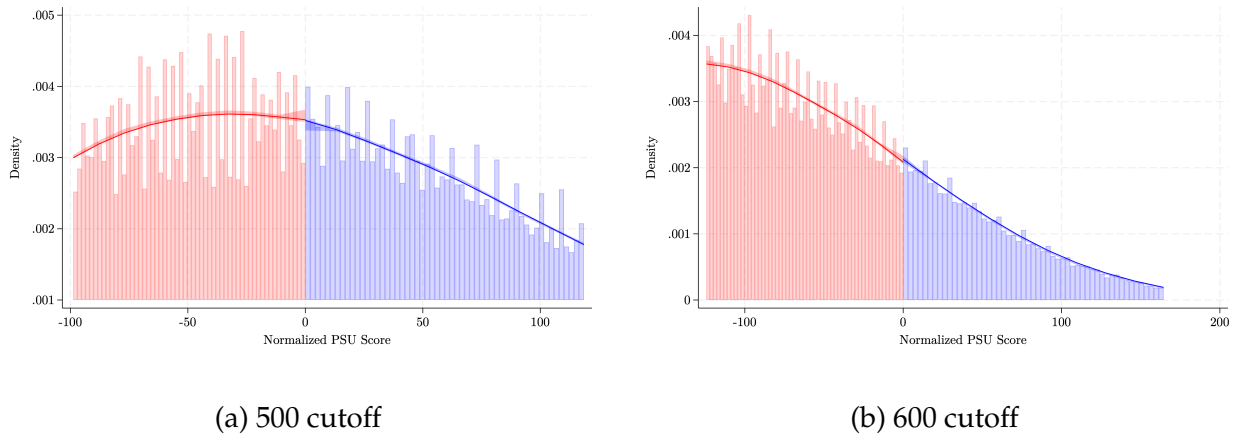
Recruiting Better Teachers?

Evidence from a Higher Education Reform in Chile

Supplemental Appendices

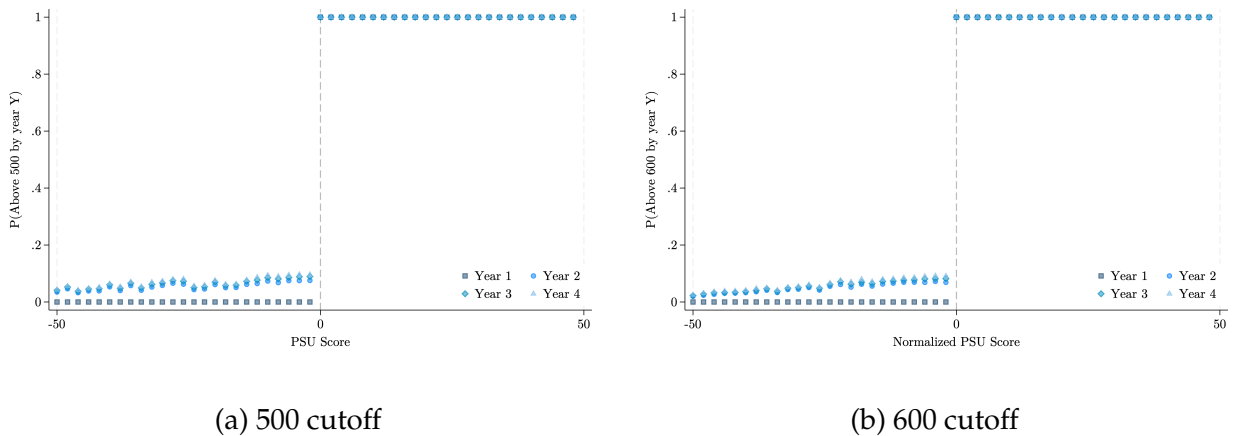
A Additional Figures

Figure A1: McCrary test



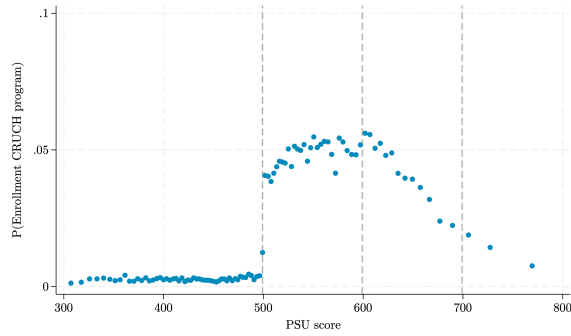
Note: The figures display the McCrary test for discontinuity in the running variable for the 500 cutoff (a) and 600 cutoff (b).

Figure A2: $P(\text{Eligible})$ by year y as a function of normalized PSU score

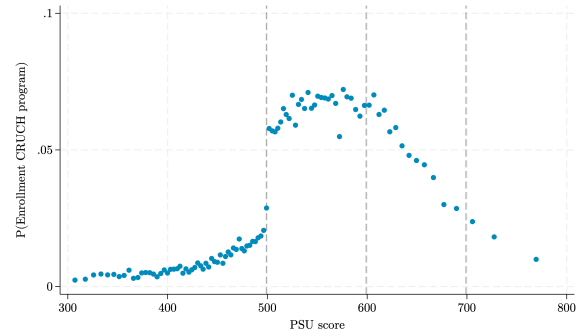


Note: The figure presents shares of individuals eligible to enroll in an eligible institution (Panel (a)) and for the BVP (Panel (b)) and as a function of the normalized PSU score. Shares are shown in a 2-point bin.

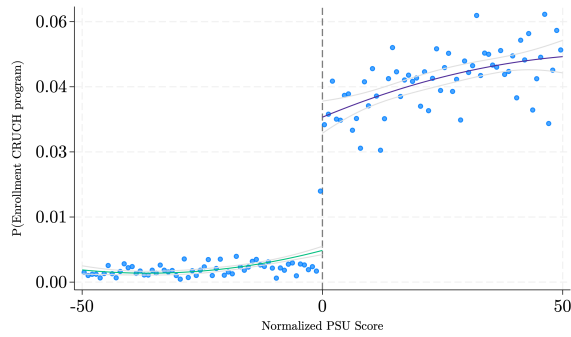
Figure A3: Enrollment in CRUCH teacher training program as a function of the PSU



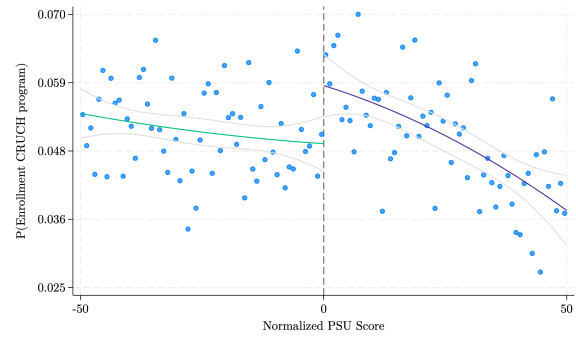
(a) Year 1



(b) Year 2



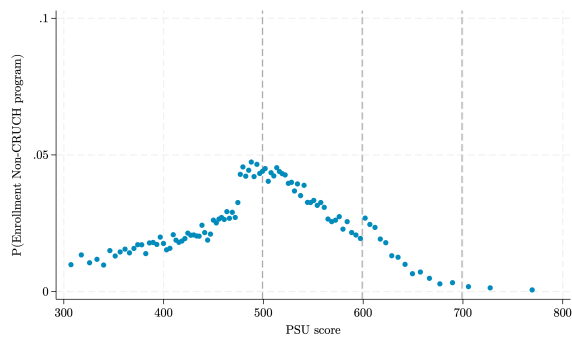
(c) 500 cutoff



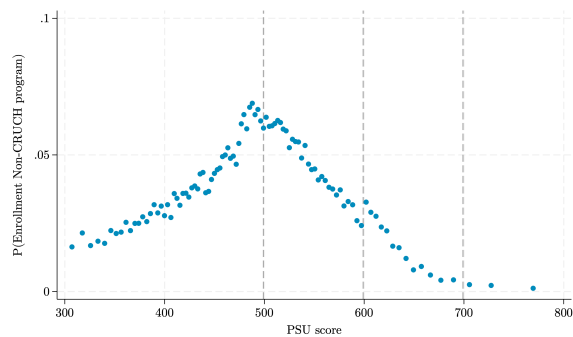
(d) 600 cutoff

Note: The figures show probabilities of enrollment in teacher training undergraduate programs in CRUCH institutions as a function of the PSU score. CRUCH institutions are the most prestigious, and almost all of them are *eligible* institutions in which to use the BVP. Figure (a) shows enrollment in the same year of the first PSU attempt; Figure (b) shows enrollment by the second year; Figures (c) and (d) show the probability of enrollment in the year of the first PSU attempt at each cutoff.

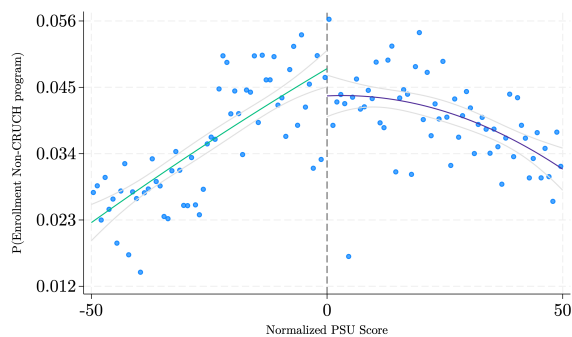
Figure A4: Enrollment in Non-CRUCH teacher training program as a function of the PSU



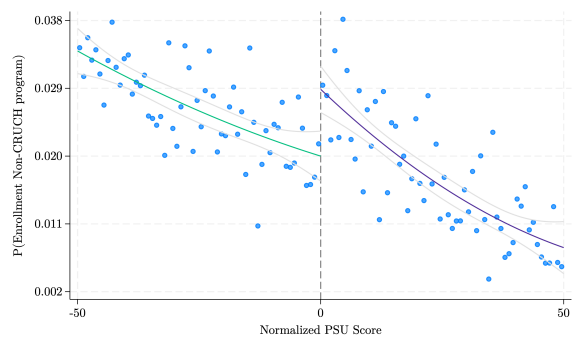
(a) Year 1



(b) Year 2



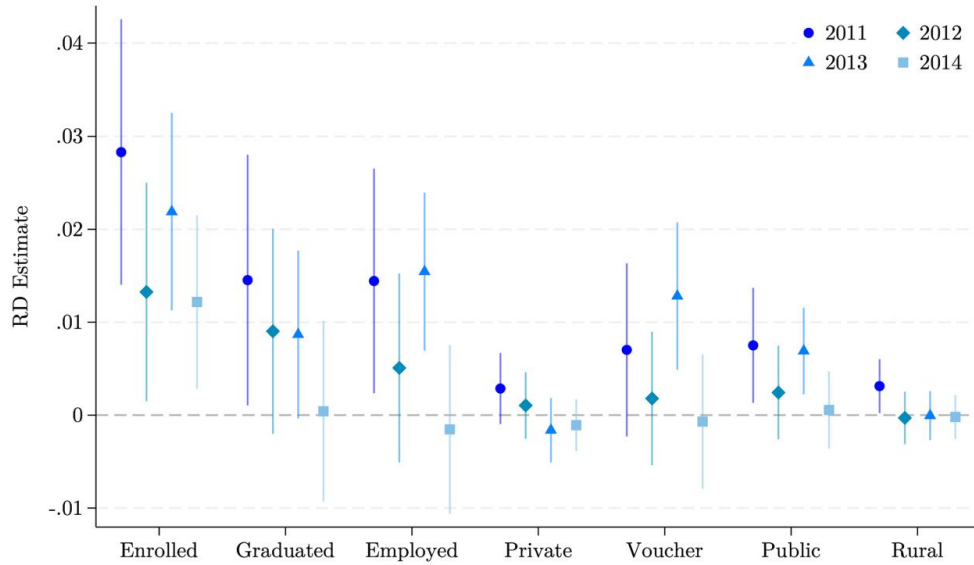
(c) 500 cutoff



(d) 600 cutoff

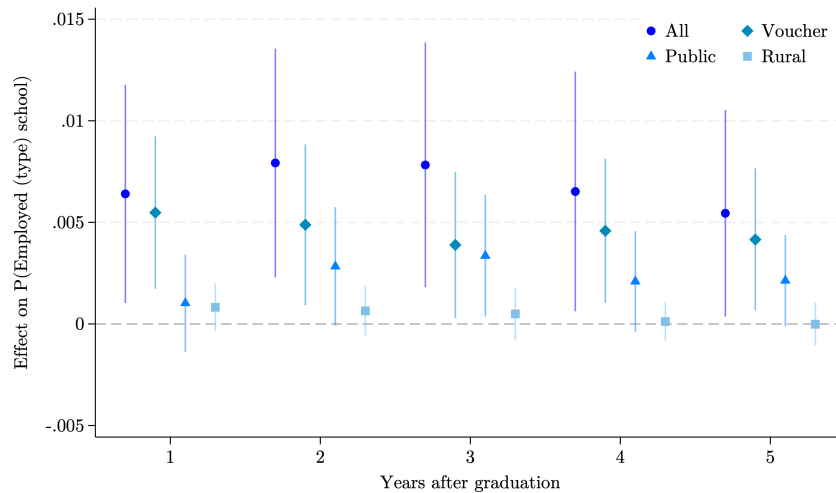
Note: The figures show probabilities of enrollment in teacher training undergraduate programs in non-CRUCH institutions as a function of the PSU score. Figure (a) shows enrollment in the same year of the first PSU attempt; Figure (b) shows enrollment by the second year; Figures (c) and (d) show the probability of enrollment in the year of the first PSU attempt at each cutoff.

Figure A5: RD estimates at the 600 cutoff by cohort of study



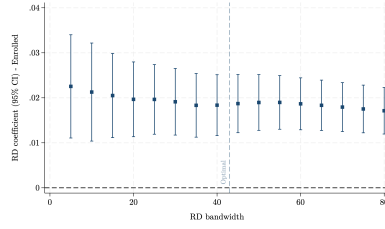
Note: The figure shows RD estimates from model (1) at the 600 cutoff, by cohorts of study. The outcomes are the same analyzed in Table 1.

Figure A6: BVP eligibility and employment at school over time

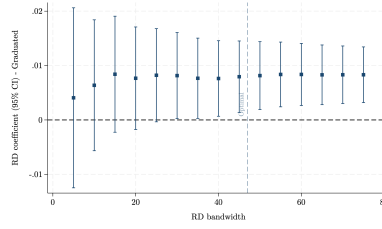


Note: The figures show RD parameters at the 600 cutoff using as outcomes the probability of being employed in a school, and the probability of being employed by type of school up to five years after graduation. For the RD results, the sample is restricted to those taking the college admissions test between 2011 and 2014.

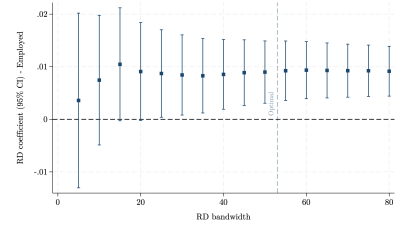
Figure A7: Sensitivity of RD estimates to different bandwidths



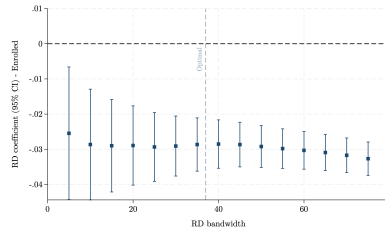
(a) Enrolled
(600 cutoff)



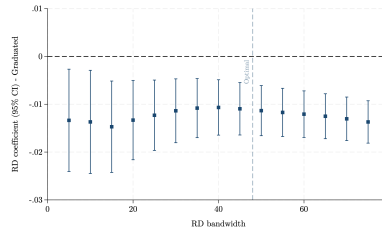
(b) Graduated
(600 cutoff)



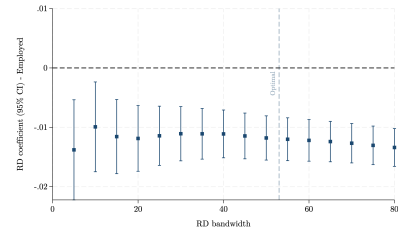
(c) Employed
(600 cutoff)



(d) Enrolled
(500 cutoff)



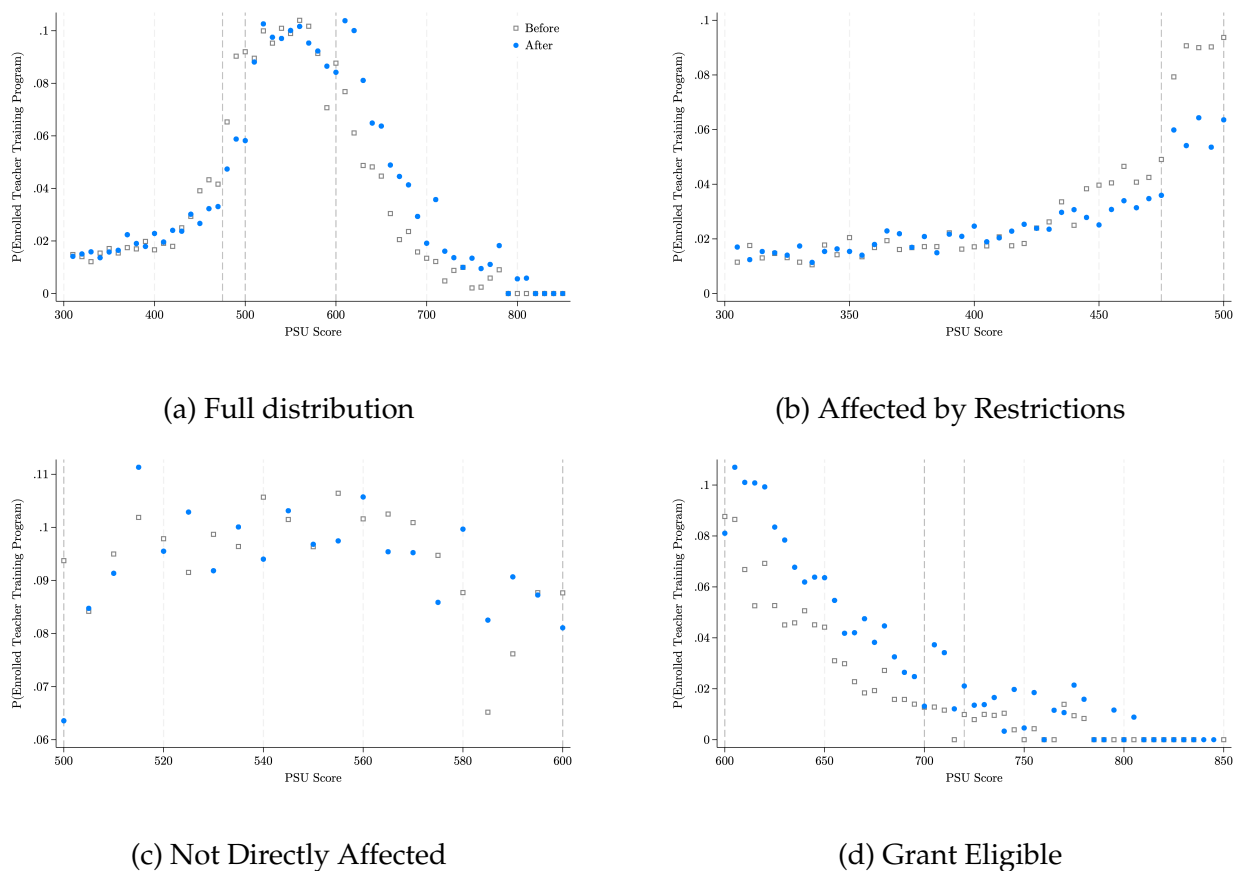
(e) Graduated
(500 cutoff)



(f) Employed
(500 cutoff)

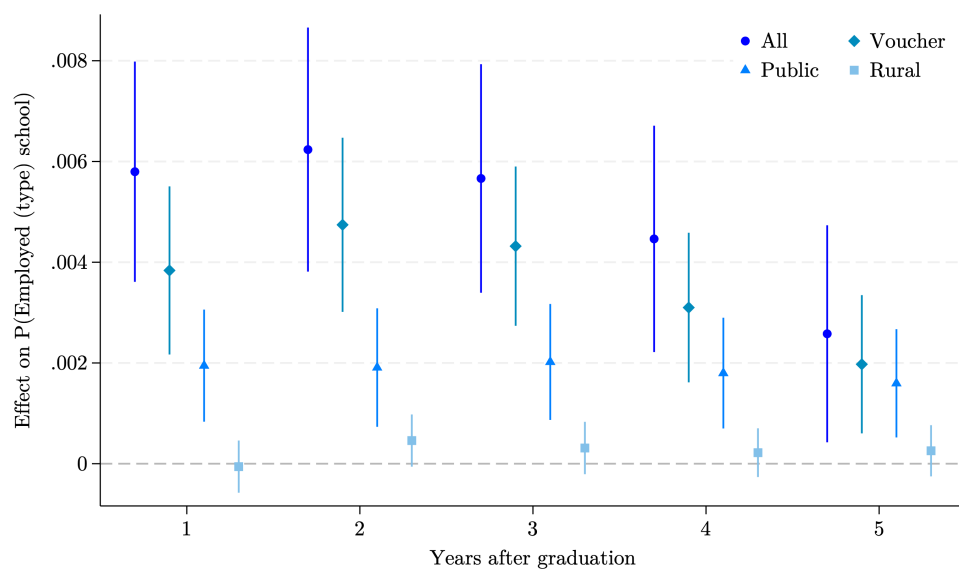
Note: The figure shows the sensitivity of our main RD estimates to different bandwidths. The estimation was performed using a triangular kernel and a polynomial of order 1. Standard errors are clustered at the PSU score level.

Figure A8: Enrollment in a teacher training program as a function of the PSU Score



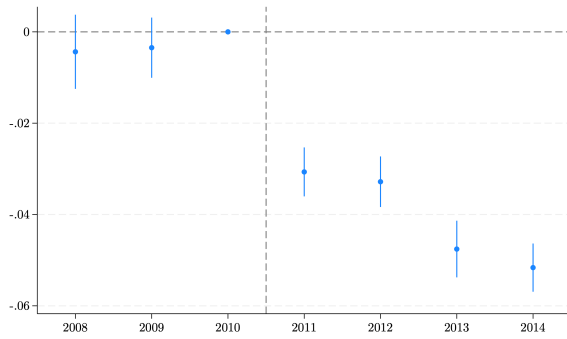
Note: The figure shows the probability of enrolling in a teacher training program one year after high school graduation, as a function of the PSU score, before and after the reform. Panel (a) plots the entire PSU distribution, Panels (b) to (d) zoom in on the specific groups that were differently affected by the reform. The gray squares represent teachers who took the PSU for the first time before the reform was implemented, and the blue dots represent those who took the PSU after the reform was implemented. The two horizontal lines represent several relevant cutoffs, namely the loan eligibility cutoff (475), the cutoff preventing students from enrolling in BVP-eligible teacher training programs (500), the cutoff defining the eligibility for the BVP scholarship (600), and the two cutoffs for additional BVP benefits (700 and 720).

Figure A9: BVP Eligibility and employment at school - Before-After analysis

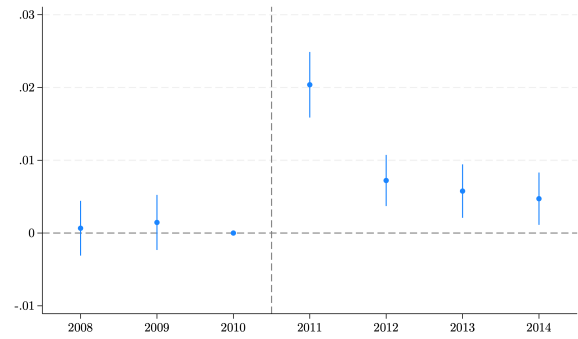


Note: The figure shows before-after difference for students scoring above 600 using as outcomes the probability of being employed in a school, and the probability of being employed by type of school up to five years after graduation. The sample is restricted to those taking the college admissions test between 2009 and 2014.

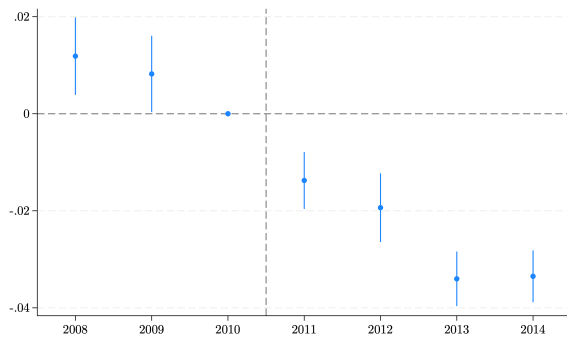
Figure A10: Trends on recruitment variables



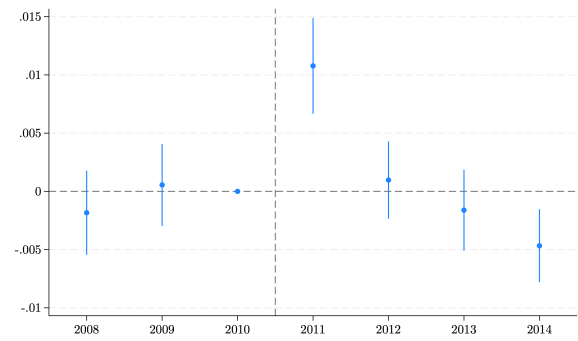
(a) Enrollment - below 500



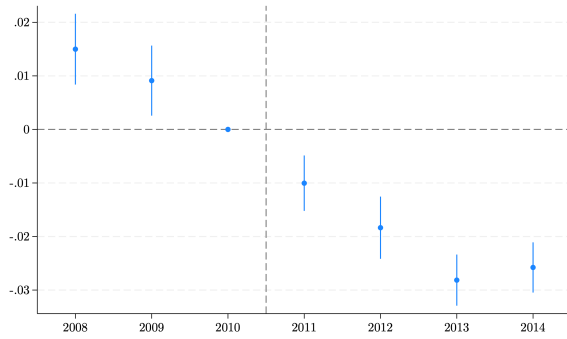
(b) Enrollment - above 600



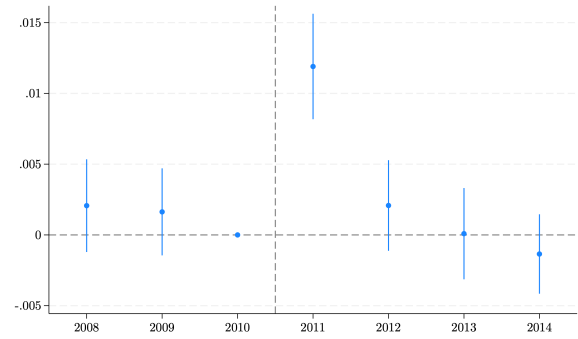
(c) Graduation - below 500



(d) Graduation - above 600



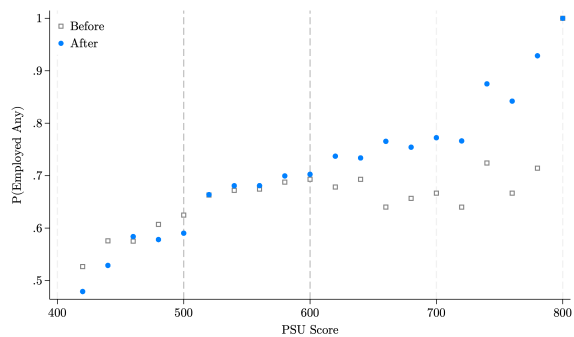
(e) Employment - below 500



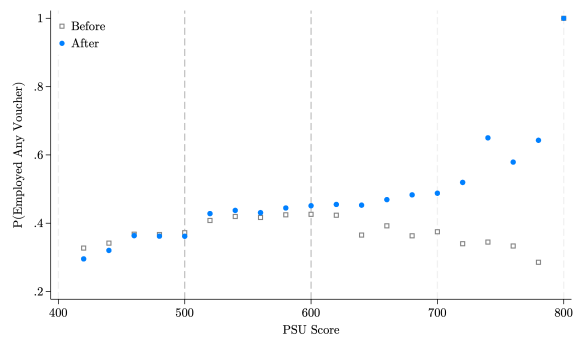
(f) Employment - above 600

Note: The figure presents the trends on our main recruitment variables of interest. On the left, the focus is on PSU takers scoring between 475 and 500, while on the right, on those scoring above 600. *Enrollment* is an indicator for individuals enrolling in a teacher training program on the year after high school graduation. *Graduation* is an indicator for individuals graduating from a teacher training program within eight years of high school graduation. *Employment* is indicator for individuals observed working in any school within 2 years of their graduation from a teacher training program.

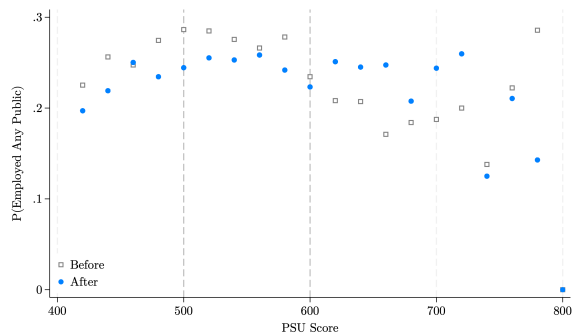
Figure A11: Employment in publicly funded schools as a function of the PSU Score



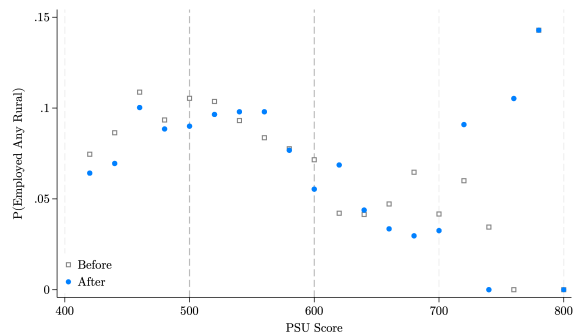
(a) Any School



(b) Voucher Schools



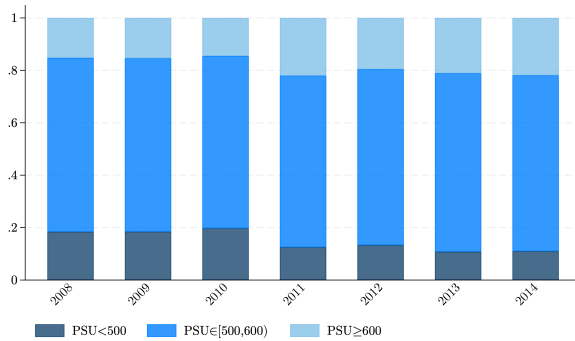
(c) Public School



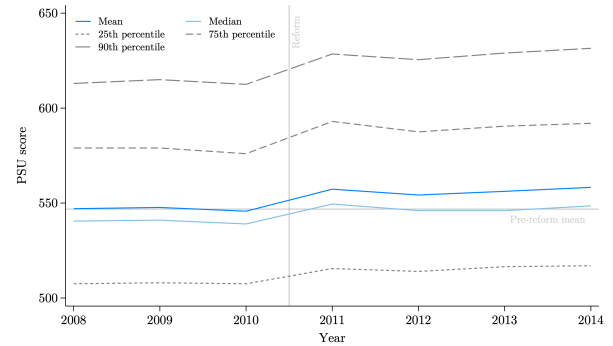
(d) Rural Schools

Note: The figure shows the probability of being employed in different types of schools in the first two years after graduation as a function of the PSU score. Panel (a) plots the probability of being employed in any school, Panel (b) plots the probability of being employed in a voucher school, Panel (c) plots the probability of being employed in a public school, and Panel (d) plots the probability of being employed in a rural school. The gray squares represent teachers who took the PSU for the first time before the reform was implemented, and the blue dots represent those who took the PSU after the reform was implemented. The two horizontal lines represent the cutoff preventing students from enrolling in BVP-eligible teacher training programs (500) and the cutoff defining the eligibility for the BVP scholarship (600).

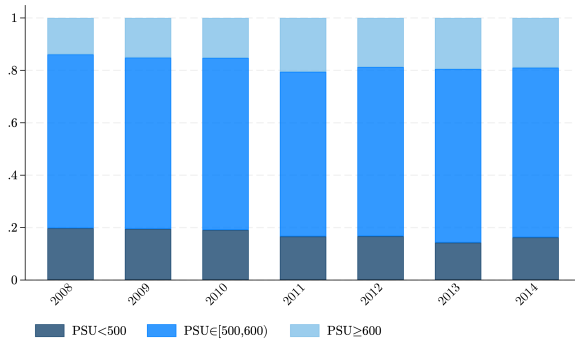
Figure A12: Trends in ability composition of enrollment and employment



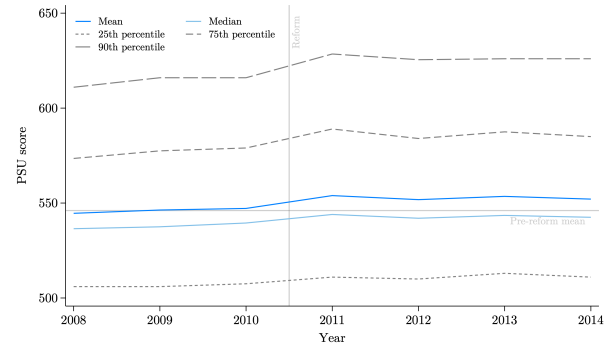
(a) Share of enrollees by score



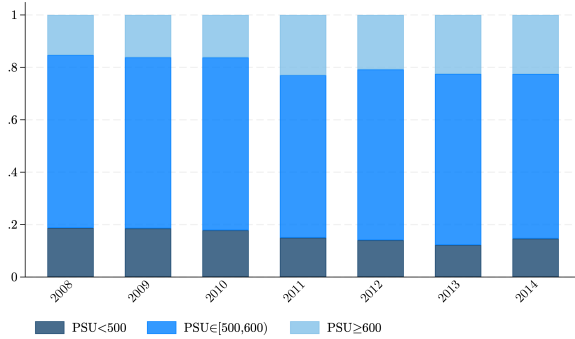
(b) Score statistics - Enrollment



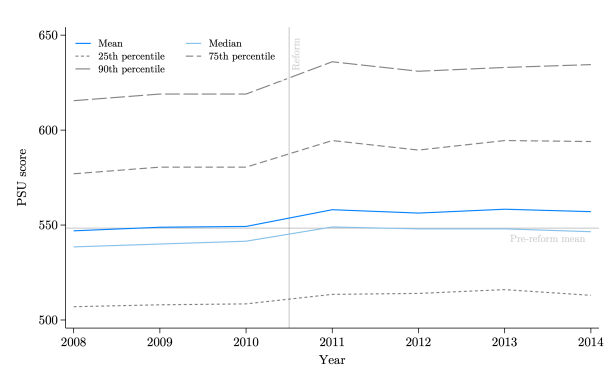
(c) Share of graduates by score



(d) Score statistics - Graduation



(e) Share of employed teachers by score

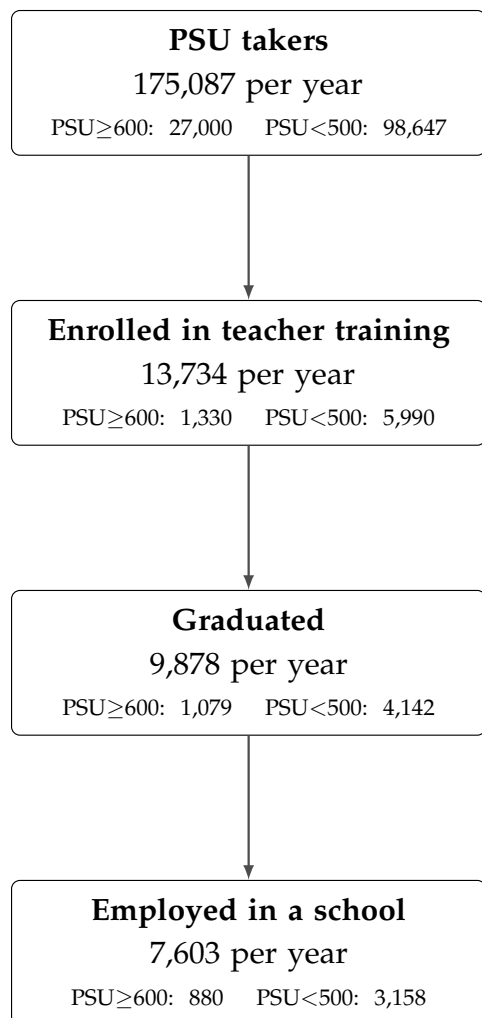


(f) Score statistics - Employment

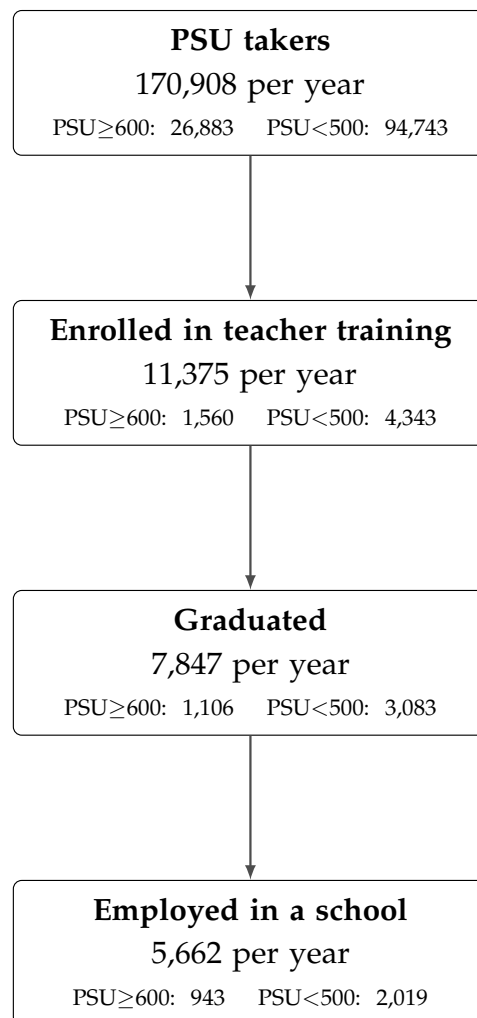
Note: The figures show the trends in PSU statistics by cohort of higher education enrollment. The two plots on the top focus on enrollment in teacher training programs. The two plots on the middle focus on graduation after 8 years from a teacher training programs. The two plots on the bottom focus on teachers employed in schools 2 years after graduation. All results are conditional on scoring above 475 points.

Figure A13: Teacher pipeline: from PSU takers to employment

Pre-reform (2009–2010)

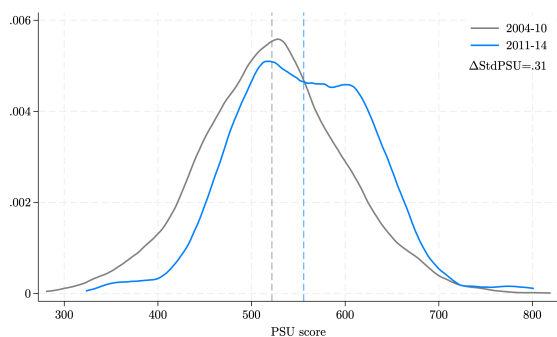


Post-reform (2011–2014)

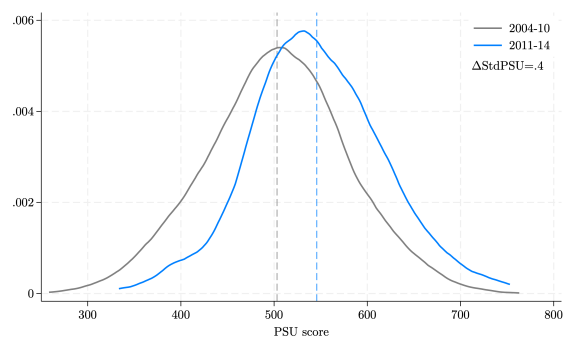


Note: The figure shows the average yearly number of individuals at each stage of the teacher pipeline, before and after the 2011 reform. Enrollment is measured within 2 years of taking the PSU; graduation within 8 years; employment in a school within 2 years of graduation. Each box also reports the number of individuals with $\text{PSU} \geq 600$ and $\text{PSU} < 500$.

Figure A14: Density of PSU of teachers before and after the reform



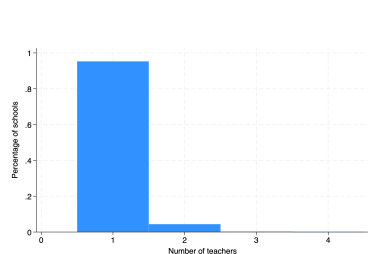
(a) Math Teachers



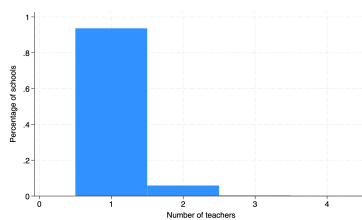
(b) Spanish Teachers

Note: The figures show the density of mathematics in Panel (a) and Spanish teachers in Panel (b) who enrolled in higher education before and after the reform and for whom we can reliably compute TVA. The dotted lines represent the mean PSU score for each group.

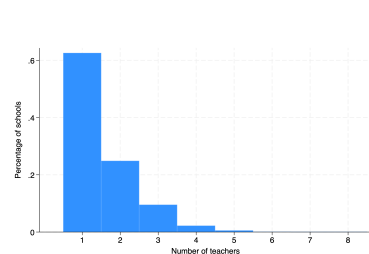
Figure A15: Variation for TVA Variance



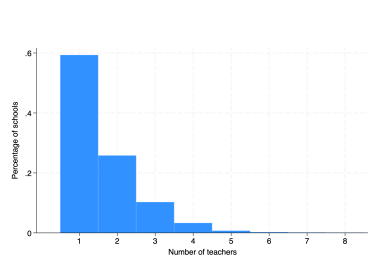
(a) # of teachers per school-year - Math



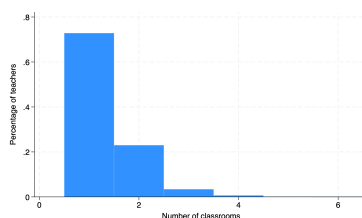
(b) # of teachers per school-year - Spanish



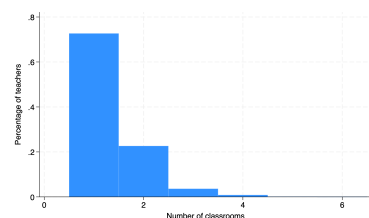
(c) # teachers per school - Math



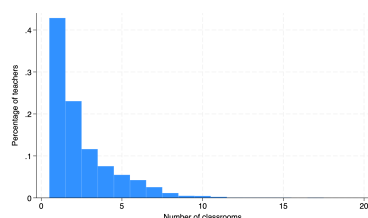
(d) # teachers per school - Spanish



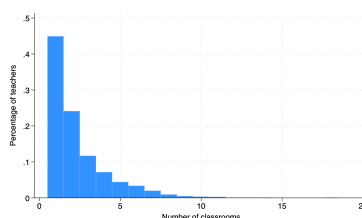
(e) # of classes per teacher-year - Math



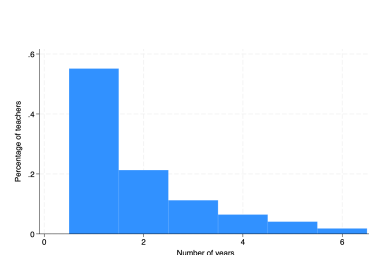
(f) # of classes per teacher-year - Spanish



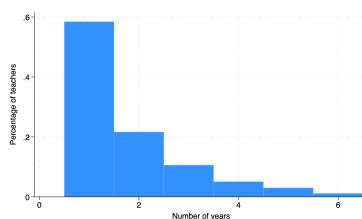
(g) # number of classes per teacher - Math



(h) # number of classes per teacher - Spanish



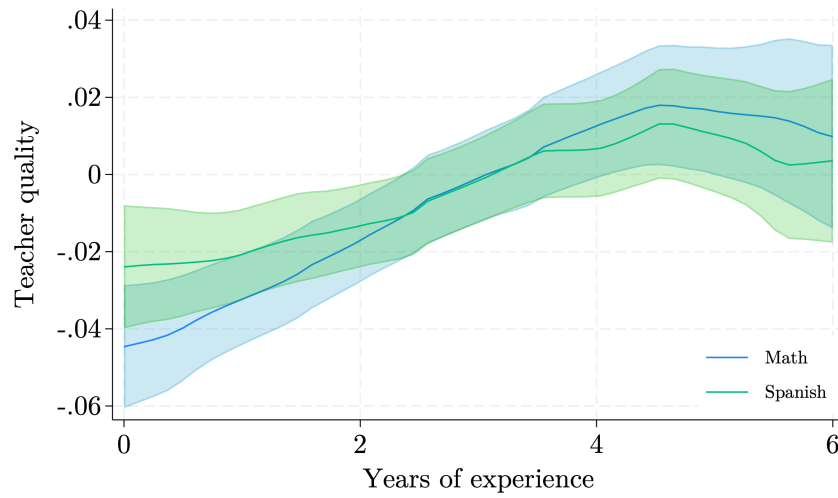
(i) # years by teacher



(j) # years by teacher

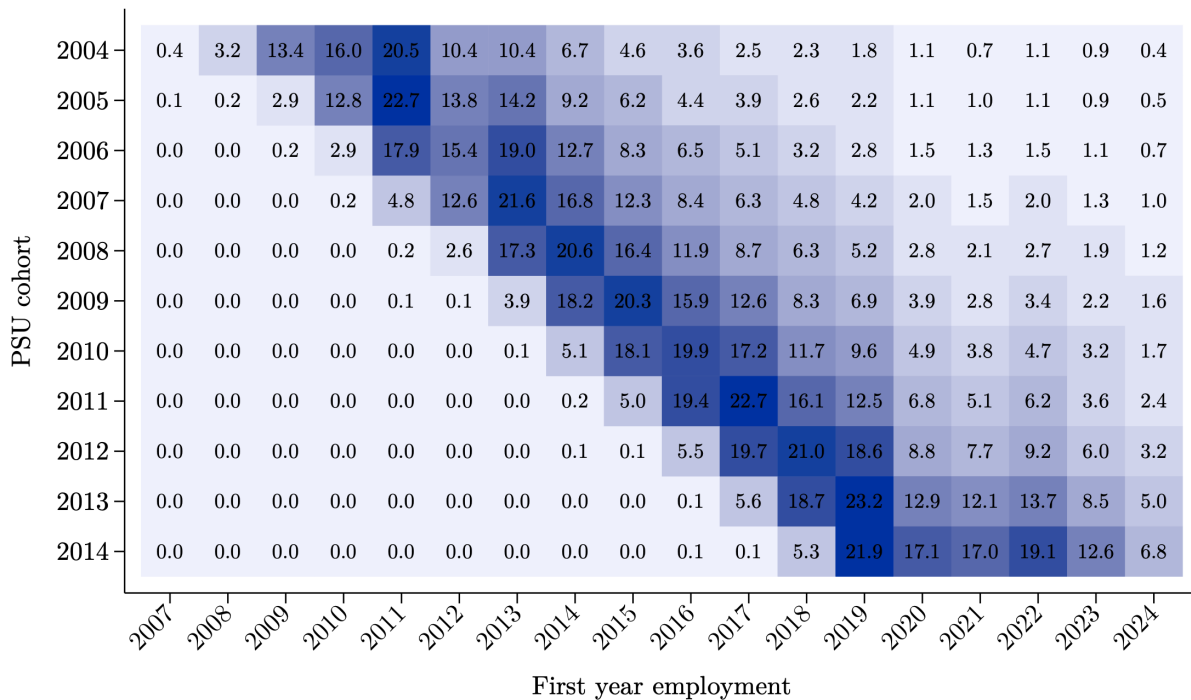
Note: Focusing on the teachers for which we can estimate class fixed effects, the figures show frequency plots by subjects. From the first to the last row, they show frequencies of teachers per school-year ((a) and (b)), and per school ((c) and (d)), frequencies of classes per teacher-year ((e) and (f)) and per school ((g) and (h)), and frequencies of years ((i) and (j)).

Figure A16: TVA estimates and experience



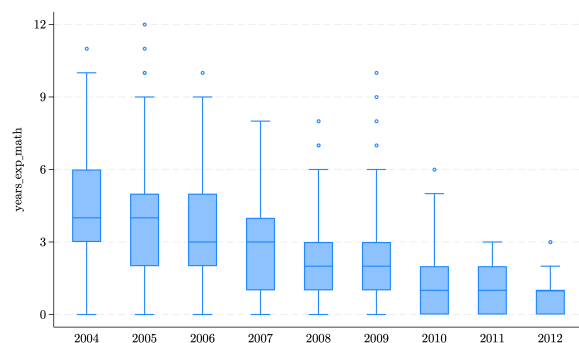
Note: The figures show the non-parametric relationship between TVA estimates and average year of experience, for mathematics and Spanish teachers.

Figure A17: Variation in the year of entry to the labor market within cohort

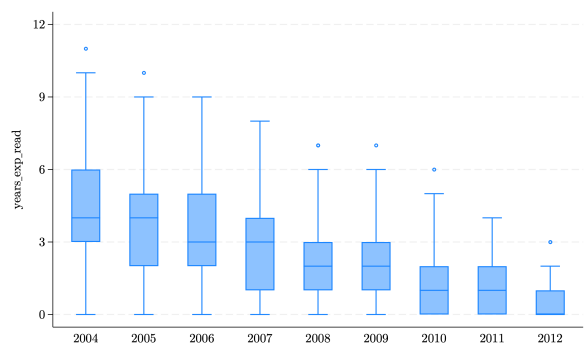


Note: The figure shows the frequency with which individuals of each cohort entered the labor market in each particular year. For illustrative purposes, cells with higher frequencies are colored in darker shades of blue.

Figure A18: Years of experience by PSU Cohort



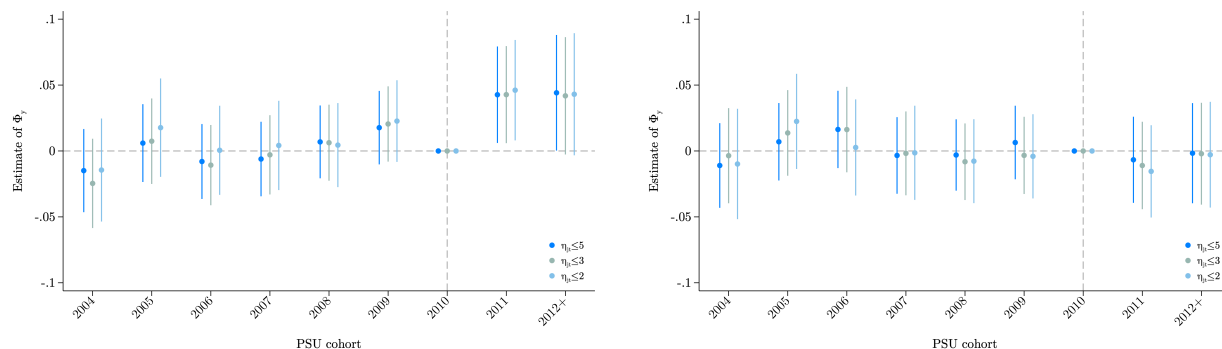
(a) Mathematics



(b) Spanish

Note: The figures show the distribution of years of experience by cohorts of study. Figure (a) refers to Mathematics teachers, while Figure (b) refers to Spanish teachers.

Figure A19: Average change in TVA by cohort - Robustness check by years of experience

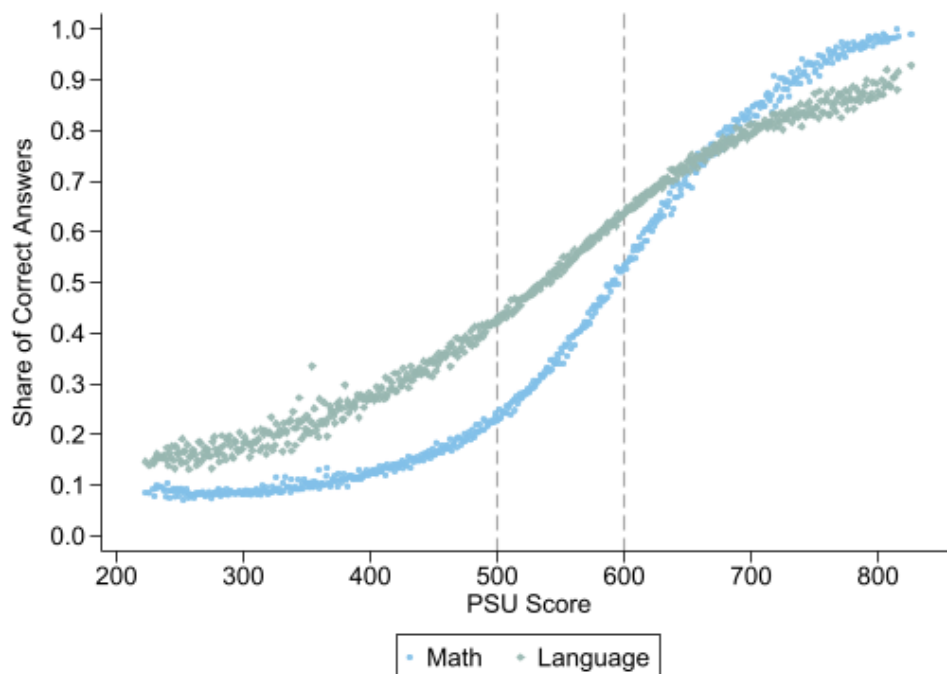


(a) Math

(b) Spanish

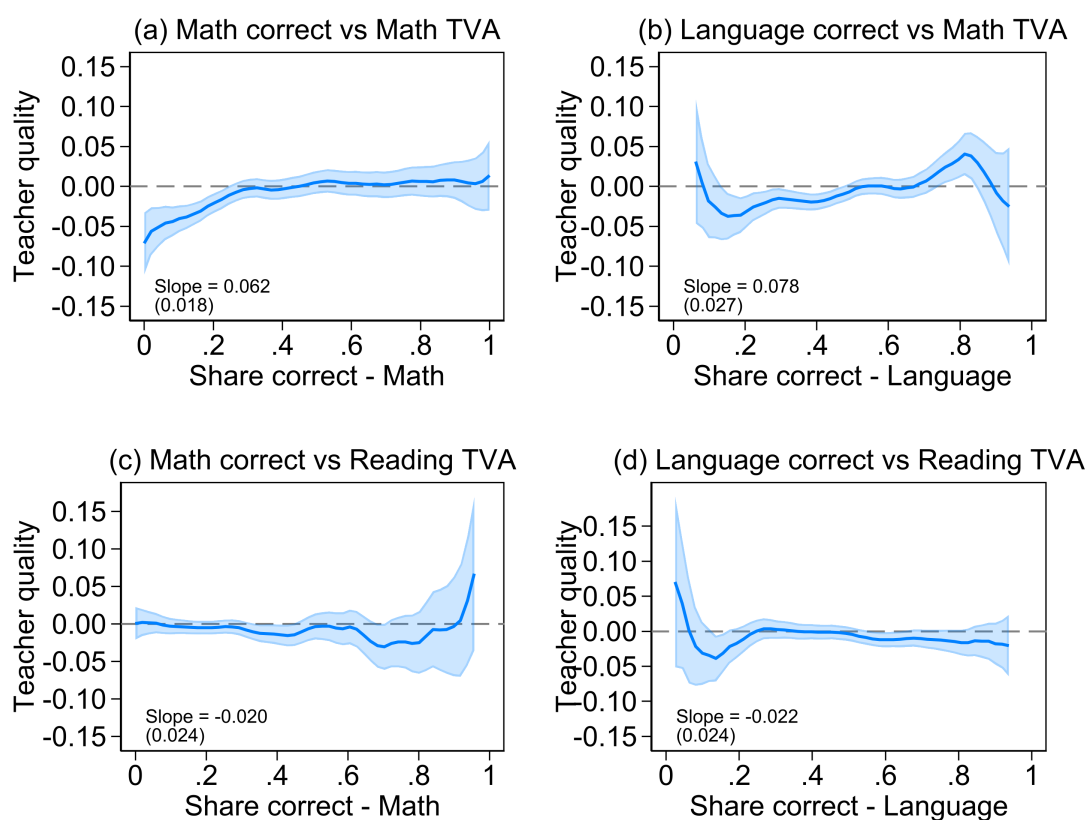
Note: The figures show OLS estimates for the coefficients Φ_y from model (6). Confidence intervals at the 10% level are displayed and computed using standard errors clustered at the teacher-by-year level. On the left panel, the focus is on math teachers, while on the right panel is on Spanish. The figures present three sets of coefficients, estimated by restricting the sample to teachers with less than five, three, and two years of experience.

Figure A20: Share of Correct Answers by PSU Score



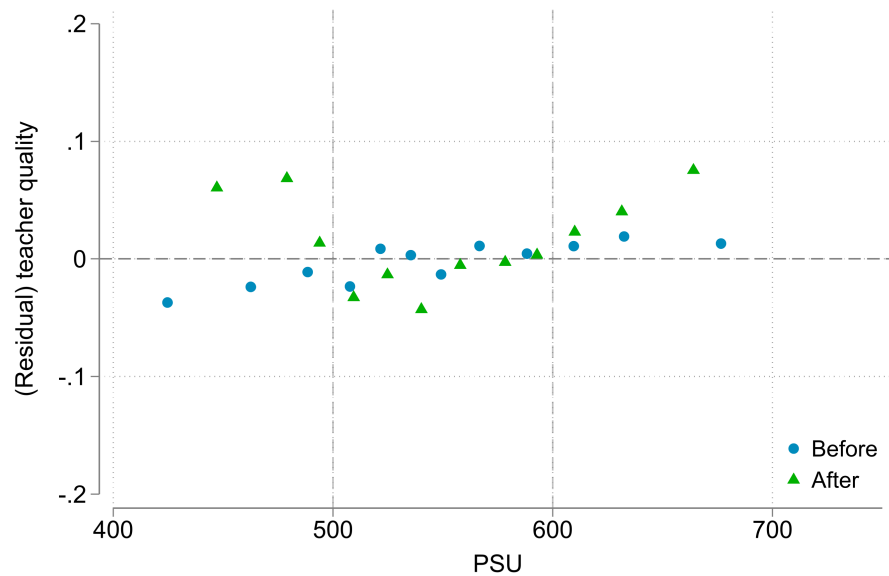
Note: Each dot represents the mean share of correct answers at a given PSU score (average of Math and Spanish), for test-takers in 2011–2014. Vertical dashed lines mark the PSU thresholds at 500 and 600.

Figure A21: Correlation Between PSU Correct Answers and Teacher Value-Added



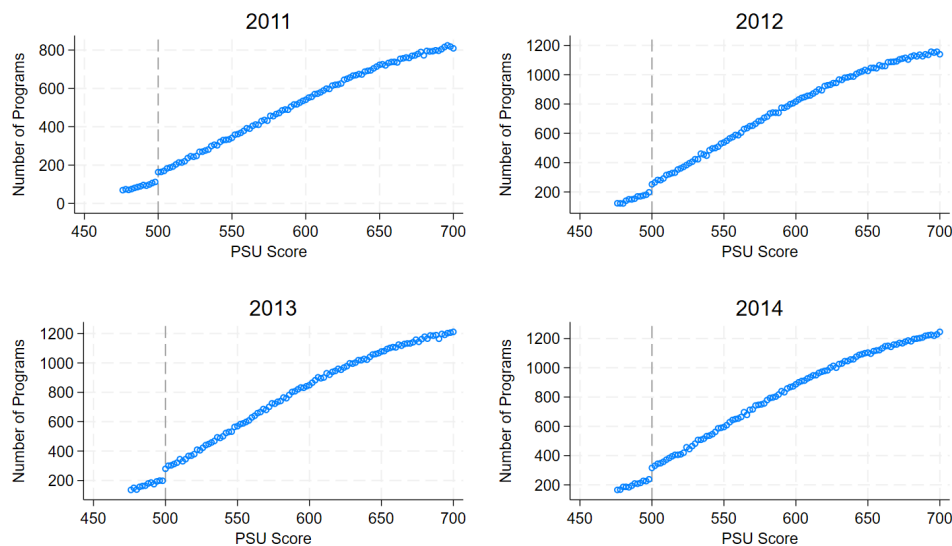
Note: The figures show the non-parametric relationship between TVA and the share of correct answers in the PSU exam, described by a local polynomial regression, for teachers that started their studies before the 2011 reform. The shaded area represents the 90% confidence interval. Panels (a) and (b) focus on mathematics teachers; panels (c) and (d) focus on Spanish teachers. The numbers in the left-bottom corner refer to the coefficient of the linear fit, with standard errors in parentheses.

Figure A22: Teacher value-added and PSU score - non-parametric (mathematics teachers)



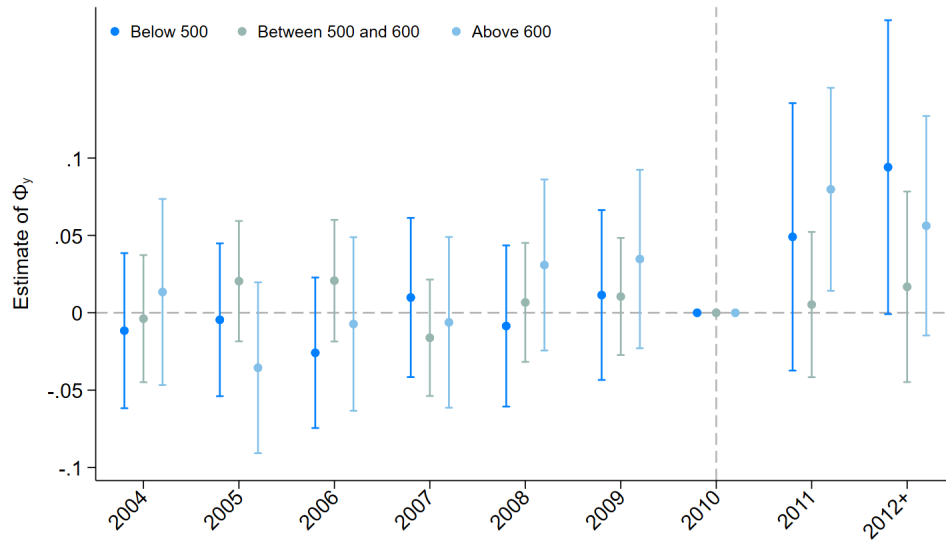
Note: The figure shows binned averages of residual teacher value-added by PSU score, for mathematics teachers who started their studies before (2004–2010, “Before”) and after (2011–2014, “After”) the reform. *(Residual) teacher quality* is the residual from a non-parametric regression of TVA on years of experience. To limit noise, we group teachers into three bins below 500, six between 500 and 600, and three above 600, chosen so that every bin contains at least 25 teachers.

Figure A23: Number of available programs in the CAS as a function of PSU



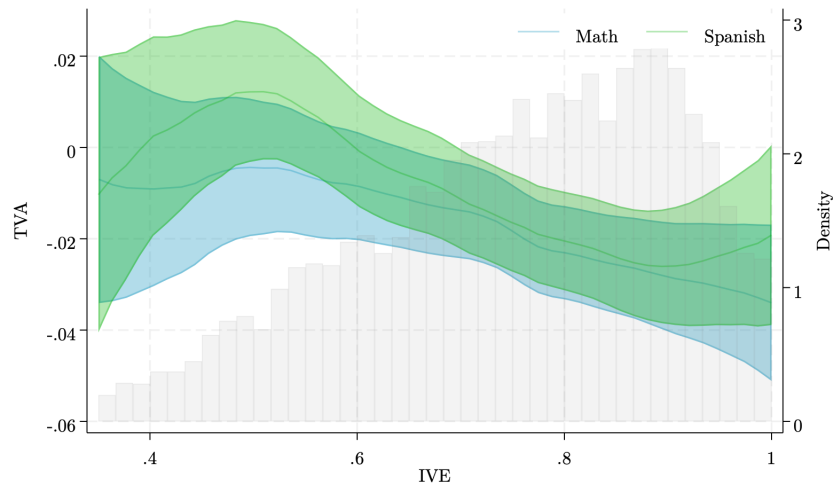
Note: The figure shows the size of the choice set of PSU takers, as a function of the PSU score. The choice set is defined as the number of higher education programs within the centralized admission system where a PSU taker can enroll.

Figure A24: Average change in TVA by cohort - Results by PSU score



Note: The figure shows OLS estimates for the coefficients Φ_y from (6) for math teachers. Confidence intervals at the 10% level are displayed and computed using standard errors clustered at the teacher-by-year level. The figures present three sets of coefficients, estimated by restricting the sample to teachers with PSU below 500, between 500 and 600, and above 600.

Figure A25: TVA estimates and IVE-SINAE



Note: The figures show the non-parametric relationship between TVA estimates and average IVE-SINAE of the schools in which teachers work. The blue figure represents the relationship for math teachers and the green figure for Spanish teachers. The gray bars represent the density distribution of the IVE-SINAE index in our sample of teachers.

B Additional Tables

Table B1: Access to financial aid based on income and cohort just below the 600 Cut-off

	2011	2012	2013	2014
Family Income				
<i>Quintile 1</i>	<i>BIC</i>	<i>BIC + JGM</i>	<i>BIC + JGM</i>	<i>BIC + JGM</i>
<i>Quintile 2</i>	<i>BIC</i>	<i>BIC + JGM</i>	<i>BIC + JGM</i>	<i>BIC + JGM</i>
<i>Quintile 3</i>	<i>FSCU + CAE</i>	<i>BIC + JGM</i>	<i>BIC + JGM</i>	<i>BIC + JGM</i>
<i>Quintile 4</i>	<i>FSCU + CAE</i>	<i>FSCU + CAE</i>	<i>FSCU + CAE</i>	<i>FSCU + CAE</i>
<i>Quintile 5</i>	<i>N.E.</i>	<i>N.E.</i>	<i>N.E.</i>	<i>N.E.</i>

Note: Displayed are the eligibility criteria in terms of family income and the cohort of study for scholarships and loans. Scholarships refer to the Bicentennial (BIC) and Juan Gómez Millas (JGM), which can be used in CRUCH and non-CRUCH institutions, respectively. Juan Gómez Millas was introduced in 2012. Loans refer to the FSCU (*Fondo Solidario de Crédito Universitario*) and CAE (*Crédito con Aval del Estado*) non-CRUCH, which can be used in CRUCH and non-CRUCH institutions, respectively. Note that the vast majority of students enrolled in CRUCH institutions rely on the grant *Bicentenario* (BIC) and the loan *Fondo Solidario de Crédito Universitario* (FSCU), covering up to 90% of tuition in CRUCH institutions. In non-CRUCH institutions, students applying for financial aid mostly rely on the loan *Crédito con Aval del Estado* (CAE), and on the scholarship *Juan Gómez Millas*, which was introduced in 2012. The former covers up to 90% of fees, while the latter is a fixed amount equal to two-thirds of the median tuition.

Table B2: Summary statistics PSU Takers

	Mean	SD	Min	Max	N
<i>Demographic</i>					
High School GPA	5.597	0.493	4	7	677,275
Female	0.529	0.499	0	1	683,633
Father High School	0.546	0.498	0	1	683,633
Mother High School	0.600	0.490	0	1	683,633
Any Parent College	0.379	0.485	0	1	564,107
# Family Members	4.561	1.608	1	30	646,982
# Family Members Working	1.232	0.782	0	30	683,633
Single Parent	0.166	0.372	0	1	640,067
Father Working	0.654	0.476	0	1	587,549
Mother Working	0.389	0.488	0	1	625,760
Applied Financial Aid	0.695	0.461	0	1	683,633
Income Quintile 1	0.354	0.478	0	1	474,858
Income Quintile 2	0.243	0.429	0	1	474,858
Income Quintile 3	0.166	0.372	0	1	474,858
Income Quintile 4	0.134	0.341	0	1	474,858
Income Quintile 5	0.103	0.304	0	1	474,858
<i>Enrollment</i>					
Enrolled HEI	0.611	0.488	0	1	683,633
University	0.388	0.487	0	1	683,633
Traditional university	0.204	0.403	0	1	683,633
Enrolled	0.045	0.208	0	1	683,633
BEA Eligibility	0.076	0.266	0	1	683,633
BVP by year 1	0.007	0.084	0	1	683,633
BVP by year 4	0.011	0.105	0	1	683,633

Note: The table shows summary statistics of demographic and variables related to enrollment decisions and relevant aid take-up. The sample consists of the universe of PSU takers from 2011 to 2014.

Table B3: Balance check

	Baseline Mean	PSU $\geq$ 600	Baseline Mean	PSU $\geq$ 500
High School GPA	5.899 (0.004)	-0.008 (0.006)	5.547 (0.003)	0.005 (0.004)
Female	0.503 (0.004)	0.001 (0.005)	0.534 (0.004)	0.001 (0.006)
Father High School	0.712 (0.004)	0.004 (0.005)	0.564 (0.002)	-0.005 (0.004)
Mother High School	0.781 (0.004)	0.009 (0.005)	0.625 (0.003)	0.002 (0.004)
Any Parent College	0.635 (0.006)	0.002 (0.008)	0.345 (0.004)	-0.008 (0.005)
# Family Members	4.444 (0.011)	-0.007 (0.016)	4.500 (0.010)	-0.006 (0.014)
# Family Members Working	1.212 (0.006)	0.016 (0.009)	1.219 (0.004)	0.003 (0.006)
Single Parent	0.131 (0.003)	0.005 (0.004)	0.176 (0.003)	0.003 (0.004)
Father Working	0.718 (0.005)	0.001 (0.006)	0.645 (0.003)	0.004 (0.004)
Mother Working	0.471 (0.004)	0.001 (0.006)	0.384 (0.002)	0.003 (0.004)
Applied Financial Aid	0.779 (0.004)	0.006 (0.005)	0.771 (0.004)	0.005 (0.005)
Income Quintile 1	0.213 (0.004)	0.006 (0.007)	0.367 (0.005)	-0.012 (0.006)
Income Quintile 2	0.224 (0.004)	-0.004 (0.005)	0.254 (0.004)	0.013 (0.005)
Income Quintile 3	0.191 (0.004)	-0.004 (0.005)	0.174 (0.003)	0.003 (0.005)
Income Quintile 4	0.197 (0.004)	-0.002 (0.005)	0.128 (0.002)	-0.002 (0.004)
Income Quintile 5	0.175 (0.003)	0.004 (0.006)	0.076 (0.002)	-0.002 (0.003)

Note: The table presents results from equation (1) using predetermined characteristics as outcome. The bandwidth is fixed in any specification to 50 for the 600 cutoff, and to 24 for the 500 cutoff. The model is estimated using a triangular weighted kernel. Standard errors are clustered at the PSU level.

Table B4: Sensitivity of RD estimates to different local polynomials

	(1) Enrolled	(2) Enrolled	(3) Graduated	(4) Graduated	(5) Employed	(6) Employed
<i>Panel A. 600 cutoff</i>						
RD Estimate	0.020*** (0.004)	0.019*** (0.004)	0.008** (0.004)	0.006 (0.004)	0.009** (0.004)	0.008* (0.004)
Polynomial order	2	3	2	3	2	3
Bandwidth	66	91	68	96	71	98
Observations	190,924	262,198	195,460	275,067	205,334	280,956
<i>Panel B. 500 cutoff</i>						
RD Estimate	-0.028*** (0.008)	-0.029*** (0.009)	-0.015*** (0.006)	-0.015** (0.006)	-0.012*** (0.004)	-0.010** (0.004)
Polynomial order	2	3	2	3	2	3
Bandwidth	24	24	24	24	24	24
Observations	113,468	113,468	113,468	113,468	113,468	113,468

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table presents the RD estimates for the effect of the reform on enrollment, graduation, and employment. The table shows Regression Discontinuity parameters where all specifications are estimated using weighted local polynomial regressions, consider high school graduates taking the PSU between 2011 and 2014, and are estimated using a polynomial of order 2 or 3. For the 600 cutoff, the RD Estimate corresponds to the effect of being marginally above 600 or marginally eligible for the scholarship; bandwidths are chosen optimally according to Calonico et al. (2019). For the 500 cutoff, the RD Estimate corresponds to the effect of being marginally below 500 or marginally affected by the restriction; bandwidths are set to 24 to avoid capturing the effect of the loan eligibility cutoff at 475 points. Standard errors are clustered at the PSU test score level and reported in parentheses. *Enrolled* is an indicator for individuals enrolling in a teacher training program in the year after high school graduation. *Graduated* is an indicator for individuals graduating from a teacher training program within eight years of high school graduation. *Employed* is an indicator for individuals observed working in any school within 2 years of their graduation from a teacher training program.

Table B5: RD estimates - cohorts 2008-2010

	(1) Enrolled	(2) Graduated	(3) Employed	(4) Private	(5) Voucher	(6) Public	(7) Rural
<i>Panel A. 600 cutoff</i>							
RD Estimate	0.001 (0.004)	-0.003 (0.004)	-0.004 (0.003)	0.000 (0.001)	-0.003 (0.003)	0.000 (0.002)	0.001 (0.001)
Bandwidth	45	45	49	59	54	45	56
Observations	93,439	94,769	102,938	122,933	112,348	93,439	117,653
<i>Panel B. 500 cutoff</i>							
RD Estimate	0.004 (0.004)	0.001 (0.003)	0.003 (0.004)	0.001 (0.001)	0.002 (0.003)	0.001 (0.002)	-0.001 (0.001)
Bandwidth	24	24	24	24	24	24	24
Observations	82,676	82,676	82,676	82,676	82,676	82,676	82,676

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table shows Regression Discontinuity parameters where all specifications are estimated using weighted local linear regressions and consider high school graduates taking the PSU before the reform, that is between 2008 and 2010. For the 600 cutoff, the RD Estimate corresponds to the effect of being marginally *above* 600 or marginally eligible for the scholarship; bandwidths are chosen optimally according to Calonico et al. (2019). For the 500 cutoff, the RD Estimate corresponds to the effect of being marginally *below* 500 or marginally affected by the restriction; bandwidths are set to 24 to avoid capturing the effect of the loan eligibility cutoff at 475 points. Standard errors are clustered at the PSU test score level and reported in parentheses. *Observations* correspond to the number of observations with non-zero weight given the chosen bandwidth. *Enrolled* is an indicator for individuals enrolling in a teacher training program in the year after high school graduation. *Graduated* is an indicator for individuals graduating from a teacher training program within eight years of high school graduation. *Employed* is an indicator for individuals observed working in any school within 2 years of their graduation from a teacher training program. *Voucher*, *Public*, and *Rural* are indicators for individuals observed working in each particular type of school within 2 years of their graduation from a teacher training program. All results are unconditional.

Table B6: Across-cohort comparison

	(1) Enrolled	(2) Graduated	(3) Employed	(4) Public	(5) Voucher	(6) Private	(7) Rural
<i>Panel A. Above 600</i>							
After Reform	0.013*** (0.002)	0.006*** (0.001)	0.006*** (0.001)	0.002*** (0.001)	0.005*** (0.001)	0.000 (0.000)	0.000 (0.000)
Observations	108,092	108,092	108,092	108,092	108,092	108,092	108,092
Baseline Mean	0.04	0.04	0.03	0.01	0.02	0.01	0.00
Percentage Change	34.3	14.4	19.4	25.4	27.7	0.9	12.9
<i>Panel B. Below 500</i>							
After Reform	-0.006*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.004*** (0.000)	-0.004*** (0.001)	-0.000** (0.000)	-0.002*** (0.000)
Observations	390,297	390,297	390,297	390,297	390,297	390,297	390,297
Baseline Mean	0.04	0.04	0.03	0.01	0.02	0.00	0.01
Percentage Change	-18.0	-16.0	-24.0	-27.3	-23.1	-19.7	-29.4
<i>Panel C. Full sample</i>							
After Reform	-0.002** (0.001)	-0.006*** (0.001)	-0.006*** (0.001)	-0.004*** (0.000)	-0.003*** (0.001)	-0.000 (0.000)	-0.001*** (0.000)
Observations	694,874	694,874	694,874	694,874	694,874	694,874	694,874
Baseline Mean	0.05	0.06	0.04	0.02	0.03	0.00	0.01
Percentage Change	-3.9	-11.3	-14.1	-20.0	-11.4	-5.5	-20.6

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table shows differences in means between outcomes for PSU takers before and after 2011, restricting the cohorts of interest to 2009 to 2012. Standard errors are clustered at the PSU test score level and reported in parentheses. The *Baseline Mean* refers to the outcome mean for students enrolling before the 2011 reform. The *Percentage Change* is computed as the coefficient over the Baseline Mean (times 100). *Enrolled* is an indicator for individuals enrolling in a teacher training program in the year after high school graduation. *Graduated* is an indicator for individuals graduating from a teacher training program within eight years of high school graduation. *Employed* is an indicator for individuals observed working in any school within 2 years of their graduation from a teacher training program. *Voucher*, *Public*, and *Rural* are indicators for individuals observed working in each particular type of school within 2 years of their graduation from a teacher training program.

Table B7: Cohorts taking the SIMCE by year

Year	Cohort										
	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
2019									8		
2018						10				6	
2017					10		8				4
2016				10				6		4	
2015			10		8		6		4		2*
2014		10		8		6		4		2*	
2013	10		8		6		4		2*		
2012						4		2*			
2011	8				4						
2010				4							
2009			4								
2008		4									
2007	4										

Note: Cohorts are identified by the year in which students start 1st grade of elementary school. The table shows in which grades different cohorts took the SIMCE. Grades 2,4,6, and 8 correspond to elementary school grades, and 10th grade corresponds to high school. The years in which students could potentially be taught by a BVP-eligible teacher have been highlighted in gray. Finally, note that second graders are only tested on their Spanish knowledge.

Table B8: Summary statistics mathematics teachers in the sample

	All			PSU < 500		PSU ≥ 600	
	(1) All	(2) Before	(3) After	(4) Before	(5) After	(6) Before	(7) After
Teacher characteristics							
Female	0.71 (0.45)	0.71 (0.46)	0.74 (0.44)	0.75 (0.43)	0.74 (0.44)	0.60 (0.49)	0.68 (0.47)
Father has college education	0.17 (0.37)	0.17 (0.37)	0.18 (0.39)	0.11 (0.31)	0.11 (0.32)	0.33 (0.47)	0.34 (0.48)
Mother has college education	0.13 (0.33)	0.13 (0.33)	0.12 (0.33)	0.09 (0.29)	0.04 (0.21)	0.22 (0.42)	0.26 (0.44)
Quintile 1 Family Income	0.44 (0.50)	0.46 (0.50)	0.41 (0.49)	0.53 (0.50)	0.44 (0.50)	0.31 (0.46)	0.30 (0.46)
Quintile 2 Family Income	0.21 (0.41)	0.20 (0.40)	0.23 (0.42)	0.22 (0.42)	0.30 (0.46)	0.21 (0.41)	0.19 (0.39)
Quintile 3 Family Income	0.15 (0.35)	0.15 (0.35)	0.15 (0.35)	0.12 (0.32)	0.10 (0.31)	0.17 (0.38)	0.19 (0.39)
Quintile 4 Family Income	0.12 (0.32)	0.11 (0.32)	0.14 (0.34)	0.09 (0.28)	0.08 (0.27)	0.17 (0.37)	0.17 (0.37)
Quintile 5 Family Income	0.08 (0.27)	0.08 (0.27)	0.08 (0.28)	0.04 (0.21)	0.08 (0.27)	0.14 (0.35)	0.16 (0.37)
Years of experience	2.55 (1.92)	2.72 (1.92)	0.83 (0.78)	2.86 (1.96)	1.01 (0.85)	2.36 (1.91)	0.79 (0.68)
Works in public school	0.34 (0.47)	0.35 (0.47)	0.30 (0.45)	0.45 (0.49)	0.44 (0.49)	0.21 (0.40)	0.23 (0.41)
Works in private school	0.07 (0.26)	0.08 (0.26)	0.05 (0.22)	0.04 (0.19)	0.03 (0.18)	0.20 (0.39)	0.08 (0.26)
Works in subsidized school	0.58 (0.48)	0.58 (0.48)	0.65 (0.47)	0.51 (0.49)	0.53 (0.49)	0.59 (0.48)	0.69 (0.45)
Works in rural school	0.17 (0.37)	0.17 (0.37)	0.16 (0.36)	0.23 (0.42)	0.28 (0.45)	0.06 (0.24)	0.06 (0.24)
# classrooms taught	2.20 (2.11)	2.29 (2.17)	1.32 (0.92)	2.25 (2.15)	1.44 (1.02)	2.01 (1.89)	1.26 (0.83)
# classrooms taught per year	1.14 (0.71)	1.15 (0.72)	1.03 (0.60)	1.11 (0.69)	1.06 (0.66)	1.17 (0.75)	1.02 (0.58)
Classroom characteristics							
Class size	18.56 (12.25)	18.49 (12.21)	19.26 (12.59)	16.84 (11.93)	15.83 (12.64)	20.85 (12.28)	21.44 (12.12)
% female students	0.42 (0.24)	0.42 (0.24)	0.43 (0.25)	0.41 (0.24)	0.42 (0.27)	0.42 (0.26)	0.43 (0.23)
% students Q1 family income	0.17 (0.21)	0.17 (0.21)	0.14 (0.19)	0.21 (0.23)	0.19 (0.22)	0.10 (0.15)	0.11 (0.16)
% students Q5 family income	0.17 (0.26)	0.16 (0.26)	0.19 (0.26)	0.11 (0.20)	0.09 (0.18)	0.28 (0.34)	0.27 (0.31)
% students with college educated mother	0.11 (0.17)	0.11 (0.17)	0.12 (0.17)	0.08 (0.13)	0.06 (0.11)	0.20 (0.25)	0.17 (0.20)
% students with college educated father	0.13 (0.20)	0.13 (0.20)	0.14 (0.20)	0.09 (0.15)	0.07 (0.14)	0.23 (0.28)	0.20 (0.24)
Average lagged SIMCE score Math	-0.03 (0.48)	-0.03 (0.48)	-0.03 (0.48)	-0.14 (0.45)	-0.16 (0.43)	0.19 (0.52)	0.10 (0.53)
Average lagged SIMCE score Spanish	0.01 (0.40)	0.01 (0.40)	0.01 (0.40)	-0.08 (0.39)	-0.09 (0.41)	0.17 (0.42)	0.11 (0.43)
Observations	4386	3981	405	1525	93	608	117

Note: The table presents the mean and standard deviation (in parenthesis) of several variables for mathematics teachers in our main TVA analysis sample. Column 1 presents the statistics for the full sample. Columns 2 and 3, divide the sample between teachers that started their studies before and after the reform. Columns 4 and 5 present the summary statistics for those scoring below 500 points. Columns 6 and 7 present the summary statistics for those scoring above 600 points.

Table B9: Summary statistics Spanish teachers in the sample

	All			PSU < 500		PSU ≥ 600	
	(1) All	(2) Before	(3) After	(4) Before	(5) After	(6) Before	(7) After
Teacher characteristics							
Female	0.81 (0.39)	0.81 (0.39)	0.84 (0.37)	0.83 (0.37)	0.85 (0.36)	0.73 (0.44)	0.80 (0.40)
Father has college education	0.16 (0.37)	0.17 (0.37)	0.13 (0.33)	0.12 (0.32)	0.06 (0.24)	0.38 (0.49)	0.27 (0.45)
Mother has college education	0.13 (0.33)	0.13 (0.33)	0.11 (0.31)	0.09 (0.29)	0.06 (0.23)	0.30 (0.46)	0.28 (0.45)
Quintile 1 Family Income	0.47 (0.50)	0.49 (0.50)	0.41 (0.49)	0.56 (0.50)	0.50 (0.50)	0.21 (0.41)	0.21 (0.41)
Quintile 2 Family Income	0.21 (0.41)	0.19 (0.39)	0.26 (0.44)	0.20 (0.40)	0.24 (0.43)	0.21 (0.41)	0.26 (0.44)
Quintile 3 Family Income	0.15 (0.35)	0.14 (0.35)	0.17 (0.37)	0.11 (0.31)	0.14 (0.35)	0.18 (0.38)	0.24 (0.43)
Quintile 4 Family Income	0.11 (0.31)	0.11 (0.31)	0.11 (0.31)	0.07 (0.25)	0.09 (0.29)	0.21 (0.41)	0.15 (0.36)
Quintile 5 Family Income	0.07 (0.26)	0.08 (0.27)	0.06 (0.24)	0.06 (0.24)	0.03 (0.17)	0.19 (0.39)	0.14 (0.35)
Years of experience	2.63 (1.97)	2.81 (1.96)	0.80 (0.76)	2.93 (2.00)	0.85 (0.86)	2.63 (1.89)	0.86 (0.78)
Works in public school	0.33 (0.46)	0.33 (0.46)	0.31 (0.46)	0.43 (0.49)	0.45 (0.50)	0.13 (0.33)	0.24 (0.42)
Works in private school	0.09 (0.28)	0.09 (0.28)	0.06 (0.23)	0.04 (0.19)	0.03 (0.16)	0.29 (0.45)	0.09 (0.29)
Works in subsidized school	0.58 (0.48)	0.57 (0.48)	0.63 (0.48)	0.53 (0.49)	0.52 (0.50)	0.58 (0.48)	0.67 (0.47)
Works in rural school	0.16 (0.36)	0.16 (0.36)	0.18 (0.38)	0.21 (0.40)	0.28 (0.45)	0.03 (0.16)	0.06 (0.23)
# classrooms taught	2.07 (1.99)	2.13 (2.05)	1.42 (1.06)	2.08 (1.99)	1.42 (1.08)	2.19 (2.11)	1.25 (0.93)
# classrooms taught per year	1.14 (0.74)	1.14 (0.74)	1.10 (0.68)	1.11 (0.72)	1.04 (0.61)	1.23 (0.81)	1.06 (0.67)
Classroom characteristics							
Class size	18.70 (12.38)	18.56 (12.33)	20.18 (12.79)	17.17 (11.90)	17.19 (12.62)	21.14 (12.77)	21.86 (12.70)
% female students	0.41 (0.24)	0.41 (0.24)	0.43 (0.24)	0.41 (0.24)	0.44 (0.24)	0.42 (0.26)	0.44 (0.25)
% students Q1 family income	0.17 (0.21)	0.17 (0.21)	0.13 (0.18)	0.21 (0.23)	0.19 (0.24)	0.07 (0.11)	0.09 (0.11)
% students Q5 family income	0.18 (0.27)	0.17 (0.27)	0.18 (0.24)	0.11 (0.20)	0.13 (0.19)	0.36 (0.37)	0.27 (0.31)
% students with college educated mother	0.12 (0.18)	0.12 (0.18)	0.11 (0.15)	0.08 (0.13)	0.07 (0.11)	0.26 (0.28)	0.17 (0.21)
% students with college educated father	0.14 (0.21)	0.14 (0.21)	0.12 (0.17)	0.09 (0.14)	0.08 (0.12)	0.30 (0.31)	0.20 (0.24)
Average lagged SIMCE score Math	-0.01 (0.49)	-0.01 (0.49)	-0.01 (0.50)	-0.12 (0.45)	-0.11 (0.47)	0.31 (0.54)	0.15 (0.52)
Average lagged SIMCE score Spanish	0.01 (0.42)	0.01 (0.42)	0.03 (0.42)	-0.08 (0.39)	-0.01 (0.41)	0.24 (0.45)	0.13 (0.46)
Observations	4966	4515	451	2145	113	472	96

Note: The table presents the mean and standard deviation (in parenthesis) of several variables for Spanish teachers in our main TVA analysis sample. Column 1 presents the statistics for the full sample. Columns 2 and 3, divide the sample between teachers that started their studies before and after the reform. Columns 4 and 5 present the summary statistics for those scoring below 500 points. Columns 6 and 7 present the summary statistics for those scoring above 600 points.

Table B10: Effect of a 1 Standard deviation increase in teacher and school effects

	With weights		Without weights	
	Math	Spanish	Math	Spanish
<i>Including controls and only contemporaneous classrooms</i>				
School	.24	.19	.24	.2
Teacher	.09	.09	.08	.08
<i>Without controls and all different classrooms</i>				
School	.22	.13	.23	.14
Teacher	.11	.11	.09	.11
<i>Without controls and only contemporaneous classrooms</i>				
School	.25	.19	.26	.21
Teacher	.11	.09	.09	.08

Table B11: Robustness - effect of the reform on TVA using different before/after specifications

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Math						
Post-reform cohort	0.036** (0.017)	0.037** (0.016)	0.035** (0.016)	0.047*** (0.017)	0.038** (0.016)	0.037** (0.016)
Observations	193143	206094	206094	206094	179445	130412
Sample Teachers	4,133	4,386	4,386	4,386	4,160	3,556
Panel B. Spanish						
Post-reform cohort	-0.001 (0.015)	-0.004 (0.014)	-0.005 (0.014)	0.007 (0.015)	-0.007 (0.015)	-0.007 (0.015)
# Students	207,617	225,584	225,584	225,584	196,655	142,267
# Teachers	4,619	4,966	4,966	4,966	4,662	3,891
Additional covariates	Yes	No	No	No	No	No
Individual covariates	Yes	No	No	No	Yes	Yes
School Covariates	Yes	Yes	Yes	No	Yes	Yes
Class covariates	Yes	Yes	No	No	Yes	Yes
Years of Experience	All	All	All	All	<6	<3

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The table shows OLS estimates from model (7). Standard errors are clustered at the teacher-by-year level, and shown in parentheses. Additional covariates include the lagged GPA of the student and the share of days in which she went to school the previous year.

Table B12: Robustness - effect of the reform on TVA for on-time-finishers

	Math teachers		Spanish teachers	
	(1)	(2)	(3)	(4)
Post-reform cohort	0.074* (0.042)	0.064** (0.026)	0.004 (0.051)	-0.000 (0.028)
# Students	22,383	39,359	26,693	44,562
# Teachers	937	1,601	1,060	1,810
Δ Std PSU	0.284	0.254	0.272	0.294
% Post-reform teachers	0.319	0.239	0.316	0.235
Years after PSU	6	7	6	7

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The table shows OLS estimates from model (7), restricting the sample only to those teachers that finish their studies on time and are first observed as teachers six (in columns (1) and (3)) or seven (in columns (2) and (4)) years after taking the PSU. Additionally, we restrict the student observations to those taught on the first year the teacher is observed. Standard errors are clustered at the teacher-by-year level, and shown in parentheses.

Table B13: Effect of the reform on TVA - Heterogeneity by vulnerability

	Math teachers		Spanish teachers	
	(1) Low vulnerability	(2) High vulnerability	(3) Low vulnerability	(4) High vulnerability
Post-reform cohort	0.035* (0.020)	0.044* (0.024)	0.004 (0.020)	-0.009 (0.022)
# Students	125,176	70,542	137,834	72,460
# Teachers	2,051	2,042	2,287	2,277
Δ Std PSU	0.45	0.27	0.47	0.45
Δ % PSU<500	-0.20	-0.15	-0.26	-0.25
Δ % PSU $\geq$ 600	0.23	0.08	0.14	0.13
Baseline % PSU<500	0.31	0.49	0.41	0.60
Baseline % PSU $\geq$ 600	0.18	0.09	0.12	0.04
Average IVE	0.63	0.88	0.63	0.88
Average very high IVE	0.38	0.69	0.37	0.69

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The table shows OLS estimates for the coefficients of interest in model (7). High vulnerability schools are defined as those with an IVE-SINAE above the median (78%). Standard errors are clustered at the teacher-by-year level, and shown in parentheses.

Table B14: Robustness - effect beyond ability - Math teachers

	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. PSU < 500</i>						
Post-reform cohort	0.058 (0.036)	0.067* (0.036)	0.061* (0.035)	0.048 (0.038)	0.062* (0.036)	0.061* (0.036)
# Students	67,992	70,092	70,092	70,092	59,507	41,529
# Teachers	1,577	1,618	1,618	1,618	1,521	1,263
<i>Panel C. PSU ∈ [500, 600)</i>						
Post-reform cohort	0.005 (0.021)	-0.000 (0.021)	-0.001 (0.021)	0.019 (0.021)	0.004 (0.021)	0.002 (0.021)
# Students	98,506	103,880	103,880	103,880	91,258	66,436
# Teachers	1,941	2,043	2,043	2,043	1,950	1,678
<i>Panel D. PSU ≥ 600</i>						
Post-reform cohort	0.054* (0.028)	0.064** (0.027)	0.063** (0.026)	0.054* (0.029)	0.065** (0.027)	0.062** (0.027)
# Students	26,645	32,122	32,122	32,122	28,680	22,447
# Teachers	615	725	725	725	689	615
Additional covariates	Yes	No	No	No	No	No
Individual covariates	Yes	No	No	No	Yes	Yes
School Covariates	Yes	Yes	Yes	No	Yes	Yes
Class covariates	Yes	Yes	No	No	Yes	Yes
Years of Experience	All	All	All	All	<6	<4

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The table shows OLS estimates from model (7). Standard errors are clustered at the teacher-by-year level, and shown in parentheses. Additional covariates include the lagged GPA of the student and the share of days in which she went to school the previous year.

Table B15: Correlation between teacher quality and inputs

	Math			Spanish		
	(1)	(2)	(3)	(4)	(5)	(6)
PSU score	0.034*** (0.008)	0.035*** (0.008)	0.024*** (0.008)	-0.013* (0.007)	-0.011 (0.008)	-0.016** (0.008)
Years of experience	0.016 (0.011)	0.017 (0.011)	0.016 (0.011)	0.032*** (0.009)	0.033*** (0.009)	0.032*** (0.009)
Choose again		0.033** (0.016)	0.037** (0.016)		0.029** (0.014)	0.031** (0.014)
Observations	812	812	812	905	905	905
R^2	0.046	0.051	0.072	0.018	0.023	0.031
Controls	No	No	Yes	No	No	Yes

Note: * $p < 0.05$, *** $p < 0.01$. The table shows the estimates of an OLS regression where we estimate the correlation of teacher quality with motivation, years of experience, and PSU score. PSU scores here are standardized to have a mean of zero and a standard deviation of one. All regressions control for a polynomial of order two on years of experience, a polynomial of order two on PSU scores. Columns 3 and 6 additionally control for the type of school the teacher works at (public, private, rural). Robust standard error reported in parentheses.

Table B16: Before-After comparison beyond academic ability - motivation

	Math teachers		Spanish teachers	
	(1)	(2)	(3)	(4)
Post-reform cohort	0.028 (0.045)		0.009 (0.043)	
Post-reform cohort $\times$ PSU < 500		0.207*** (0.074)		-0.009 (0.067)
Post-reform cohort $\times$ PSU $\in [500, 600)$		0.010 (0.065)		0.005 (0.057)
Post-reform cohort $\times$ PSU ≥ 600		-0.070 (0.071)		0.055 (0.079)
Observations	1079	1079	1210	1210
Baseline mean	0.573		0.572	

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The table shows OLS estimates for the coefficients of a post-reform cohort indicator, in a model that additionally controls for a polynomial of order two on (standard) PSU score, as well as fixed effects for years of experience, gender, year, public, private, and rural schools. Robust standard errors are reported in parentheses.

Table B17: Beyond ability - adjusting for higher education and school characteristics

	Baseline	Program characteristics			School characteristics			All
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Full sample</i>								
Post-reform cohort	0.034** (0.016)	0.035** (0.016)	0.033** (0.016)	0.033** (0.016)	0.035** (0.016)	0.029* (0.016)	0.030* (0.016)	0.029* (0.016)
<i>Panel B. By score</i>								
Post-reform cohort $\times$ PSU < 500	0.065* (0.036)	0.067* (0.036)	0.063* (0.036)	0.065* (0.036)	0.066* (0.036)	0.065* (0.035)	0.066* (0.035)	0.065* (0.035)
Post-reform cohort $\times$ PSU $\in [500, 600)$	0.001 (0.021)	0.002 (0.021)	-0.001 (0.021)	-0.001 (0.021)	0.002 (0.021)	-0.005 (0.021)	-0.004 (0.021)	-0.006 (0.020)
Post-reform cohort $\times$ PSU ≥ 600	0.065** (0.026)	0.065** (0.026)	0.066** (0.026)	0.064** (0.026)	0.067** (0.026)	0.057** (0.025)	0.058** (0.025)	0.057** (0.025)
# Students	206,094	206,094	206,094	206,094	206,094	206,094	206,094	206,094
# Teachers	4,386	4,386	4,386	4,386	4,386	4,386	4,386	4,386
PSU	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
HE program type	No	No	Yes	Yes	No	No	No	Yes
Program size & peer PSU	No	Yes	No	Yes	No	No	No	Yes
Colleague characteristics	No	No	No	No	No	Yes	Yes	Yes
IVE-SINAE	No	No	No	No	Yes	No	Yes	Yes
All controls	No	No	No	No	No	No	No	Yes

Note: * $p < 0.1$ ** $p < 0.05$, *** $p < 0.01$. The table shows OLS estimates for the coefficients of interest in model (7). We control for a polynomial of order two of PSU score. Program characteristics (HE program type, program size and average peer PSU) are the ones described in Appendix F.1. Standard errors are clustered at the teacher-by-year level, and shown in parentheses.

C Recruitment - Additional Results

In this section, we present additional results related to the section on recruitment.

Characterizing enrollees Appendix Table B2 presents the results of performing our RD analysis, but using observable characteristics as dependent variables. We observe that marginally incentivized students ($PSU \geq 600$) are similar to their peers right below the threshold. The only noteworthy difference relates to the family income quintile. Students who are enrolled in teacher training programs just above the scholarship cutoff are more likely to come from poorer families, compared to those just below. Similarly, we observe that students enrolling in teacher training programs right below the 500 cutoff come from richer families.

Where do they come from? Students scoring above 600 select different *majors* within the pool of teaching colleges. When dividing degrees by the grade levels they prepare individuals to teach (e.g., secondary or primary school), we see that they are more (less) likely to choose a high-school (primary school) career (see Table C19). The same holds true for those marginally eligible at the 500 cutoff. Finally, regarding the counterfactual college career path, we show in Figure C26 that BVP takers would not have opted to forgo enrolling in college. The effect on general enrollment is estimated to be close to zero at the cutoff. Students at the 500 cutoff are instead more likely to enroll. Therefore, some students who see their choice set reduced decide not to pursue a higher education career.

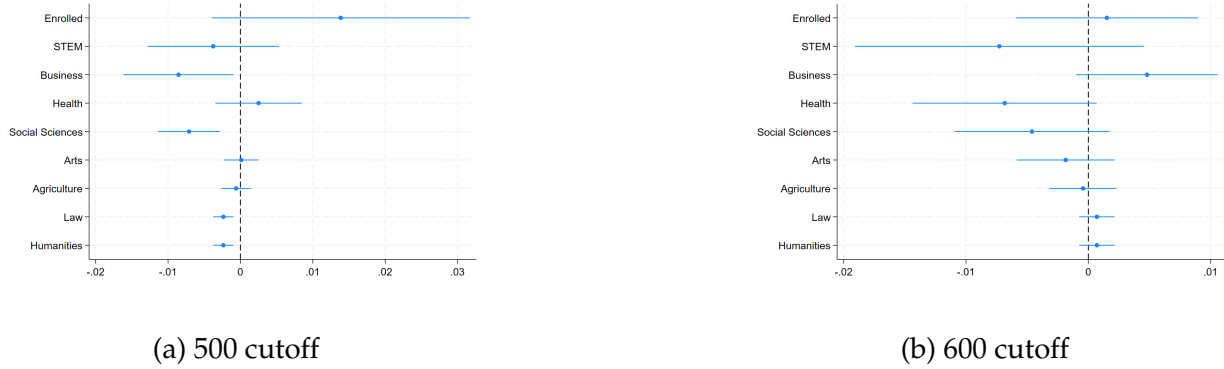
Do they choose different types of teacher training programs? It is difficult to infer from the name of the program in which students enroll whether this choice preconditions them to become mathematics or Spanish teachers. Therefore, in the recruitment section, we cannot directly test whether students are more likely to select into teaching math or Spanish. What we can test, however, is whether they choose programs that naturally track into teaching at the pre-school, primary, or high-school level. For a subset of programs, it is still not possible to determine the eventual teaching track. In Table C19, we examine whether, conditional on enrolling in a teacher training program, marginally eligible scholarship recipients are more likely to select one type of program over another. The results indicate that the distribution of program types changes slightly above the cutoff: enrollment in high-school-oriented programs increases, while enrollment in primary school programs decreases.

Table C18: Characteristics enrollees

	(1) PSU $\leq$ 500	(2) PSU $\geq$ 600
High School GPA	-0.072*** (0.023) [5.590]	-0.031** (0.015) [5.868]
Female	-0.008 (0.035) [0.671]	-0.021 (0.017) [0.624]
Father High School	0.054** (0.023) [0.516]	0.044** (0.020) [0.655]
Mother High School	0.041** (0.016) [0.617]	-0.002 (0.018) [0.746]
Any Parent College	0.053** (0.025) [0.302]	0.020 (0.019) [0.527]
# Family Members	0.126** (0.062) [4.366]	0.038 (0.052) [4.471]
Single Parent	0.018 (0.016) [0.171]	0.014 (0.012) [0.135]
Father Working	0.005 (0.024) [0.638]	0.001 (0.017) [0.691]
Mother Working	0.005 (0.022) [0.391]	0.025 (0.018) [0.426]
Income Quintile 1	-0.029* (0.016) [0.369]	0.048** (0.022) [0.259]
Income Quintile 2	-0.015 (0.018) [0.290]	-0.004 (0.016) [0.240]
Income Quintile 3	-0.004 (0.019) [0.167]	-0.040** (0.016) [0.210]
Income Quintile 4	0.017 (0.017) [0.124]	-0.014 (0.017) [0.192]
Income Quintile 5	0.031*** (0.009) [0.050]	0.010 (0.012) [0.099]

Note: The table presents regression discontinuity parameters using predetermined characteristics as outcomes. The bandwidth is fixed to 24 for the 500 cutoff, to 50 for the 600 cutoff, and the model is estimated using a triangular weighted kernel for all specifications. Standard errors are clustered at the PSU level and are shown in parentheses. Baseline means are presented below the standard errors, in brackets.

Figure C26: Effect on general enrollment and enrollment in other fields



Note: All outcomes are binary indicators. All specifications are estimated using weighted local linear regressions. Bandwidths are chosen optimally according to Calonico et al. (2019). Standard errors are clustered at the PSU test score level.

Table C19: Categories of programs within teacher colleges

	(1) Preschool	(2) Primary	(3) High School	(4) Other
<i>Panel A. 600 cutoff</i>				
RD Estimate	-0.010 (0.010)	-0.022* (0.013)	0.039** (0.018)	-0.005 (0.023)
Baseline Mean	0.061	0.123	0.337	0.475
Bandwidth	43	48	58	47
Observations	9,184	10,391	12,472	10,076

Note: ** $p < 0.05$, *** $p < 0.01$. The Table presents estimated discontinuity in the relationship between PSU and outcomes of interest at the 600 and 500 cutoffs. The sample is composed of individuals who enrolled in a teacher college, and the outcomes are binary indicators for enrolling in a program that leads to teaching in preschool, elementary school, and high school. We add an indicator for when the program does not specify the specialization (others).

D Cost-Benefit Analysis

This section provides a back-of-the-envelope cost-benefit analysis of the reform, focusing on mathematics teachers, the only subject for which the reform produced a measurable increase in teacher value-added.

Cost. The BVP covers the full tuition of scholarship recipients for the duration of their teacher training program. Using administrative data on tuition fees and program duration, we estimate an average annual tuition of approximately 2.05 million CLP and an average program duration of 4.6 years, yielding a total cost of approximately 9.4 million CLP for a recipient who completes the program.

The scholarship is paid during enrollment, however, regardless of whether the recipient graduates and is eventually employed as a teacher, so the cost of producing one employed teacher exceeds the tuition of a single completed degree. The teacher pipeline in Appendix Figure A13 quantifies this: among post-reform $PSU \geq 600$ entrants, of the 1,560 who enroll in teacher training each year, 1,106 graduate (71%) and 943 are employed in a school (60% of those who enroll). We account for the scholarship spending on enrollees who do not become teachers as follows. Graduates receive the full 9.4 million CLP; dropouts, assumed to leave uniformly over the program, receive on average half (4.7 million CLP). The expected tuition per enrollee is therefore $0.71 \times 9.4 + 0.29 \times 4.7 \approx 8.0$ million CLP, and since only 60% of enrollees end up employed, the cost per employed teacher is $8.0/0.60 \approx 13.3$ million CLP. Finally, since the 0.038 SD TVA effect is computed across all post-reform math teachers, of whom only 29% are BVP recipients (117 out of 405), the cost per teacher for producing this average effect is $13.3M \times 117/405 \approx 3.84$ million CLP.

Benefit. We do not observe a link between teacher value-added and earnings in Chile, so we transfer the elasticity estimated by Chetty et al. (2014) and apply it to Chilean lifetime earnings. Chetty et al. estimate that a 1 SD increase in teacher value-added raises a student's annual earnings, measured at age 28, by 1.34%; assuming this percentage gain persists over the working life, it raises the present value of lifetime earnings by the same 1.34%. In our data a 1 SD increase in teacher value-added corresponds to a 0.10 SD increase in student test scores (the standard deviation of teacher effects, Table 2), so our ITT effect of 0.038 SD amounts to $0.038/0.10 = 0.38$ SD of teacher value-added, i.e. a $0.38 \times 1.34\% = 0.51\%$ increase in the present value of the lifetime earnings of the average student taught by a post-reform teacher. The earnings return to cognitive skills may differ between Chile and the US, so we adjust for this using the returns to numeracy estimated by

Hanushek et al. (2015), which are lower at the OECD average than in the US (18% versus 28% per SD); scaling by $18/28 = 0.643$ gives a $0.51\% \times 0.643 = 0.33\%$ gain in lifetime earnings.

To value this gain we need the present value of a Chilean student's lifetime earnings. Following the procedure in Chetty et al. (2014), who build a cross-sectional age-earnings profile from a random sample of the US population, we construct the analogous profile for Chile from the 2013 CASEN survey, the wave closest to the period of our cost figures. For each single year of age we compute mean annual labor income (monthly corrected labor income times twelve, counting individuals with no labor income as zero). The profile over ages 28–65 is estimated from a nationally representative sample of 100,680 individuals. We then take the present value at age 12 (the mean age at which students are taught) of this profile, letting earnings grow at 2% per year and discounting at rate r :

$$PV(r) = \sum_{a=28}^{65} \text{earn}(a) \frac{(1.02)^{a-12}}{(1+r)^{a-12}}, \quad (11)$$

where $\text{earn}(a)$ is mean annual labor income at age a , in 2013 CLP. This gives a present value of lifetime earnings of 118.2, 65.5, and 29.4 million CLP at discount rates of 3%, 5%, and 8%, respectively. The benefit per student is 0.33% of this present value, $b(r) = 0.0033 \times PV(r)$, e.g. approximately 215,000 CLP at the 5% rate.

Career benefits. We assume each teacher teaches approximately $n = 20$ students per year, consistent with our summary statistics. Because cohorts are taught at different points in time, we value costs and benefits at a common date, the year the teacher enters the profession (the scholarship is paid over the earlier training years, so we compound it forward to that date). The total career benefit per teacher is:

$$B = \sum_{t=1}^T \frac{n \times b(r) \times \delta(t)}{(1+r)^t}, \quad (12)$$

where $b(r)$ is the per-student benefit defined above, T is the number of years the teacher remains in the profession, and $\delta(t)$ governs the persistence of the per-cohort effect over the career. The discount rate r enters both through $b(r)$, which converts each student's future earnings gains into a present value, and through the $(1+r)^t$ term, which discounts the benefits generated by later cohorts of students back to the teacher's entry date.

We report two complementary sets of scenarios. First, over a full $T = 30$ -year career, we vary the persistence of the effect through $\delta(t)$: (i) *constant* ($\delta(t) = 1$), where the 0.038

SD effect is permanent; (ii) *linear fade* ($\delta(t) = 1 - t/30$), where it declines to zero over 30 years; and (iii) *exponential fade* ($\delta(t) = 0.5^{t/10}$), where it halves every 10 years. Second, holding the effect constant ($\delta(t) = 1$), we vary teacher retention, considering $T = 5, 10$, and 15 years.

Results. Table D20 presents the benefit-cost ratio (BCR). At a 5% discount rate, the BCR is 15.3 under a constant effect over a full career and 9.2 and 7.7 under linear and exponential fade; restricting attention to teachers who stay 5, 10, or 15 years gives BCRs of 4.3, 7.7, and 10.4. The program passes a social cost-benefit test in every scenario and at all three discount rates: even in the most conservative case, an 8% discount rate with only 5 years of retention, the BCR is 1.7.

These estimates are conservative in several respects. They exclude non-earnings benefits documented by Chetty et al. (2014) (higher college attendance, lower teen pregnancy rates), potential TVA growth with experience, and spillovers to non-cognitive skills. On the other hand, they rely on transferring a US causal elasticity to Chile, which remains a limitation. Implicit in this argument is the assumption that what matters for earnings is the impact of a one-standard-deviation increase in TVA, regardless of the variability of TVA. This may be problematic because our estimated standard deviation of teacher effects (0.10) is smaller than the one implicit in Chetty et al. (2014) (0.14), so a one-standard-deviation improvement is associated with a smaller gain in student outcomes in our setting. In Table D21 we account for this by scaling the benefits by the ratio of the two standard deviations. Practically, the benefit-cost ratios continue to exceed one in all scenarios.

Table D20: Cost-benefit analysis

PV per student		30-year career			Retention		
Discount rate	(lifetime)	Constant	Linear fade	Exp. fade	5 years	10 years	15 years
<i>Per teacher, million CLP (PV per student in first column)</i>							
3%	118.2	151.7	84.3	70.6	35.4	66.0	92.4
5%	65.5	65.9	39.6	33.2	18.6	33.1	44.5
8%	29.4	21.7	14.3	12.1	7.7	12.9	16.5
<i>Benefit-cost ratio</i>							
3%		36.9	20.5	17.2	8.6	16.1	22.5
5%		15.3	9.2	7.7	4.3	7.7	10.4
8%		4.7	3.1	2.6	1.7	2.8	3.6

Note: The table presents a cost-benefit analysis of the reform for mathematics teachers, entirely in Chilean pesos. The benefit applies the teacher value-added earnings elasticity of Chetty et al. (2014) (1.34% of lifetime earnings per 1 SD of teacher value-added), scaled by OECD-average returns to cognitive skills (Hanushek et al., 2015), to the present value of a Chilean student's lifetime earnings. The latter is built from a cross-sectional age-earnings profile (mean annual labor income by age, including non-earners) estimated from the 2013 CASEN survey using national expansion factors, discounted over ages 28–65 with 2% annual wage growth. Career benefits assume 20 students per year, discounted to the teacher's entry date. The "30-year career" columns hold the teacher for a full career and vary the persistence of the per-cohort value-added effect: constant (permanent), linear fade (declines to zero over 30 years), and exponential fade (halves every 10 years). The "Retention" columns hold the effect constant and vary how long the teacher remains in the profession (5, 10, or 15 years). The cost per teacher (3.84 million CLP, compounded to the entry date) accounts for the ITT nature of the estimates (only 29% of post-reform math teachers received BVP) and for the enrollment-to-employment pipeline: the scholarship is paid to all enrollees, but only about 60% become employed teachers, and dropouts (assumed uniform over the program) receive partial tuition.

Table D21: Benefit-cost ratios adjusted for the dispersion of teacher value-added

Discount rate	30-year career			Retention		
	Constant	Linear fade	Exp. fade	5 years	10 years	15 years
3%	26.4	14.7	12.3	6.2	11.5	16.1
5%	11.0	6.6	5.5	3.1	5.5	7.4
8%	3.4	2.2	1.9	1.2	2.0	2.6

Note: Benefit-cost ratios computed as in Table D20, but scaling the per-student benefit by the ratio of our estimated standard deviation of teacher effects to the one implicit in Chetty et al. (2014) (0.10/0.14). All other assumptions are unchanged.

E Theoretical Framework

We develop a simple framework to formalize how the reform may affect average teacher performance, or productivity. At its core, the policy leverages admission test scores under the assumption that academic ability is an important predictor of teaching effectiveness. While this channel captures a direct form of selection, it is unlikely to be the only one at play. Individuals whose decision to enter the teaching profession responds to the reform may differ along other unobserved traits that also influence teaching quality. Depending on how these traits correlate with academic ability and with the decision to teach, the reform's overall impact on productivity may be amplified or attenuated.

Setup. We model entry into teaching as a Roy-style occupational choice. The framework is deliberately minimal to make the mapping from primitives to composition effects transparent. It follows the classic occupational-choice logic of Roy (1951) and the more recent “motivated agents” literature (Besley and Ghatak, 2005; Dal Bó et al., 2013), adapted to teacher recruitment.

Productivity for an individual with academic ability A in teaching and in the outside (non-teaching) sector is given by

$$\begin{aligned} P_T &= \kappa_T A + \epsilon_T, \\ P_O &= \kappa_O A + \epsilon_O. \end{aligned} \tag{13}$$

Productivity depends on academic ability (A), measured in our setting by the observed PSU test score; a teaching residual ability (ϵ_T), capturing effectiveness as a teacher beyond A ; and an outside-sector residual ability (ϵ_O). The coefficients κ_T and κ_O measure how strongly academic ability translates into productivity in each sector.

The utilities from choosing teaching versus the outside sector are

$$\begin{aligned} U_T &= \bar{w}_T + \epsilon_M + r_T P_T + S(A), \\ U_O &= \bar{w}_O + r_O P_O. \end{aligned} \tag{14}$$

Here, $\bar{w}_T$ and $\bar{w}_O$ denote average wages in the teaching and outside sectors, respectively, and r_O is a pay-productivity slope, i.e. the return to a unit of productivity in the outside sector. We normalize $r_T = 0$.²⁶ The utility from teaching, U_T , also depends on a

²⁶We set $r_T = 0$ to simplify the exposition. This is empirically reasonable for Chile during the studied period: when students made their enrollment decisions, teachers in public schools were paid according to a wage scale that was almost entirely determined by seniority, so r_T is plausibly close to zero. Wage dispersion

preference-for-the-job parameter (ϵ_M), or intrinsic motivation for teaching. We assume that $(\epsilon_M, \epsilon_T, \epsilon_O)$ follow a joint distribution that allows for correlation across the three dimensions.

The policy term $S(A)$ captures in reduced form how the policy alters the net payoff from choosing a teaching career based on test scores. In our setting, $S(A) > 0$ for $A \geq 600$, as the scholarship directly lowers the cost of enrollment in a teacher training program, while $S(A) < 0$ for $A < 500$, as restricted access to teacher training programs reduces the expected quality of the best available match between the individual and the remaining institutions.

A student of academic ability A observes $(\epsilon_T, \epsilon_O, \epsilon_M)$ and chooses teaching if $U_T \geq U_O$. Thus, our model nests two strands of literature: (i) selection when outside wages are more responsive to productivity than public-sector wages (Bacolod, 2007; Tincani, 2021), and (ii) selection with non-pecuniary “mission” motives (ϵ_M) that may be correlated with productivity (Dal Bó et al., 2013; Ashraf et al., 2020). Collecting constants and terms that depend only on A , define

$$K(A) := (\bar{w}_T - \bar{w}_O) - r_O \kappa_O A, \quad (15)$$

and the selection index

$$Z := -r_O \epsilon_O + \epsilon_M, \quad (16)$$

so that

$$U_T - U_O = K(A) + Z + S(A). \quad (17)$$

An individual enters the teaching profession if and only if $Z \geq -K(A) - S(A) = C(A)$, where $C(A)$ represents the choice cutoff. Hence, the decision is driven by two forces: (i) the outside sector pulls away individuals whose productivity is well rewarded outside ($-r_O \epsilon_O$), and (ii) the motivation (ϵ_M) which directly raises the payoff from teaching.

Characterizing entrants and stayers. From a baseline without $S(A)$, the introduction of the policy term $S(A)$ shifts the choice cutoff $C(A)$ in the selection index $Z = -r_O \epsilon_O + \epsilon_M$ for the targeted ranges of academic ability. A scholarship ($S(A) > 0$) lowers the cutoff, attracting individuals who, in the absence of $S(A)$, would have fallen just below the threshold. Conversely, a reduction of the available choice set of university programs ($S(A) < 0$) raises the cutoff, retaining only a stricter upper tail of the distribution of Z .

in voucher schools was somewhat higher but still considerably lower than in the private sector (Tincani, 2021). Allowing $r_T > 0$ would add a positive “own-productivity” term to $\text{Cov}(\epsilon_T, Z)$ without altering the qualitative two-channel interpretation.

Under the assumption that $(\epsilon_T, \epsilon_O, \epsilon_M)$ are jointly normally distributed, the expected change in ϵ_T when $S(A)$ is introduced can be written as

$$\Delta \mathbb{E}[\epsilon_T] = \frac{\text{Cov}(\epsilon_T, Z)}{\text{Var}(Z)} \Delta \mathbb{E}[Z], \quad (18)$$

where Δ denotes “post-reform minus pre-reform” for the selected A group. The shift in the cutoff $C(A)$ implies $\Delta \mathbb{E}[Z] < 0$ with the introduction of the scholarship and $\Delta \mathbb{E}[Z] > 0$ under the case of choice set reduction. Thus, the sign of $\Delta \mathbb{E}[\epsilon_T]$ is determined by the covariance $\text{Cov}(\epsilon_T, Z)$, which can be rewritten as:

$$\text{Cov}(\epsilon_T, Z) = \sigma_T (\rho_{T,M} \sigma_M - r_O \rho_{T,O} \sigma_O), \quad (19)$$

where $\sigma_T^2 = \text{Var}(\epsilon_T)$, $\sigma_O^2 = \text{Var}(\epsilon_O)$, $\sigma_M^2 = \text{Var}(\epsilon_M)$ and $\rho_{.,.}$ denote the corresponding correlations.

There are two channels that determine the sign of $\text{Cov}(\epsilon_T, Z)$. The first is an *outside pull* channel: if the traits that make someone a good teacher are also rewarded in the outside sector (i.e., large r_O and positive correlation $\rho_{T,O}$), then $\text{Cov}(\epsilon_T, Z)$ is pulled downward. The second is a *motivation* channel: if intrinsic motivation to teach is positively related to teaching effectiveness, then $\text{Cov}(\epsilon_T, Z)$ is pulled upward.

When the cutoff $C(A)$ is lowered (so $\Delta \mathbb{E}[Z] < 0$), a scholarship increases the average ϵ_T among entrants precisely when the outside pull dominates the motivation channel (i.e., when $r_O \rho_{T,O} \sigma_O > \rho_{T,M} \sigma_M$). The intuition is straightforward: in the absence of the policy, if the outside pull channel dominates, individuals who self-select into teaching are more likely to be those with a lower ϵ_T . Thus, when $S(A)$ is introduced, the marginal entrant is *positively selected* on ϵ_T .²⁷ By contrast, when the cutoff $C(A)$ is higher, average quality rises if the motivation channel dominates (i.e., $\rho_{T,M} \sigma_M > r_O \rho_{T,O} \sigma_O$).

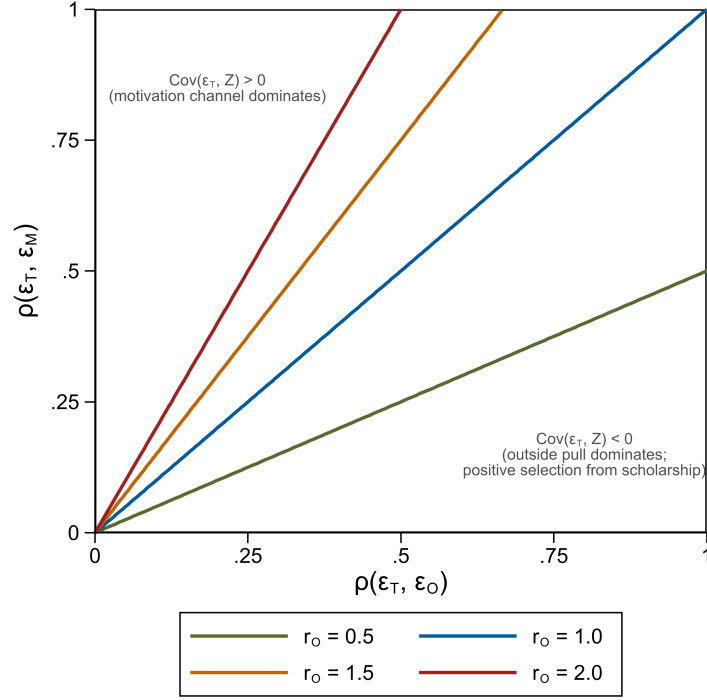
Figure E27 plots the line $\rho_{T,M} = r_O \rho_{T,O}$, that divides the regions where $\text{Cov}(\epsilon_T, Z)$ is positive and negative, for different values of r_O .

Overall change in value-added. What are the implications for the overall change in P_T ? In general, it reflects two components: (i) the direct effect of shifting the distribution of A , which raises productivity through the $\kappa_T A$ term, and (ii) the change in residual teaching quality, ϵ_T , which depends on how the reform alters the composition of entrants.

For example, when the motivation channel prevails, scholarships increase the share of high- A individuals, but at the same time the marginal entrants are negatively selected on

²⁷This intuition closely parallels that in Borjas (1987). See also Bacolod (2007) for an application of the model proposed by Borjas (1987) to selection into teaching.

Figure E27: The line $\rho_{T,M} = r_O \rho_{T,O}$ where $\text{Cov}(\epsilon_T, Z) = 0$



Note: Each line traces $\rho_{T,M} = r_O \rho_{T,O}$, along which $\text{Cov}(\epsilon_T, Z) = 0$, for one r_O . Below the line, a scholarship induces positive selection on ϵ_T . Normalization: $\sigma_T = \sigma_M = \sigma_O = 1$; r_O on the figure absorbs the variance ratio σ_O/σ_M .

ϵ_T . In this case the reform generates an academic ability–motivation trade-off: average value-added rises through stronger academic ability, yet falls at the margin due to lower residual quality. By contrast, when the policy relies on disincentives, the marginal exit is negatively selected on ϵ_T , so the remaining teachers combine higher academic ability with stronger residual quality, and P_T increases unambiguously.

Proofs and additional results

Proof of equation (9). Distributional assumption. Let $(\epsilon_T, \epsilon_O, \epsilon_M)$ be jointly normally distributed with mean zero and some positive definite covariance matrix Σ . Define the selection index

$$Z := -r_O \epsilon_O + \epsilon_M.$$

Because Z is a linear combination of jointly normal variables, (ϵ_T, Z) is jointly normal. Let $\mu_T := \mathbb{E}[\epsilon_T] = 0$, $\mu_Z := \mathbb{E}[Z] = 0$, $\sigma_Z^2 := \text{Var}(Z)$, and $\sigma_{TZ} := \text{Cov}(\epsilon_T, Z)$.

Policy, cutoff, and selection event. Fix an ability bin A so that $K(A)$ is constant within the

bin. The policy $S(A)$ shifts the cutoff in the selection rule:

$$\text{choose teaching} \iff Z \geq C(A) := -K(A) - S(A).$$

Let $c_0 := -K(A)$ be the pre-reform cutoff and $c_1 := C(A)$ the post-reform cutoff (with $c_1 \neq c_0$ whenever $S(A) \neq 0$). For any cutoff c , denote the selected set by $\{Z \geq c\}$ and write

$$\mathbb{E}_c[\cdot] := \mathbb{E}[\cdot \mid Z \geq c].$$

Step 1: Conditional mean of a bivariate normal is linear. Since (ϵ_T, Z) is jointly normal,

$$\mathbb{E}[\epsilon_T \mid Z = z] = \mu_T + \frac{\text{Cov}(\epsilon_T, Z)}{\text{Var}(Z)}(z - \mu_Z) = \frac{\sigma_{TZ}}{\sigma_Z^2}z,$$

where we used $\mu_T = \mu_Z = 0$. This follows from the least-squares projection property of the multivariate normal.

Step 2: Law of iterated expectations under selection. Applying the tower property,

$$\mathbb{E}_c[\epsilon_T] = \mathbb{E}\left[\mathbb{E}[\epsilon_T \mid Z] \mid Z \geq c\right] = \mathbb{E}\left[\frac{\sigma_{TZ}}{\sigma_Z^2}Z \mid Z \geq c\right] = \frac{\sigma_{TZ}}{\sigma_Z^2}\mathbb{E}_c[Z].$$

Therefore, for any two cutoffs c_0, c_1 ,

$$\mathbb{E}_{c_1}[\epsilon_T] - \mathbb{E}_{c_0}[\epsilon_T] = \frac{\sigma_{TZ}}{\sigma_Z^2} \left(\mathbb{E}_{c_1}[Z] - \mathbb{E}_{c_0}[Z] \right).$$

□

Remark. *Explicit form for $\mathbb{E}_c[Z]$.* Because $Z \sim \mathcal{N}(0, \sigma_Z^2)$, the truncated normal mean is

$$\mathbb{E}_c[Z] = \sigma_Z \lambda(\alpha)$$

with

$$\alpha := \frac{c}{\sigma_Z} \quad \text{and} \quad \lambda(\alpha) := \frac{\phi(\alpha)}{1 - \Phi(\alpha)},$$

where ϕ and Φ are the standard normal PDF and CDF. This confirms that changing the cutoff c , via the introduction of $S(A)$, changes $\mathbb{E}_c[Z]$ and hence $\mathbb{E}_c[\epsilon_T]$, exactly in proportion to $\frac{\text{Cov}(\epsilon_T, Z)}{\text{Var}(Z)}$.

Prediction on changes in motivation

The theoretical framework also yields predictions for changes in ϵ_M , which follow the same logic and are likewise governed by the outside-option parameter r_O . In fact, equation (9) can be rewritten replacing ϵ_T with ϵ_M .

$$\Delta\mathbb{E}[\epsilon_M] = \frac{\text{Cov}(\epsilon_M, Z)}{\text{Var}(Z)} \Delta\mathbb{E}[Z], \quad (20)$$

In this case, the statistic that would determine the sign of $\Delta\mathbb{E}[\epsilon_M]$ would be $\text{Cov}(\epsilon_M, Z) = \sigma_M - r_O \rho_{M,O} \sigma_O$. The key difference is that motivation enters the index Z directly: its "own" term σ_M contributes one-for-one rather than being scaled by $\rho_{T,M}$. By contrast, the pull-out force on motivation operates through the correlation between motivation and outside productivity, $\rho_{M,O}$. Under the plausible assumption $\rho_{M,O} < \rho_{T,O}$, it is easy to show that the framework delivers a sharp prediction whenever $\text{Cov}(\epsilon_T, Z) > 0$. First,

$$\text{Cov}(\epsilon_T, Z) > 0 \iff r_O < \frac{\rho_{T,M}}{\rho_{T,O}} \cdot \frac{\sigma_M}{\sigma_O}.$$

Second,

$$\text{Cov}(\epsilon_M, Z) > 0 \iff r_O < \frac{1}{\rho_{M,O}} \cdot \frac{\sigma_M}{\sigma_O}.$$

Since $\rho_{M,O} < \rho_{T,O}$, it follows that

$$\frac{\rho_{T,M}}{\rho_{T,O}} \leq \frac{1}{\rho_{T,O}} \leq \frac{1}{\rho_{M,O}},$$

and therefore

$$r_O < \frac{\rho_{T,M}}{\rho_{T,O}} \cdot \frac{\sigma_M}{\sigma_O} \implies r_O < \frac{1}{\rho_{M,O}} \cdot \frac{\sigma_M}{\sigma_O}.$$

Hence $\text{Cov}(\epsilon_T, Z) > 0$ implies $\text{Cov}(\epsilon_M, Z) > 0$. When $\text{Cov}(\epsilon_T, Z) < 0$, the condition only implies

$$r_O > \frac{\rho_{T,M}}{\rho_{T,O}} \cdot \frac{\sigma_M}{\sigma_O},$$

which does not determine the sign of $\text{Cov}(\epsilon_M, Z)$. The prediction for motivation is therefore ambiguous in this case.

F The Role of Peers and Programs

A potential concern is that the increase in mathematics teacher quality we document could reflect changes in the higher-education programs themselves rather than changes in who selects into teaching. If the reform also altered the composition of teacher-training programs—for example, attracting more able peers—or if the quality of training programs improved over time independently of the reform, one might worry that these program-level changes are partly responsible for the effect we observe in TVA. In this section, we directly investigate this concern.

F.1 Peer Effects Within Programs

Changes in program characteristics around the reform. We document how the characteristics of the higher-education programs attended by mathematics teachers changed around the reform. We construct these measures by linking each teacher to administrative graduation records from the Ministry of Education, which also record each individual's year of enrollment, and identifying the cohort of students who enrolled in the same program in the same year. We focus on five characteristics: the average pre-college test score of the peers (peers' PSU score), the size of the program cohort (number of students who enrolled in the same program in the same year), the share of peers without a recorded test score, and the type of institution (CRUCH public versus CRUCH private). Table F22 reports pre- and post-reform averages for the full sample (Panel A) and separately for the three PSU subgroups defined in the main analysis (Panels B–D). For the full sample, the average peer PSU rises by 25 points, and program size shrinks by 7 graduates after the reform.

Within-program identification. To assess whether these changes in program characteristics marginally contribute to the documented effects in teacher quality, we use the standard within-group across-cohort design typically employed in the peer-effects literature. One of the first implementations of this approach is Hoxby (2000), which exploits idiosyncratic year-to-year variation in the composition of peers within the same group to identify peer effects on student achievement. In our setting, the relevant group is the higher-education program. We define a program as the combination of higher-education institution, campus location, program of study, and teaching schedule. The identifying variation arises from comparing students taught by teachers who enrolled in the same program in different years and were therefore exposed to different peer cohorts. Table F23 reports the distribution of teachers in our sample by the number of cohorts in which their

Table F22: Peer and program characteristics: pre- vs post-reform

	Pre-reform	Post-reform	Diff.	(s.e.)
<i>Panel A. Full sample</i>				
Average peer PSU	518.22	543.71	25.48***	(3.16)
Program cohort size	42.80	35.64	-7.16***	(1.37)
Share of peers w/ missing PSU	0.15	0.03	-0.11***	(0.01)
CRUCH public	0.32	0.26	-0.06***	(0.02)
CRUCH private	0.23	0.27	0.04*	(0.02)
<i>Panel B. PSU < 500</i>				
Average peer PSU	429.15	433.49	4.35	(5.87)
Program cohort size	43.73	29.87	-13.86***	(1.93)
Share of peers w/ missing PSU	0.21	0.08	-0.13***	(0.01)
CRUCH public	0.21	0.09	-0.12***	(0.03)
CRUCH private	0.11	0.07	-0.04	(0.03)
<i>Panel C. PSU 500–599</i>				
Average peer PSU	492.49	494.06	1.56	(4.85)
Program cohort size	40.41	32.82	-7.59***	(1.67)
Share of peers w/ missing PSU	0.10	0.02	-0.08***	(0.01)
CRUCH public	0.38	0.31	-0.07**	(0.03)
CRUCH private	0.29	0.25	-0.04	(0.03)
<i>Panel D. PSU ≥ 600</i>				
Average peer PSU	554.17	562.71	8.54	(9.65)
Program cohort size	47.80	44.74	-3.06	(3.56)
Share of peers w/ missing PSU	0.10	0.01	-0.09***	(0.01)
CRUCH public	0.42	0.31	-0.12***	(0.04)
CRUCH private	0.39	0.47	0.08*	(0.05)

Note: Pre- and post-reform averages of peer and program characteristics for mathematics teachers in the TVA estimation sample. “Pre-reform” refers to teachers in cohorts up to 2010; “Post-reform” refers to cohorts 2011 and later. The unit of observation is the teacher. Peer characteristics are computed within the program-cohort cell. Standard errors of the difference in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

program is observed. The vast majority of teachers belong to programs observed in many cohorts, providing variation for the within-program comparison.

Table F23: Variation used in the within-program design

Number of cohorts the teacher's program is observed in	Share
1	0.08
2-3	0.07
4-6	0.10
7-9	0.75

Note: The table reports the distribution of mathematics teachers in our sample by the number of cohorts in which their program is observed.

Specification. We estimate the following regression at the student level:

$$Y_{ipsjgtc} = \lambda_0 + \lambda_1 Y_{i,t-2} + \gamma_1 \mathbf{X}_{it} + \gamma_2 \mathbf{C}_{ct} + \gamma_3 \mathbf{S}_{st} + \gamma_4 \mathbf{T}_j + \alpha_t + \mu_g + \delta_p + \theta_{c(j)} + \beta P_{j,c(j)} + \eta_{jt} + \varepsilon_{ipsjgtc}, \quad (21)$$

where $Y_{ipsjgtc}$ is the (standardized) mathematics test score of student i , in school s , classroom c , taught by teacher j who enrolled in program p , in grade g and year t . $\mathbf{X}_{it}$, $\mathbf{C}_{ct}$ and $\mathbf{S}_{st}$ are the same student-, classroom- and school-level covariates as in the value-added model in equation (4). $\mathbf{T}_j$ is a vector of teacher controls (the teacher's own PSU score) and η_{jt} controls non-parametrically for teachers' years of experience. α_t and μ_g are year and grade fixed effects. The new terms δ_p and $\theta_{c(j)}$ are program and enrollment-cohort fixed effects. $P_{j,c(j)}$ is the peer or program characteristic of interest measured at the program-cohort cell of teacher j . The coefficient β therefore captures the marginal effect of changing peer characteristics within the same program across cohorts. Average peer PSU is standardized, so the coefficient measures the effect of a one-standard-deviation increase in peer PSU. As in the main analysis, we weight observations by the inverse of the number of students taught by each teacher and cluster standard errors at the teacher-by-year level.

Results. Table F24 reports the estimates. None of the peer or program characteristics has a statistically significant effect on student test scores, and the magnitudes of the estimates are very small. We interpret this as evidence that the changes in peer composition documented in Table F22 are unlikely to be responsible for the observed improvements in mathematics teacher quality.

Table F24: Effect of program-level peer characteristics on student math scores

	(1)	(2)	(3)	(4)
Average peer PSU (std)	-0.0033 (0.0226)			-0.0040 (0.0225)
Program cohort size		0.0001 (0.0002)		0.0001 (0.0002)
Share missing PSU			0.0171 (0.0453)	0.0164 (0.0456)
# Students	205,871	205,871	205,871	205,871
# Teachers	4,382	4,382	4,382	4,382

Note: OLS estimates from equation (21). The dependent variable is the standardized mathematics test score. The unit of observation is the student. Each peer variable is included separately in columns (1)–(3); column (4) includes all three peer variables jointly. All regressions absorb program fixed effects, enrollment-cohort fixed effects, year fixed effects, grade fixed effects, individual demographic fixed effects, and school-type fixed effects. They also include the teacher’s own PSU score (linear and squared), years of experience (non-parametrically), and the standard student, class, and school controls used in the main TVA estimation. Average peer PSU is standardized using the national PSU scale (mean 500, sd 100), so the coefficient measures the effect of a one-standard-deviation increase in peer PSU. Observations are weighted by the inverse of the number of times a teacher appears in the data. Standard errors clustered at the teacher-by-year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

F.2 Did Training Quality Improve?

The within-program peer-effects analysis above rules out that changes in peer composition explain the TVA gains we document for math teachers. A distinct but related concern is that the quality of teacher-training programs may have improved over time—whether as a consequence of the reform or due to independent trends or shocks. Simply comparing average program quality before and after the reform (e.g., estimating a program-level value added) would not resolve this concern, because any such measure would confound changes in training quality with changes in the composition of students enrolling.

To isolate the role of training quality, we exploit the fact that our theoretical framework does not predict selection effects for teachers with PSU scores in the 500–599 range. In fact, the reform did not change the incentives for these individuals to enter teaching, and our main results confirm that the average post-reform effect on TVA for this group is essentially zero (Table 5). If training quality improved at certain programs (or those programs experienced other contemporaneous shocks), however, *all* teachers at those programs would be affected, including those in the 500–599 PSU range. This group, therefore, provides a natural placebo sample for testing the training-quality hypothesis.

We further check that this null result does not mask heterogeneity across program types. We focus on two subgroups of programs. First, we restrict to BVP-eligible programs,

defined as programs with at least one graduate who has ever received the BVP scholarship. These programs experienced the largest change in the composition of incoming students after the reform; therefore, if training quality improved in those programs, we may worry that what we observe could partially reflect trends or shocks in program quality. Second, we split by CRUCH and non-CRUCH institutions. We estimate the same value-added specification as in our main analysis (Table 5), restricting the sample to mathematics teachers with PSU scores in the 500–599 range. Table F25 reports the results: column (1) is the full 500–599 benchmark, column (2) restricts to BVP-eligible programs, and columns (3) and (4) split by CRUCH and non-CRUCH institutions. The null result manifests in all subgroups: none of the coefficients is statistically distinguishable from zero, and they are close to zero in magnitude.

Table F25: Placebo test: post-reform TVA changes for 500–599 PSU teachers, by training program type

	Full sample (500–599 PSU) (1)	BVP-eligible (2)	CRUCH (3)	Non-CRUCH (4)
Post-reform	0.001 (0.021)	–0.007 (0.023)	0.010 (0.030)	–0.014 (0.032)
# Students	103,880	80,563	67,571	36,309
# Teachers	2,043	1,571	1,344	699
of which post-reform	195	168	107	88

Note: Each column reports the coefficient on a post-reform indicator from a separate regression estimated on the subsample of mathematics teachers with PSU scores in 500–599. Column (1) uses the full 500–599 subsample. Column (2) restricts to BVP-eligible programs: a program is BVP-eligible if at least one of its graduates ever received the BVP scholarship. Columns (3) and (4) restrict to CRUCH and non-CRUCH institutions, respectively. The specification follows the main TVA estimation (Table 5) and includes the same student, classroom, and school controls, plus the teacher’s own PSU score (linear and squared) and years of experience (non-parametrically). Observations are weighted by the inverse of the number of students taught by each teacher. Standard errors clustered at the teacher-by-year level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

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